HIKITA'S $\star$-PRODUCT ON SYMMETRIC FUNCTIONS: PIERI RULES, DOMINANCE SUPPORT, AND THE $(s-1)$-ADIC VALUATION

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Abstract. Hikita introduced a two-parameter deformation $\star$ of the product of symmetric functions, obtained by transporting the ordinary product through the level-one polynomial representation of the affine Hecke algebra. We give an explicit rule for $e_k \star (e_{a_1} \cdots e_{a_\ell})$ for all k and ℓ, in generating-function form; its case $\ell = 1$ is a Pieri rule for $e_k \star e_r$ generalising Hikita's rule for $e_1 \star e_r$. Writing $e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_\mu c_{\lambda\mu}(s,t) e_\mu$, we show that $c_{\lambda\mu}$ is supported exactly on the dominance up-set of λ with s-adic valuation $n(\mu)$, and that the $(s-1)$-adic valuation of $c_{\lambda\mu}$ is $\ell(\lambda) - \kappa(\lambda,\mu)$, where κ is the maximal number of blocks into which λ and μ split compatibly. Leading coefficients factor over blocks into connected pieces; for a full merge the connected piece is a connected-graph cumulant. The coefficients are invariant, up to a power of s, under complementation in a box, and all second-order leading coefficients are given in closed form via a residue formula for Green polynomials at two-part classes. Finally, we give a self-contained proof that $\star$ is the ordinary product transported by an operator diagonal on Macdonald polynomials, an identity implicit in work of Di Francesco and Kedem.

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1. Introduction

Hikita [Hik25] defined a commutative, associative product $\star$ on symmetric functions over $\mathbb{Q}(q,t)$ by transporting the ordinary product through the level-one polynomial representation of

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the affine Hecke algebra. We write $s := q^{-1}$. At $s = 1$, $\star$ is the ordinary product. Hikita's Pieri rule [Hik25, Thm. 3.12] reads, in our normalisation,

$$e_1 \star e_r = (1 - s)[r + 1]_t \, e_{r+1} + s \, e_1 e_r.$$

Di Francesco and Kedem [DFK19, §8.3] observed that the associated t-deformed graded characters lose Schur positivity at generic t. Chen, Lu and Ruan [CLR26] identify the derived Hall algebra of the Jordan quiver with a ring of double symmetric functions with parameter t, sending the derived Hall basis to double Hall–Littlewood functions, for which they also prove Pieri rules; this is a related Hall-algebra product, and we do not compare it with $\star$ here. This paper is about the structure that survives: explicit multiplication rules, triangularity, and valuations.

Pieri rules. The operator $e_k \star (\cdot)$ is a sum over k-subsets (Theorem 2.5); up to normalisation it is the generalised Macdonald operator $M_{k;1}$ of Di Francesco and Kedem [DFK19] (Remark 2.6),

$$e_k \star F = \sum_{|A|=k} \prod_{i \in A,\, j \notin A} \frac{x_i - t x_j}{x_i - x_j} \prod_{i \in A} x_i \;\cdot F\big|_{x_i \mapsto s x_i \;\; (i \in A)}.$$

From it we derive a closed generating function for $e_k \star (e_{a_1} \cdots e_{a_\ell})$ for all k, ℓ (Theorem 3.2). Since the e_a generate, this determines $e_k \star F$ for every F. The case $\ell = 1$ is a Pieri rule (Corollary 3.3):

$$e_k \star e_r = \sum_{b=0}^{k} s^b \, F_{k-b}(t^{r-b}) \, e_b \, e_{r+k-b},$$

with explicit polynomials $F_n(w)$, and at $t = 0$ the coefficients form a truncated geometric distribution (Corollary 3.7). In the generating function, the ℓ columns interact only through a pairwise cross kernel, and the proof closes with a single residue identity (Lemma 3.10).

Dominance and valuations. Write $e^\star_\lambda := e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu$. A termwise degree count on the subset formula shows that $c_{\lambda\mu} \in \mathbb{Q}[s,t]$ is supported exactly on the dominance upset of λ, with s-adic valuation exactly $n(\mu)$ (Theorem 4.1). In the rescaled basis $b_\mu = s^{n(\mu)} e_\mu$, the operator $e_k \star (\cdot)$ reduces modulo s to Hall–Littlewood multiplication by $t^{-\binom{k}{2}} e_k$, so the s-leading coefficients form the classical $e \to$Hall–Littlewood transition matrix (Theorem 5.1 and Corollary 5.2).

At $s = 1$, $\star$ is the ordinary product, and the variable s enters the subset formula only through an Euler operator; so the $(s-1)^p$ Taylor piece of $e_k \star (\cdot)$ is a differential operator of order p. The first-order term is a symmetric biderivation with an explicit matrix (Theorem 7.1 and Proposition 7.2), and an order-p piece can glue at most p existing blocks of parts. This gives the *block valuation law*

$$v_{s-1}(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu) \qquad (\mu \trianglerighteq \lambda,\ \mu \neq \lambda),$$

where κ is the maximal number of compatible blocks (Theorem 7.6); the lower bound holds for all t, and exactness is settled at $t = 0$, where no cancellation occurs. The leading coefficients are sums over merge histories with explicit weights (Theorems 7.9 and 7.10). For a full merge the lead is $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} \sum_{H \text{ connected}} \prod_{ij \in H} (t^{\lambda_i \lambda_j} - 1)$ (Theorem 8.4), whose graph half is due to Dołęga [Doł19], and every lead is a sum of products of connected leads (Theorems 8.7 and 8.8). The coefficients satisfy $c_{N^\ell - \lambda, N^\ell - \mu} = s^{N\binom{\ell}{2} - (\ell-1)|\lambda|} c_{\lambda\mu}$ (Theorem 9.1). Finally, every leading coefficient at valuation 2 is given in closed form (Theorem 10.7 and Corollary 10.9), via a constant-term formula and a residue evaluation of Green polynomials at two-part classes (Theorems 10.2 and 10.3).

A Macdonald-diagonal transport. The product $\star$ is the ordinary product transported by the operator $\mathcal{N}$ that multiplies the Macdonald polynomial $P_\nu(x;s,t^{-1})$ by $t^{n(\nu)}s^{n(\nu')}$; equivalently, $e_k\star(\cdot)=t^{-\binom{k}{2}}\mathcal{N}e_k\mathcal{N}^{-1}$, a finite-variable form of a ∇-conjugation (Theorem 6.2). This identity is implicit in [DFK19] and we claim no novelty for it. We give a self-contained proof via a nonsymmetric Gaussian, whose engine is the commutator $[\sum_j Y_j, x_1]=(s-1)x_1Y_1$ for Cherednik's untwisted operators (Lemma 6.6 and Proposition 6.7). No other result of the paper depends on Theorem 6.2.

Organisation. Section 2 sets up Hikita's product and proves the subset formula. Section 3 contains the Pieri package. Section 4 proves dominance support, and Section 5 identifies the $s\to 0$ edge. Section 6 proves the ∇-transport identity, using the facts about Cherednik operators collected in Appendix A. Sections 7 and 8 treat the expansion at $s=1$, Section 9 the box symmetry and the linear coefficient, and Section 10 the second-order leading coefficients.

2. Hikita's product and the subset formula

2.1. **Notation.** For $m\ge 0$ let $\Lambda_m:=\mathbb{Q}(s,t)[x_1,\dots,x_m]^{S_m}$ and let Λ be the ring of symmetric functions over $\mathbb{Q}(s,t)$. We write e_r for the elementary symmetric functions ($e_0=1$, $e_r=0$ for $r<0$), $e_\lambda=e_{\lambda_1}e_{\lambda_2}\cdots$, and

$$E(z):=\prod_{i=1}^m(1+x_iz)=\sum_{r\ge0}e_rz^r,\qquad Q(y):=\frac{E(-ty)}{E(-y)}=\prod_{i=1}^m\frac{1-tx_iy}{1-x_iy}=\sum_{n\ge0}q_ny^n,$$

so that the q_n are the one-row Hall–Littlewood functions of [Mac95, III (2.10)]. For a set B of indices, $E(x_B;z):=\prod_{i\in B}(1+x_iz)$, and $\hat{x}_i$ denotes the variables $x_1,\dots,x_m$ with x_i omitted. We put

$$a_{ij}:=\frac{x_i-tx_j}{x_i-x_j},\qquad c_A:=\prod_{i\in A,\ j\notin A}a_{ij},\qquad x_A:=\prod_{i\in A}x_i\qquad(A\subseteq[m]).$$

Further, $[n]_q=(1-q^n)/(1-q)$ and $[n]=[n]_t$, $[n]_t!=[1][2]\cdots[n]$, $(a;t)_n=\prod_{i=0}^{n-1}(1-at^i)$, $\varphi_r(t)=\prod_{a=1}^r(1-t^a)$, $n(\lambda)=\sum_i(i-1)\lambda_i$, $m_v(\lambda)=\#\{i:\lambda_i=v\}$, and $\trianglerighteq$ is dominance order. $P_\lambda(x;t)$ and $Q_\lambda=b_\lambda(t)P_\lambda$, $b_\lambda(t)=\prod_{v\ge1}\varphi_{m_v(\lambda)}(t)$, are the Hall–Littlewood functions of [Mac95, III §2]; $K_{\lambda\mu}$ are the Kostka numbers and $K_{\lambda\mu}(t)$ the Kostka–Foulkes polynomials, $s_\lambda=\sum_\mu K_{\lambda\mu}(t)P_\mu(x;t)$ [Mac95, III (2.6)]. We use two scalar products on Λ: the Hall pairing $\langle p_\lambda,p_\mu\rangle=\delta_{\lambda\mu}z_\lambda$ [Mac95, I §4] and the Hall–Littlewood pairing $\langle p_\lambda,p_\mu\rangle_t=\delta_{\lambda\mu}z_\lambda\prod_i(1-t^{\lambda_i})^{-1}$ [Mac95, III §4]. They are different: an undecorated bracket $\langle\cdot,\cdot\rangle$ is always the Hall pairing.

2.2. **The level-one representation and Hikita's product.** The symmetric group S_m acts on $\mathbb{Q}(s,t)[x_1,\dots,x_m]$ by permuting variables; s_i is the transposition of x_i and x_{i+1}. Define operators on $\mathbb{Q}(s,t)(x_1,\dots,x_m)$ by

$$T_if:=t\,s_if+(t-1)\,\frac{x_{i+1}}{x_i-x_{i+1}}\,(s_if-f)\qquad(1\le i\le m-1),$$
$$\pi f(x_1,\dots,x_m):=x_1\,f(x_2,\dots,x_m,sx_1),$$
$$Y_i:=t^{m-i}\,T_{i-1}\cdots T_1\,\pi\,T_{m-1}^{-1}\cdots T_i^{-1}\qquad(1\le i\le m).$$

The T_i are Demazure–Lusztig operators; they preserve polynomials, satisfy $(T_i-t)(T_i+1)=0$ and the braid relations, and $T_if=tf$ whenever $s_if=f$. For $1\le k\le m$ put

$$e_k(Y):=\sum_{b_1<\cdots<b_k}Y_{b_1}Y_{b_2}\cdots Y_{b_k},$$

each product taken in increasing order of indices from left to right. (The Y_i commute (Proposition 6.7, Remark A.3), but the proof of Theorem 2.5 does not use this.) Hikita's product $F\star G$ is

defined by transporting the product through the map $F(Y) \mapsto F(Y) \bullet 1$; by [Hik25, Lemma 3.1, Cor. 3.9] this map is a linear isomorphism compatible with $m \to m-1$, so $\star$ is well defined on Λ [Hik25, Cor. 3.10], and it is commutative and associative [Hik25, §3.2, after Def. 3.4]. We use $\star$ only through the following consequence of the definitions [Hik25, Def. 3.4, Lemma 3.3]: for symmetric F,

$$e_k \star F = t^{-\binom{k}{2}}\, e_k(Y)\, F. \tag{2.1}$$

2.3. **Coxeter preliminaries.** We use standard facts about S_m as a Coxeter group with generators $s_1, \dots, s_{m-1}$ [Hum90, §§1.10, 5.12], [BB05, §2.4]. Words act as compositions of operators, the rightmost factor first.

- (H1) For a reduced word $w = s_{i_1} \cdots s_{i_\ell}$ the operator $T_w := T_{i_1} \cdots T_{i_\ell}$ depends only on w, and $T_u T_v = T_{uv}$ whenever $\ell(uv) = \ell(u) + \ell(v)$.
- (H2) For $J \subseteq \{s_1, \dots, s_{m-1}\}$ let W_J be the parabolic subgroup and $W^J = \{w : \ell(ws) > \ell(w)\ \forall s \in J\}$. Every w factors uniquely as $w = w^J w_J$ with $w^J \in W^J$, $w_J \in W_J$, and $\ell(w) = \ell(w^J) + \ell(w_J)$.
- (H3) For $K \subseteq J$, multiplication $W^J \times (W_J)^K \to W^K$ is a length-additive bijection.
- (H4) If $T_j H = tH$ for all $s_j \in J$, then $T_v H = t^{\ell(v)} H$ for all $v \in W_J$.
- (H5) $\pi T_j = T_{j+1} \pi$ for $1 \le j \le m-2$.

(H3) is standard: if $u \in W^J$, $v \in (W_J)^K$ and $s \in K$, then $\ell(uvs) = \ell(u) + \ell(vs) = \ell(uv) + 1$; conversely, for $x \in W^K$ write $x = x^J x_J$, and $x_J s < x_J$ for some $s \in K$ would give $\ell(xs) < \ell(x)$. (H4) is immediate from (H1). For (H5), $\pi s_j f = x_1 f(x_2, \dots, x_{j+2}, x_{j+1}, \dots, s x_1) = s_{j+1} \pi f$ because $j+2 \le m$, and π turns multiplication by $x_{j+1}/(x_j - x_{j+1})$ into multiplication by $x_{j+2}/(x_{j+1} - x_{j+2})$.

Let $J_n := \{s_i : i \ne n\}$, so $W_{J_n} = S_n \times S_{m-n}$, and the coset wW_{J_n} is determined by $D = w([n])$. Its minimal element w^D is increasing on $[n]$ and on $[n+1, m]$, with $\ell(w^D) = \sum_l (d_l - l)$ for $D = \{d_1 < \cdots < d_n\}$. The word

$$w(D) := (s_{d_1 - 1} \cdots s_1)(s_{d_2 - 1} \cdots s_2) \cdots (s_{d_n - 1} \cdots s_n)$$

sends $l \mapsto d_l$ for $l \le n$: the blocks with index $l' > l$ fix l, block l sends l to d_l, and the blocks $l' < l$ only move points $\le d_{l'} < d_l$. It has length $\le \sum_l (d_l - l)$, so $w(D) = w^D$ and the word is reduced. We write $T_{w(D)}$ for the corresponding operator and

$$\sigma^{(k)} := \sum_{|D| = k} T_{w(D)}.$$

2.4. **The parabolic kernel.**

Lemma 2.1. *Let $0 \le n \le m$, $1 \le c < d_1 < \cdots < d_n \le m$, and let H satisfy $T_j H = tH$ for all $j \ne n$. Then*

$$(T_c T_{c+1} \cdots T_{m-1})^{-1}\, T_{w(D)} H = t^{\,n-(m-c)}\, T_{w(D-1)} H, \qquad D - 1 := \{d_1 - 1, \dots, d_n - 1\}.$$

Proof. Let $x := s_c s_{c+1} \cdots s_{m-1}$; it is reduced of length $m - c$, sends $m \mapsto c$ and $i \mapsto i+1$ for $c \le i \le m-1$, and fixes $i < c$. Let $y := w(D-1)$; the word is valid since $d_l - 1 \ge c + l - 1 \ge l$, and $\ell(y) = \ell(w^D) - n$. As $m \notin D - 1$, the value m is the largest on the increasing block $[n+1, m]$, so $y(m) = m$. Since x is increasing on $[1, m-1]$, the inversions of y among positions $< m$ survive in xy; at position m, $xy(m) = c$ is inverted with exactly those $i < m$ having $y(i) \ge c$, and there are $m - c$ of them. Hence $\ell(xy) = \ell(y) + m - c$. Finally $xy([n]) = x(D-1) = D$ because every $d_l - 1 \ge c$, so $xy = w^D v$ with $v \in W_{J_n}$ and $\ell(v) = \ell(xy) - \ell(w^D) = m - c - n$. By (H1) and (H4), $T_x T_y H = T_{xy} H = T_{w^D} T_v H = t^{m-c-n} T_{w^D} H$. Apply T_x^{-1}. □

Proposition 2.2 (Parabolic kernel)**.** *For symmetric F and $1 \le b_1 < \cdots < b_k \le m$, $Y_{b_1} Y_{b_2} \cdots Y_{b_k} F = t^{\binom{k}{2}}\, T_{w(\{b_1,\dots,b_k\})}\, \pi^k F$. Hence*

$$e_k(Y) F = t^{\binom{k}{2}}\, \sigma^{(k)} \pi^k F.$$

Proof. Note that $\pi^n F = x_1 \cdots x_n\, F(x_{n+1},\dots,x_m, sx_1,\dots,sx_n)$ is symmetric in $x_1,\dots,x_n$ and in $x_{n+1},\dots,x_m$, so $T_j \pi^n F = t\pi^n F$ for $j \neq n$. We induct on the number of factors, the empty product being trivial. Let $c < d_1 < \cdots < d_n$ and assume $Y_{d_1} \cdots Y_{d_n} F = t^{\binom{n}{2}} T_{w(D)} H$ with $H = \pi^n F$. By Lemma 2.1,

$$Y_c\, t^{\binom{n}{2}} T_{w(D)} H = t^{\binom{n}{2}} t^{m-c}\, T_{c-1} \cdots T_1\, \pi\, (T_c \cdots T_{m-1})^{-1} T_{w(D)} H = t^{\binom{n}{2}+n}\, T_{c-1} \cdots T_1\, \pi\, T_{w(D-1)} H.$$

All indices in the word $w(D-1) = \prod_l (s_{d_l-2} \cdots s_l)$ lie in $[1, m-2]$, so by (H5), $\pi T_{w(D-1)} = \prod_l (T_{d_l-1} \cdots T_{l+1})\, \pi$. Now $T_{c-1} \cdots T_1 \prod_l (T_{d_l-1} \cdots T_{l+1}) = T_{w(\{c\}\cup D)}$, $\pi H = \pi^{n+1} F$, and $\binom{n}{2} + n = \binom{n+1}{2}$. □

2.5. **The k-set kernel.** Let $\sigma_{[l,m]} := \sum_{a=l}^{m} T_{a-1} \cdots T_l$ be the partial symmetriser on $x_l,\dots,x_m$.

Lemma 2.3. *Let F be symmetric in $x_2,\dots,x_m$, with coefficients in any field containing $\mathbb{Q}(t)$, and let $F^{(i)}$ be F with x_1 and x_i swapped. Then $\sigma_{[1,m]} F = \sum_{i=1}^{m} F^{(i)} \prod_{j\neq i} a_{ij}$.*

Proof. Induction on m; $m = 1$ is trivial. Since $s_1 F = F^{(2)}$, a direct computation gives $T_1 F = a_{21} F^{(2)} + (a_{12} - 1) F^{(1)}$. Write $\sigma_{[1,m]} = 1 + \sigma_{[2,m]} T_1$. The function $G := T_1 F$ is symmetric in $x_3,\dots,x_m$, so by induction (over the field with x_1 adjoined) $\sigma_{[2,m]} G = \sum_{i\ge 2} G|_{x_2 \leftrightarrow x_i} \prod_{j\ge 2,\, j\neq i} a_{ij}$. Under $x_2 \leftrightarrow x_i$, $F^{(2)} \mapsto F^{(i)}$, $F^{(1)}$ is fixed, $a_{21} \mapsto a_{i1}$ and $a_{12} \mapsto a_{1i}$. Hence

$$\sigma_{[1,m]} F = \sum_{i\ge 2} F^{(i)} \prod_{j\neq i} a_{ij} + F^{(1)} \Big[1 + \sum_{i\ge 2} (a_{1i} - 1) \prod_{j\ge 2,\, j\neq i} a_{ij} \Big].$$

The function $R(x) := \prod_{j\ge 2} (x - t x_j)/(x - x_j)$ tends to 1 at ∞ and has simple poles at $x = x_i$ with residues $(1-t) x_i \prod_{j\ge 2, j\neq i} a_{ij}$; so $R(x) = 1 + \sum_{i\ge 2} \frac{(1-t)x_i}{x - x_i} \prod_{j\ge 2, j\neq i} a_{ij}$. Evaluating at $x = x_1$ and using $(1-t)x_i/(x_1 - x_i) = a_{1i} - 1$ shows that the bracket equals $R(x_1) = \prod_{j\ge 2} a_{1j}$. □

Proposition 2.4 (k-set kernel)**.** *Let G be symmetric in $x_1,\dots,x_k$ and in $x_{k+1},\dots,x_m$, and let $G^{(A)}$ denote G with the variables x_A in the first k slots and $x_{[m]\setminus A}$ in the last $m-k$. Then*

$$\sigma^{(k)} G = \sum_{|A|=k} c_A\, G^{(A)}.$$

Proof. Let $K_l := \{s_{l+1},\dots,s_{m-1}\}$. The minimal coset representatives of $W_{K_{l-1}}/W_{K_l}$ are $s_{a-1} \cdots s_l$ $(l \le a \le m)$, so applying (H3) along $K_k \subset K_{k-1} \subset \cdots \subset K_0$ gives $\sum_{u\in W^{K_k}} T_u = \sigma_{[1,m]} \sigma_{[2,m]} \cdots \sigma_{[k,m]}$. Applying (H3) to $K_k \subset J_k$, with $(W_{J_k})^{K_k} = S_k$, gives $\sum_{u\in W^{K_k}} T_u = \sigma^{(k)} \sum_{v\in S_k} T_v$. For G as in the statement, (H4) gives $\sum_{v\in S_k} T_v G = [k]_t!\, G$, so

$$\sigma_{[1,m]} \cdots \sigma_{[k,m]} G = [k]_t!\, \sigma^{(k)} G. \tag{2.2}$$

On the other hand, if G is merely symmetric in $x_{k+1},\dots,x_m$, then iterating Lemma 2.3 gives

$$\sigma_{[1,m]} \cdots \sigma_{[k,m]} G = \sum_{(a_1,\dots,a_k)\ \text{distinct}} G(x_{a_1},\dots,x_{a_k}; \text{rest})\ \omega(a), \qquad \omega(a) := \prod_{l=1}^{k} \prod_{j\notin\{a_1,\dots,a_l\}} a_{a_l j}. \tag{2.3}$$

Indeed, by induction on k in the variables $x_2,\dots,x_m$, $H := \sigma_{[2,m]} \cdots \sigma_{[k,m]} G$ is the analogous sum over distinct tuples in $[2,m]$; it is symmetric in $x_2,\dots,x_m$, since permuting those variables

permutes the terms; and Lemma 2.3 applied to H, followed by $x_1 \leftrightarrow x_{a_1}$, turns tuples in $[2,m]$ into tuples in $[m] \setminus \{a_1\}$.

Now let G be symmetric in both groups. Then $G(x_{a_1},\dots,x_{a_k};\text{rest}) = G^{(A)}$ with $A = \{a_1,\dots,a_k\}$, and

$$\omega(a) = c_A \prod_{1\le l<l'\le k} a_{a_l a_{l'}}. \tag{2.4}$$

By the Poincaré identity $\sum_{w\in S_k} w(\prod_{i<j\le k} a_{ij}) = [k]_t!$ [Mac95, III (1.4)] (which is also (2.2)=(2.3) at $m=k$, $G=1$), summing over the orderings of A gives $[k]_t!\, c_A G^{(A)}$. Compare with (2.2). □

Theorem 2.5 (Subset formula)**.** *For every $m \ge 0$, $k \ge 1$ and $F \in \Lambda_m$,*

$$e_k \star F = t^{-\binom{k}{2}} e_k(Y) F = E_k F := \sum_{|A|=k} c_A\, x_A\, F\big|_{x_i \mapsto s x_i \ (i\in A)}.$$

Proof. By Propositions 2.2 and 2.4 with $G = \pi^k F$, since $(\pi^k F)^{(A)} = x_A\, F|_{x_i\mapsto s x_i\,(i\in A)}$; the first equality is (2.1). For $m<k$ both sides vanish. □

Remark 2.6. For $k=1$ the subset formula gives back Hikita's rule $e_1 \star e_r = (1-s)[r+1]e_{r+1} + s e_1 e_r$ [Hik25, Thm. 3.12]; see Example 3.5. Hikita's explicit $\star$-multiplication rule is this case $k=1$, and he does not pursue general k; the subset form of the $\star$-multiplication operator for $k \ge 2$ is not in [Hik25]. By contrast, his Lemma 3.3, which is the input to (2.1), holds for all r: it computes the image of $e_r(Y)$ for every r, via the image of $\prod_{i\in I} Y_i$ for every subset I. Proposition 2.4 is the level-one form of the classical expression of the Hecke symmetriser as $\sum_w w \circ \prod_{i<j} a_{ij}$, restricted to minimal coset representatives.

The operator E_k is not new: with the dictionary of Remark 6.4, $E_k = t^{k(m-k)} M_{k;1}$, where $M_{k;1}$ is, in our notation, the generalised Macdonald operator $\mathcal{M}_{k;1}$ of [DFK19, (1.5)]. Di Francesco and Kedem [DFK19, Def. 2.6, Thm. 2.17] obtain the operators $M_{k;n}$ (our notation) as the restriction to symmetric functions of the k-th elementary symmetric function of the twisted Cherednik operators $Y_{i,n} = (X_1\cdots X_{i-1})^{-n} Y_i (X_1\cdots X_i)^n$, by conjugating Macdonald's formula for $e_k(Y)$ with an X-Gaussian. On symmetric functions, Theorem 2.5 is therefore the case $n=1$, for Hikita's level-one operators, which are twisted by multiplication by the x_i ($\pi = x_1\pi_\circ$ in the notation of Section 6). Hikita notes, right after his definition [Hik25, (3)], that his Y_i corresponds to Y_i^{-1} in the usual DAHA convention. Operator by operator, the match is with the twist $n=-1$, and with a different dictionary. Take the (q,t) of [DFK19] to be (s^{-1},t) and $\theta = t^{1/2}$. Then $Y_i = \theta^{m-1}(Y_{i,-1})^{-1}$ for every i, since $x_1\cdots x_{i-1}$ commutes with T_j for $j \ge i$ and $x_1\cdots x_i$ commutes with T_j for $j<i$. No such identity holds with $n=1$. The two pictures agree on symmetric functions. Inverting the commuting family turns e_k into e_{m-k}/e_m, and a direct computation from [DFK19, (1.5)] gives $\mathcal{M}^{-1}_{m;-1}\mathcal{M}_{m-k;-1} = q^{-k} t^{k(m-k)}\, \mathcal{M}_{k;1}|_{(q,t)\mapsto(q^{-1},t^{-1})}$ (DFK's parameters). This identity is not new: it is the case $n=-1$ of $\mathcal{M}^{-1}_{m;n}\mathcal{M}_{m-k;n} = q^{kn} t^{k(m-k)}\, \mathcal{M}_{k;-n}|_{(q,t)\mapsto(q^{-1},t^{-1})}$, which follows from [DFK19, (6.7)] (the dual operators $\bar{\mathcal{D}}_\alpha = \mathcal{D}^{q^{-1},t^{-1}}_{\alpha;0}$ are $e_m(Y_{\cdot,0})^{-1} e_{m-\alpha}(Y_{\cdot,0})$) by conjugating with γ^n as in [DFK19, Lemma 2.16] and converting with [DFK19, Thm. 2.17]; we include the direct check for completeness. This change of dictionary leads back to the one in Remark 6.4. Combined with [DFK19, Thm. 2.17], it gives a second proof of Theorem 2.5, with the same scalar. We do not use [DFK19] here: the proof above is direct, and it does not use that the Y_i commute.

Lemma 2.7. $E_k(1) = \sum_{|A|=k} c_A x_A = e_k$.

Proof. Grouping $w \in S_m$ by $w(\{1,\dots,k\}) = A$ and applying the Poincaré identity to $S_k \times S_{m-k}$ shows that $\sum_{|A|=k} c_A x_A$ is the Hall–Littlewood function $P_{(1^k)}(x;t)$ as defined in [Mac95, III (2.1)–(2.2)], and $P_{(1^k)} = e_k$ [Mac95, III (2.8)]. □

Lemma 2.8 (Stability)**.** *For $F \in \Lambda_m$, $(E_k F)|_{x_m=0} = E_k(F|_{x_m=0})$, the right side computed in $m-1$ variables. Hence the e-coefficients of any expression built from the E_k and 1 are independent of m once m is at least the degree.*

Proof. Terms with $m \in A$ vanish because of the factor x_A; for $m \notin A$, $a_{im}|_{x_m=0} = 1$. Restriction $\Lambda_m \to \Lambda_{m-1}$ maps $e_\nu \mapsto e_\nu$ and is injective in degrees $\le m-1$. □

2.6. **Two consequences of the kernel.** The next two lemmas drive every computation in Section 3. For a function φ of one variable put $S_k^\varphi(x_1,\dots,x_m) := \sum_{|A|=k} c_A \prod_{a\in A} \varphi(x_a)$.

Lemma 2.9 (Peeling one variable)**.** *For $m \ge 1$ and $k \ge 1$,*

$$[k]\, S_k^\varphi(x) = \sum_{i=1}^{m} \varphi(x_i) \prod_{j\neq i} a_{ij}\; S_{k-1}^\varphi(\hat{x}_i).$$

Proof. By (2.4) and the Poincaré identity, $[k]_t!\, S_k^\varphi = \sum_a \omega(a) \prod_l \varphi(x_{a_l})$, the sum over distinct k-tuples. Separate $a_1 = i$: $\omega(a) = \prod_{j\neq i} a_{ij} \cdot \omega'(a_2,\dots,a_k)$, where ω' is the weight of (2.3) for the variables $\hat{x}_i$. Summing over $(a_2,\dots,a_k)$ gives $[k-1]_t!\, S_{k-1}^\varphi(\hat{x}_i)$. □

Lemma 2.10 (Residue form of the HL kernel)**.** *Let $\mathbb{K} \supseteq \mathbb{Q}(t)$ be a field and $H \in \mathbb{K}(v)$ with $H(0) = 0$, such that $H(v)/v$ has only simple poles, at points $p \in \mathbb{K}$, and $H(v)/v = O(1/v)$ at ∞. Then in $\mathbb{K}(x_1,\dots,x_m)$*

$$(1-t)\sum_{i=1}^{m} H(x_i) \prod_{j\neq i} a_{ij} = -\sum_{p} \operatorname*{Res}_{v=p} f - \operatorname*{Res}_{v=\infty} f, \qquad f(v) := \frac{H(v)}{v} \prod_{j=1}^{m} \frac{v - t x_j}{v - x_j}.$$

Proof. f is regular at $v = 0$, its residue at $v = x_i$ is $(1-t)H(x_i)\prod_{j\neq i} a_{ij}$, and the sum of all residues of a rational function is zero. □

For $H(v) = v^n$, $n \ge 1$, this is $(1-t)\sum_i x_i^n \prod_{j\neq i} a_{ij} = q_n$ [Mac95, III (2.9)]. Note that $\prod_j (v - t x_j)/(v - x_j) = Q(1/v)$ and $Q(-a) = E(ta)/E(a)$.

3. The Pieri package

3.1. **The coefficients.** For $j \ge 0$ put

$$\alpha_j := \prod_{i=1}^{j} \frac{s - t^i}{1 - t^i} \quad (\alpha_{-1} := 0), \qquad c(n,j) := \frac{(s;t)_{n-j}}{(t;t)_{n-j}} (\alpha_j - s\, t^{n-j} \alpha_{j-1}) \quad (0 \le j \le n),$$

and $c(n,j) := 0$ otherwise. Let $F_n(w) := \sum_{j=0}^{n} c(n,j) w^j$. For instance $F_0 = 1$ and $F_1(w) = (1-s)(1-tw)/(1-t)$.

Lemma 3.1. *Let $G(x) := \sum_{n\ge 0} \frac{(s;t)_n}{(t;t)_n} x^n$, $C_j(x) := \sum_n c(n,j) x^n$, and $D_j := \alpha_j - s\alpha_{j-1}$. Then*

$$(1-x)G(x) = (1-sx)G(tx), \qquad C_j(x) = D_j\, x^j\, G(tx)\, \frac{1 - s t^{-j} x}{1 - x},$$

$D_0 = 1$, $D_j = \alpha_{j-1} t^j (s-1)/(1-t^j) \neq 0$ for $j \ge 1$, and $(1-t^j)D_j = (ts - t^j) D_{j-1}$ for $j \ge 1$.

Proof. Write $g_n = (s;t)_n/(t;t)_n$. The first identity says $g_n - g_{n-1} = t^n g_n - st^{n-1}g_{n-1}$, i.e. $(1-t^n)g_n = (1-st^{n-1})g_{n-1}$. By definition $C_j(x) = x^j(\alpha_j G(x) - s\alpha_{j-1}G(tx)) = x^j G(tx)(\alpha_j(1-sx) - s\alpha_{j-1}(1-x))/(1-x)$. The bracket is linear in x. It equals D_j at $x=0$, and at $x = t^j/s$ it equals $\alpha_j(1-t^j) - \alpha_{j-1}(s-t^j) = 0$ (for $j=0$ use $\alpha_{-1}=0$). So it equals $D_j(1-st^{-j}x)$. The formula for D_j follows from $\alpha_j = \alpha_{j-1}(s-t^j)/(1-t^j)$, and the recursion from it and the same relation one step down (for $j=1$ use $D_0 = 1$). □

3.2. **Statements.** Fix $\ell \geq 1$ and indeterminates $z_1, \dots, z_\ell$. For integers $i, j \geq 0$ let

$$K_{ij}(z,w) := \prod_{p=0}^{i-1} \frac{t^p z - sw}{t^p z - t^j w} \prod_{r=0}^{j-1} \frac{sz - t^r w}{t^i z - t^r w},$$

and note $K_{ij}(z,w) = K_{ji}(w,z)$. For $I = (i_1, \dots, i_\ell) \in \mathbb{Z}_{\geq 0}^{\ell}$ with $|I| = S$ put $K_I := \prod_{c<c'} K_{i_c i_{c'}}(z_c, z_{c'})$ and

$$V_I^{(n)} := K_I \, s^{(\ell-1)(n-S)} \sum_{n_1+\cdots+n_\ell = n} t^{-\sum_{c\neq c'}(n_c - i_c)i_{c'}} \prod_{c=1}^{\ell} t^{-n_c i_c}\, c(n_c, i_c)\, z_c^{-n_c},$$

with $V_I^{(n)} := 0$ if some $i_c < 0$. Since $c(n_c, i_c) = 0$ for $n_c < i_c$, $V_I^{(n)} = 0$ unless $S \leq n$.

Theorem 3.2 (ℓ-column rule)**.** *For all $m, k \geq 0$ and $\ell \geq 1$, in $\Lambda_m \otimes \mathbb{Q}(s,t)(z_1, \dots, z_\ell)$,*

$$\mathcal{E}_k := \sum_{a \in \mathbb{Z}_{\geq 0}^{\ell}} z_1^{a_1} \cdots z_\ell^{a_\ell}\; e_k \star (e_{a_1} \cdots e_{a_\ell}) = \sum_{b=0}^{k} \sum_{I} \left(s^\ell t^{-|I|}\right)^b V_I^{(k-b)}\; e_b \prod_{c=1}^{\ell} E(t^{i_c} z_c).$$

In particular the right-hand side is a polynomial in $z_1, \dots, z_\ell$, and $e_k \star (e_{a_1} \cdots e_{a_\ell})$ is its coefficient of z^a. The coefficients do not depend on m, so the identity holds in Λ.

Corollary 3.3 (Pieri rule for $e_k \star e_r$)**.** *For all $k \geq 1$ and $r \geq 0$,*

$$e_k \star e_r = \sum_{b=0}^{k} s^b\, F_{k-b}(t^{r-b})\, e_b\, e_{r+k-b}.$$

Equivalently, $\sum_r z^r\, e_k \star e_r = \sum_{j+b\leq k} s^b t^{-kj} c(k-b, j)\, e_b\, z^{b-k} E(t^j z)$.

Proof. Take $\ell = 1$ in Theorem 3.2: $K_I = 1$, and $V_{(j)}^{(n)} = t^{-nj} c(n,j) z^{-n}$, so the right-hand side is the stated generating function. Extract $[z^r]$ using $[z^r] z^{b-k} E(t^j z) = t^{j(r+k-b)} e_{r+k-b}$. □

For $r < k$, distinct b may give the same monomial $e_b e_{r+k-b}$; the identity is between the sums. Since $\star$ is commutative, applying Corollary 3.3 to $e_r \star e_k$ shows that the support of $e_k \star e_r$ is contained in $\{e_b e_{r+k-b} : 0 \leq b \leq \min(k,r)\}$ (see also Theorem 4.1).

Remark 3.4 (Forced roots of F_n). For $k \geq 1$ and $1 \leq N \leq k$,

$$\sum_{b=0}^{k-N} s^b\, F_{k-b}(t^{-N-b})\, e_b\, e_{k-N-b} = 0 \qquad \text{in } \Lambda.$$

Indeed, the left side of the generating function in Corollary 3.3 is a power series in z, so its coefficient of z^{-N} vanishes; on the right, $[z^{-N}] z^{b-k} E(t^j z) = t^{j(k-N-b)} e_{k-N-b}$, which is zero for $b > k-N$, and summing over j gives the displayed sum. The case $N = k$ is $F_k(t^{-k}) = 0$: the linear factor $1 - t^n w$ divides $F_n(w)$ for every $n \geq 1$. It is also the case $m = 0$ of Theorem 3.2 at $\ell = 1$, where $\mathcal{E}_k = 0$ and $\mathcal{T}_k = z^{-k} F_k(t^{-k})$. For $N < k$ these are $k-1$ further relations among the F_n. This corollary was suggested by Clio (review of 2026-10-15), who checked it for $k \leq 6$; we checked it independently for $k \leq 8$ and every N, with $e_1, e_2, \dots$ independent. The

quotient $F_n(w)/(1-t^n w)$ has a conjectural ${}_2\phi_1$ form, $\frac{(s;t)_n}{(t;t)_n}\,{}_2\phi_1(t^{1-n}, t/s; t^{1-n}/s; t; w)$, which has only been checked by computer ($n \le 6$ by us, $n \le 14$ by Clio).

Example 3.5 ($k = 1$)**.** $c(1,0) = \frac{1-s}{1-t}$ and $c(1,1) = \frac{t(s-1)}{1-t}$, so $F_1(t^r) = (1-s)[r+1]$ and Corollary 3.3 is Hikita's rule $e_1 \star e_r = (1-s)[r+1]e_{r+1} + s e_1 e_r$ [Hik25, Thm. 3.12].

Example 3.6 ($k = 2$)**.** One computes $F_2(w) = \dfrac{(1-s)(1-t^2w)(1-st+(s-t)w)}{(1-t)(1-t^2)}$, so for all $r \ge 0$

$$e_2 \star e_r = \frac{(1-s)(1-t^{r+2})(1-st+(s-t)t^r)}{(1-t)(1-t^2)}\, e_{r+2} + s(1-s)[r]\, e_1 e_{r+1} + s^2 e_2 e_r.$$

For $r = 0$ this is $e_2 \star 1 = e_2$.

Corollary 3.7 (The case $t = 0$)**.** *For $k \ge 1$, $r \ge 0$, put $\mu = \min(k, r)$ and $M = \max(k, r)$. The coefficients of $e_k \star e_r$ are regular at $t = 0$, and*

$$(e_k \star e_r)\big|_{t=0} = \sum_{b=0}^{\mu-1} (1-s)s^b\, e_b\, e_{r+k-b} + s^\mu\, e_\mu e_M.$$

For $s \in [0,1]$ the weights $(1-s)s^b$ ($b < \mu$) and s^μ are the law of $\min(G, \mu)$ for a geometric variable G with $P(G = b) = (1-s)s^b$, and the monomials $e_b e_{r+k-b}$ ($b < \mu$), $e_\mu e_M$ are pairwise distinct. So at $t = 0$, $e_k \star e_r$ is a truncated-geometric average of ordinary products.

Corollary 3.8 (Two columns; bounded support)**.** *For $\ell = 2$, Theorem 3.2 reads*

$$\sum_{a,b\ge 0} z^a w^b\; e_k \star (e_a e_b) = \sum_{b_0,i,j} s^{2b_0} t^{-b_0(i+j)}\, V_{ij}^{(k-b_0)}\, e_{b_0} E(t^i z) E(t^j w),$$

$$V_{ij}^{(n)} = K_{ij}(z,w) \sum_{n_1+n_2=n} s^{n-i-j}\, t^{-(n_1-i)j-(n_2-j)i-n_1 i-n_2 j}\, c(n_1, i)\, c(n_2, j)\, z^{-n_1} w^{-n_2}.$$

In general, $e_k \star (e_{a_1} \cdots e_{a_\ell})$ is a linear combination of products of at most $\ell + 1$ elementary symmetric functions.

Proof. The first part is the case $\ell = 2$. For the second, expand both sides of Theorem 3.2 in the field of iterated Laurent series $\mathbb{Q}(s,t)((z_\ell))\cdots((z_1))$. The left side is a polynomial, so its expansion is itself. On the right, each term is $R(z)\, e_b \prod_c E(t^{i_c} z_c)$ with R rational and free of x, and its coefficient of z^a is a finite $\mathbb{Q}(s,t)$-combination of products $e_b \prod_{c=1}^{\ell} e_{n_c}$. □

3.3. **The recursion.** By Theorem 2.5, $e_k \star (e_{a_1} \cdots e_{a_\ell}) = \sum_{|A|=k} c_A x_A \prod_c e_{a_c}(x_{A^c}, s x_A)$, and $\sum_a z^a \prod_c e_{a_c}(x_{A^c}, s x_A) = \prod_c \prod_{j\notin A}(1 + x_j z_c) \prod_{i\in A}(1 + s x_i z_c)$. Hence

$$\mathcal{E}_k(x) = \prod_c E(z_c) \cdot S_k^{\varphi}(x), \qquad \varphi(u) := u \prod_{c=1}^{\ell} \frac{1 + suz_c}{1 + uz_c}. \tag{3.1}$$

Since $\varphi(x_i) \prod_c E(z_c) = x_i \prod_c (1 + s x_i z_c) \cdot \prod_c E(\hat{x}_i; z_c)$, Lemma 2.9 gives, for $m \ge 1$ and $k \ge 1$,

$$[k]\, \mathcal{E}_k(x) = \sum_{i=1}^{m} x_i \prod_{c=1}^{\ell} (1 + s x_i z_c) \prod_{j\ne i} a_{ij}\; \mathcal{E}_{k-1}(\hat{x}_i), \tag{3.2}$$

where $\mathcal{E}_{k-1}(\hat{x}_i)$ denotes $\mathcal{E}_{k-1}$ computed in the $m-1$ variables $\hat{x}_i$. For $m = 0$ we have $\mathcal{E}_0 = 1$ and $\mathcal{E}_k = 0$ for $k \ge 1$.

3.4. **The step map.** Fix I and put $\gamma_c := t^{i_c} z_c$, $C := s^\ell t^{-|I|} = s^\ell \prod_c z_c / \prod_c \gamma_c$, and

$$\mathcal{F}(v) := \frac{\prod_{c=1}^{\ell}(v - sz_c)}{v\prod_{c=1}^{\ell}(v-\gamma_c)} = \frac{C}{v} + \sum_{c=1}^{\ell} \frac{\eta_c}{v-\gamma_c}, \qquad \eta_c := \frac{\prod_{c'}(\gamma_c - sz_{c'})}{\gamma_c \prod_{c'\neq c}(\gamma_c - \gamma_{c'})}.$$

Lemma 3.9 (Step map)**.** *For $b \geq 0$, in $m \geq 0$ variables,*

$$(1-t)\sum_i x_i \prod_c (1+sx_iz_c)\prod_{j\neq i} a_{ij}\; e_b(\hat{x}_i)\prod_c E(\hat{x}_i;\gamma_c) = C(1-t^{b+1})\, e_{b+1}\prod_c E(\gamma_c) \\ - \sum_{n=0}^{b} e_n \sum_{c=1}^{\ell} \eta_c\, \gamma_c^{n-b-1}\big(E(t\gamma_c) - t^n E(\gamma_c)\big)\prod_{c'\neq c} E(\gamma_{c'}).$$

Proof. For $m = 0$ both sides vanish: the left side is an empty sum, and on the right $e_n = 0$ for $n \geq 1$ and $E \equiv 1$, so the first term carries $e_{b+1} = 0$ and the $n = 0$ term carries $E(t\gamma_c) - E(\gamma_c) = 0$. Let $m \geq 1$. Write $u = x_i$. Then $e_b(\hat{x}_i) = [y^b]E(y)/(1+uy)$ and $E(\hat{x}_i;\gamma) = E(\gamma)/(1+\gamma u)$, so the left side is $\prod_c E(\gamma_c)\cdot[y^b]\,E(y)\,(1-t)\sum_i H(x_i;y)\prod_{j\neq i} a_{ij}$ with

$$H(v;y) := \frac{v\prod_c(1+svz_c)}{(1+yv)\prod_c(1+\gamma_c v)}$$

(the sum over i is finite and each term is regular at $y = 0$, so $[y^b]$ commutes with it). Apply Lemma 2.10 over $\mathbb{K} = \mathbb{Q}(s,t,z,y)$. The poles of H/v are simple, at $v = -1/a$ for $a \in \{y, \gamma_1, \dots, \gamma_\ell\}$, and $H(v)/v \sim C/(yv)$ at ∞, which contributes C/y. At $v = -1/a$ the residue of f is $\rho_a\, E(ta)/E(a)$, where

$$\rho_a = \frac{\prod_c(a - sz_c)}{a\prod_{a'\neq a}(a-a')},$$

as one sees from $\prod_c(1 - sz_c/a) = a^{-\ell}\prod_c(a - sz_c)$, $\prod_{a'\neq a}(1 - a'/a) = a^{-\ell}\prod_{a'\neq a}(a-a')$ and $Q(-a) = E(ta)/E(a)$. Thus $\rho_y = \mathcal{F}(y)$ and $\rho_{\gamma_c} = -\eta_c/(y-\gamma_c)$, and

$$(1-t)\sum_i H(x_i;y)\prod_{j\neq i} a_{ij} = \frac{C}{y} - \mathcal{F}(y)\frac{E(ty)}{E(y)} + \sum_c \frac{\eta_c}{y-\gamma_c}\,\frac{E(t\gamma_c)}{E(\gamma_c)}.$$

Multiply by $E(y)\prod_c E(\gamma_c)$ and write $E(y) = \sum_n e_n y^n$, $E(ty) = \sum_n t^n e_n y^n$. The coefficient of e_n is $[y^{b-n}]$ of the Laurent expansion at $y = 0$ of

$$\Big(\frac{C}{y} - t^n\mathcal{F}(y)\Big)\prod_c E(\gamma_c) + \sum_c \frac{\eta_c}{y-\gamma_c}E(t\gamma_c)\prod_{c'\neq c} E(\gamma_{c'}) \\ = \frac{C(1-t^n)}{y}\prod_c E(\gamma_c) + \sum_c \frac{\eta_c}{y-\gamma_c}\big(E(t\gamma_c) - t^n E(\gamma_c)\big)\prod_{c'\neq c} E(\gamma_{c'}),$$

by the partial fraction expansion of $\mathcal{F}$. Expanding $1/(y-\gamma) = -\sum_{p\geq 0} y^p \gamma^{-p-1}$ gives the claim: $n = b+1$ picks the pole at 0, $n \leq b$ picks the second sum, and $n > b+1$ contributes nothing. □

3.5. **Proof of Theorem 3.2.** Write $\mathcal{T}_k$ for the right-hand side of Theorem 3.2, so $\mathcal{T}_k = \sum_{b,I} W_{b,I}^{(k)}\, e_b \prod_c E(t^{i_c} z_c)$ with $W_{b,I}^{(k)} = C_I^b V_I^{(k-b)}$, $C_I := s^\ell t^{-|I|}$; $\epsilon_1, \dots, \epsilon_\ell$ are the unit vectors of $\mathbb{Z}^\ell$.

Reduction to an index identity. We show the step identity

$$\sum_i x_i \prod_c (1+sx_iz_c)\prod_{j\neq i} a_{ij}\,\mathcal{T}_{k-1}(\hat{x}_i) = [k]\,\mathcal{T}_k(x) \qquad (m \geq 0,\ k \geq 1) \tag{3.3}$$

(for $m=0$ the left side is empty). Given (3.3), the theorem follows by induction on k for all m at once: $\mathcal{E}_0=\prod_c E(z_c)=\mathcal{T}_0$ because $V_0^{(0)}=1$; for $m=0$, (3.3) says $\mathcal{T}_k=0=\mathcal{E}_k$; for $m\geq 1$, (3.2), the induction hypothesis in the variables $\hat{x}_i$, and (3.3) give $[k]\mathcal{E}_k=[k]\mathcal{T}_k$.

Apply Lemma 3.9 to each term of $\mathcal{T}_{k-1}(\hat{x}_i)$. The term $e_{b'}\prod_c E(t^{i_c}z_c)$ on the right receives three kinds of contributions: from $(b'-1,I)$ via the pole at 0; from (b,I), $b\geq b'$, via $-t^{b'}E(\gamma_c)$; and from $(b,I-\epsilon_c)$, $b\geq b'$, via $E(t\gamma_c')$ with $\gamma_c'=t^{i_c-1}z_c$. Writing η_c' for η_c computed at $I-\epsilon_c$, $n:=k-b'$, $p:=b-b'$, and using $C_{I-\epsilon_c}=tC_I$, the coefficient of $e_{b'}\prod_c E(t^{i_c}z_c)$ in $(1-t)$ times the left side of (3.3) is $C_I^{b'}$ times

$$(1-t^{b'})V_I^{(n)}+t^{b'}\sum_{p\geq 0}C_I^p\Big[V_I^{(n-1-p)}\sum_c\eta_c\gamma_c^{-p-1}-t^p\sum_c V_{I-\epsilon_c}^{(n-1-p)}\eta_c'\,(t^{i_c-1}z_c)^{-p-1}\Big].$$

So (3.3) follows from the index identity

$$\sum_{p\geq 0}C_I^p\Big[V_I^{(n-1-p)}\sum_c\eta_c\gamma_c^{-p-1}-t^p\sum_c V_{I-\epsilon_c}^{(n-1-p)}\eta_c'\,(t^{i_c-1}z_c)^{-p-1}\Big]=(1-t^n)V_I^{(n)}\qquad(n\geq 0),\tag{3.4}$$

because then the coefficient is $C_I^{b'}(1-t^{b'+n})V_I^{(n)}=(1-t^k)W_{b',I}^{(k)}$. (No uniqueness of such expansions is needed: the two sides agree term by term.) This applies verbatim at $m=0$, where Lemma 3.9 holds by the remark at the start of its proof: every equality used is an identity between expressions in the e_b and $E(\cdot)$ valid in every Λ_m, and specialising to $m=0$ ($e_b\mapsto 0$ for $b\geq 1$, $E\mapsto 1$) is a ring homomorphism. There the left side of (3.3) is empty, so the case $m=0$ of (3.3) is not a formality: it is the vanishing of $\mathcal{T}_k|_{m=0}=\sum_I V_I^{(k)}$, which for $\ell=1$ reads $z^{-k}F_k(t^{-k})=0$ (Remark 3.4). We thank Clio for pointing out that this case must be argued (review of 2026-10-15).

Generating function. Let $\mathcal{V}_I(x):=\sum_{n\geq 0}V_I^{(n)}x^n$. The p-sums in (3.4) are convolutions, and $\sum_{p\geq 0}C^p\gamma^{-p-1}x^{p+1}=x/(\gamma-Cx)$. So (3.4) for all n is equivalent to

$$\mathcal{V}_I(x)\,U(x)-\mathcal{V}_I(tx)=-\sum_c\eta_c'\,\frac{x}{t^{i_c-1}z_c-C_Itx}\,\mathcal{V}_{I-\epsilon_c}(x),\qquad U(x):=1-\sum_c\frac{\eta_c x}{\gamma_c-C_Ix}.\tag{3.5}$$

Put $X_c:=C_Ix/(sz_c)$ and $A_c:=t^{i_c}$. Summing over (n_c) in the definition of $V_I^{(n)}$, the exponent of t attached to column c is $-n_c|I|+i_c(|I|-i_c)$, so

$$\mathcal{V}_I(x)=K_I\,s^{-(\ell-1)|I|}\,t^{|I|^2-\sum_c i_c^2}\prod_c C_{i_c}(X_c),\tag{3.6}$$

and passing from I to $I-\epsilon_c$ replaces every $X_{c'}$ by $tX_{c'}$.

Step A: U is a product. By the partial fraction expansion of $\mathcal{F}$, $U(x)=x\,\mathcal{F}(C_Ix)$. Since $C_Ix-sz_c=-sz_c(1-X_c)$, $C_Ix-\gamma_c=-\gamma_c(1-sX_c/A_c)$ and $C_I\prod_c\gamma_c=s^\ell\prod_c z_c$,

$$U(x)=\prod_{c=1}^{\ell}\frac{1-X_c}{1-sX_c/A_c}.$$

Step B: normalisation. Divide (3.5) by $\mathcal{N}:=K_I\,s^{-(\ell-1)|I|}t^{|I|^2-\sum i_c^2}\prod_c D_{i_c}X_c^{i_c}G(t^2X_c)$, which is legitimate since $D_j\neq 0$. By Lemma 3.1 and $G(tX)/G(t^2X)=(1-stX)/(1-tX)$, with $u_c:=stX_c$:

- column c of $\mathcal{V}_IU$ contributes $\frac{(1-stX_c)(1-sX_c/A_c)}{(1-tX_c)(1-X_c)}\cdot\frac{1-X_c}{1-sX_c/A_c}=\frac{1-u_c}{1-tX_c}$, and column c of $\mathcal{V}_I(tx)$ contributes $\frac{A_c-u_c}{1-tX_c}$;
- in the c-th term on the right (present only if $i_c\geq 1$), the columns $c'\neq c$ contribute $\frac{A_{c'}-u_{c'}}{1-tX_{c'}}$; column c contributes $\frac{D_{i_c-1}}{D_{i_c}}\frac{A_c}{t}X_c^{-1}\frac{1-st^2X_c/A_c}{1-tX_c}$ with $D_{i-1}/D_i=(1-A)/(ts-A)$; the prefactors contribute $s^{\ell-1}t^{-2(|I|-i_c)}$; and $\frac{x}{t^{i_c-1}z_c-C_Itx}=\frac{stX_c}{C_I(A_c-st^2X_c)}$.

It remains to compute $\Phi_c := \eta'_c K_{I-\epsilon_c}/K_I$. The diagonal factor of η'_c is $(\gamma'_c - sz_c)/\gamma'_c = (A_c - st)/A_c$. For $c' \neq c$ put $z = z_c$, $w = z_{c'}$, $i = i_c$, $j = i_{c'}$, using $K_{ij}(z,w) = K_{ji}(w,z)$ to place c in the first slot. Then

$$\frac{t^{i-1}z - sw}{t^{i-1}z - t^j w} \cdot \frac{K_{i-1,j}(z,w)}{K_{ij}(z,w)} = \prod_{r=0}^{j-1} \frac{t^i z - t^r w}{t^{i-1}z - t^r w} = t^j \, \frac{t^i z - w}{t^i z - t^j w},$$

the last product telescoping after writing $t^i z - t^r w = t(t^{i-1}z - t^{r-1}w)$. With $\rho_{cc'} := z_c/z_{c'} = X_{c'}/X_c$ we get

$$\Phi_c = \frac{A_c - st}{A_c} \prod_{c' \neq c} A_{c'} \frac{A_c \rho_{cc'} - 1}{A_c \rho_{cc'} - A_{c'}}.$$

Multiplying everything, $\frac{stX_c}{C_I(A_c - st^2 X_c)} \cdot \frac{A_c}{t} X_c^{-1}(1 - st^2 X_c/A_c) = s/C_I$, then $\frac{s}{C_I} s^{\ell-1} t^{-2(|I|-i_c)} = t^{2i_c - |I|} = A_c / \prod_{c' \neq c} A_{c'}$, and $(A_c - st)/(ts - A_c) = -1$. Hence, after division by $\mathcal{N}$ and multiplication by $\prod_c (1 - tX_c)$, (3.5) becomes

$$\prod_c (1 - u_c) - \prod_c (A_c - u_c) - \sum_c (1 - A_c) \prod_{c' \neq c} \frac{A_c X_{c'} - X_c}{A_c X_{c'} - A_{c'} X_c} (A_{c'} - u_{c'}) = 0. \tag{3.7}$$

(For $i_c = 0$ the c-th term is absent, consistently with the factor $1 - A_c = 0$.)

Step C: the closing identity. Put $\lambda_c := A_c/u_c$ and $\mu_c := 1/u_c$. Since $X_c \propto u_c$, $\frac{A_c X_{c'} - X_c}{A_c X_{c'} - A_{c'} X_c} = \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}$, and $(A_{c'} - u_{c'})/u_{c'} = \lambda_{c'} - 1$, $(1 - A_c)/u_c = \mu_c - \lambda_c$. Dividing (3.7) by $\prod_c u_c$ turns it into the case of Lemma 3.10 below. As rational functions of x, the $\lambda_c = t^{i_c} z_c/(tC_I x)$ are pairwise distinct and different from 1. Hence (3.5) holds, so (3.4) holds for every n, which proves (3.3) and the theorem. □

Lemma 3.10. *For indeterminates $\lambda_1, \ldots, \lambda_\ell, \mu_1, \ldots, \mu_\ell$,*

$$\prod_c (\mu_c - 1) - \prod_c (\lambda_c - 1) = \sum_{c=1}^{\ell} (\mu_c - \lambda_c) \prod_{c' \neq c} (\lambda_{c'} - 1) \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}.$$

Proof. The rational function

$$g(w) = \prod_c (w - \mu_c) \Big/ \Big((w-1) \prod_c (w - \lambda_c) \Big)$$

has residue $\frac{\lambda_c - \mu_c}{\lambda_c - 1} \prod_{c' \neq c} \frac{\lambda_c - \mu_{c'}}{\lambda_c - \lambda_{c'}}$ at λ_c, residue $\prod_c (1 - \mu_c)/\prod_c (1 - \lambda_c)$ at 1, and residue -1 at ∞. The residues sum to zero; multiply by $\prod_c (\lambda_c - 1)$. □

Remark 3.11. For $\ell = 1$, Lemma 3.10 is the tautology $(\mu - 1) - (\lambda - 1) = \mu - \lambda$, and the proof above reduces to a one-index recursion for $c(n,j)$. For $\ell = 2$ it is the polynomial identity that closes the two-column case. The cross kernel K_I is forced by the pairwise splitting of η'_c in Step B.

3.6. **Proof of Corollary 3.7.** At $t = 0$, $(t;t)_n = 1$, so every α_j and $c(n,j)$ is regular at $t = 0$; $(s;t)_n|_{t=0} = 1 - s$ for $n \geq 1$; hence $F_n(0) = c(n,0)|_{t=0} = 1 - s$ for $n \geq 1$, and $F_0 = 1$.

Case $r \geq k$. Every $b \in [0,k]$ has $r - b \geq 0$, so each term $s^b F_{k-b}(t^{r-b}) = s^b \sum_j c(k-b,j) t^{j(r-b)}$ of Corollary 3.3 is regular at $t = 0$. If $b < r$, the terms with $j \geq 1$ vanish at $t = 0$, leaving $(1 - s)s^b$ for $b < k$ and s^k for $b = k$. If $b = r$, then $b = k = r$ and the term is $s^k F_0(1) = s^k$. Summing gives the claim with $\mu = k$.

Case $r < k$. By commutativity of $\star$ [Hik25], $e_k \star e_r = e_r \star e_k$. For $r = 0$ the claim is $e_k \star 1 = e_k$; for $r \geq 1$ apply the first case to (r,k). □

In the (k,r)-expansion of Corollary 3.3 with $r<k$, the terms with $b>r$ carry negative powers $t^{j(r-b)}$; the argument shows that these cancel once each monomial $e_b e_{r+k-b}$ is collected with its partner $b'=r+k-b$.

Remark 3.12 (Other forms and earlier cases). A second, independent proof of the Pieri rule, in the form $e_k \star e_r = \sum_y s^y \mathcal{C}_{k-y,r-y} e_{k+r-y} e_y$ with $\mathcal{C}_{a,b} = (-1)^b \frac{[a+b]}{[a]} \pi_a(b)$, is given in Section 9 (Corollary 9.6); it goes through the plethystic linear coefficient of Theorem 9.5 and the Column Lemma. The two forms agree after collecting terms. The kernel form of $e_k(Y)$ on symmetric functions is classical in spirit (Macdonald's D-operators); what is new here is the level-one transfer π^k, the explicit coefficients, and the ℓ-column rule.

4. Dominance support

For a partition λ write

$$e^\star_\lambda := e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu,$$

with E_k as in Theorem 2.5. The proofs below apply the operators E_{λ_i} in an arbitrary order, so they use neither commutativity nor associativity of $\star$.

Theorem 4.1 (Dominance support; the $s=0$ valuation). *$e^\star_\lambda = s^{n(\lambda)} e_\lambda + \sum_{\mu \rhd \lambda} c_{\lambda\mu} e_\mu$ with $c_{\lambda\mu} \in \mathbb{Q}[s,t]$ and $c_{\lambda\mu}|_{s=1} = 0$. Moreover $c_{\lambda\mu} \neq 0$ for every $\mu \trianglerighteq \lambda$, and the s-adic valuation of $c_{\lambda\mu}$ over $\mathbb{Q}(t)$ is exactly $n(\mu)$. At $t=0$ the s-leading coefficients $[s^{n(\mu)}]c_{\lambda\mu}$ are all 1: in the basis $b_\mu = s^{n(\mu)} e_\mu$, $e^\star_\lambda$ at $s=t=0$ is the zeta function of dominance order.*

Corollary 4.2 (Operator form). *For $k \geq 1$ and every partition μ, $e_k \star e_\mu \in s^{\sum_i \min(\mu_i,k)} e_{\mu\cup k} + \operatorname{span}\{e_\nu : \nu \rhd \mu \cup k\}$.*

We work in Λ_m with $m \geq n = |\lambda|$; by Lemma 2.8 this loses nothing.

4.1. Degrees and dominance.

Lemma 4.3. *Let $m \geq n$ and $\rho \vdash n$. Then $\operatorname{span}\{e_\nu : \nu \trianglerighteq \rho'\} = \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ in the degree-n part of Λ_m.*

Proof. By [Mac95, I (6.6), (6.7)], $e_\nu = \sum_\kappa M_{\nu\kappa} m_\kappa$, where $M_{\nu\kappa}$ is the number of 0-1 matrices with row sums ν and column sums κ; $M_{\nu\kappa} \neq 0$ forces $\kappa \trianglelefteq \nu'$, and $M_{\nu\nu'} = 1$. Since $\nu \trianglerighteq \rho' \iff \nu' \trianglelefteq \rho$ [Mac95, I (1.11)], the left side is contained in the right. Both have dimension $\#\{\kappa \vdash n : \kappa \trianglelefteq \rho\}$, because the e_ν, $\nu \vdash n$, are independent for $m \geq n$. □

For $S \subseteq [m]$ and nonzero $R \in \mathbb{Q}(s,t)(x)$, let $\deg_S R$ be the degree in u of $R|_{x_i \mapsto u x_i\,(i\in S)}$ (numerator degree minus denominator degree). It is additive on products and satisfies $\deg_S(R_1+R_2) \leq \max(\deg_S R_1, \deg_S R_2)$. For a polynomial P, $\deg_S P$ is the largest value of $\sum_{i\in S} \alpha_i$ over the monomials x^α of P; for symmetric P it depends only on $j=|S|$, and we write $D_j(P)$. Since the m_κ have disjoint supports and $D_j(m_\kappa) = \kappa_1 + \cdots + \kappa_j$:

Lemma 4.4. *A homogeneous $P \in \Lambda_m$ of degree $|\rho|$ lies in $\operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ if and only if $D_j(P) \leq \rho_1 + \cdots + \rho_j$ for all j.*

Lemma 4.5 (Degree step). *For symmetric F and every j, $D_j(E_k F) \leq D_j(F) + \min(k,j)$.*

Proof. Fix $|S| = j$ and a term $R_A = c_A\, x_A\, F|_{x_i \mapsto s x_i\,(i\in A)}$ of Theorem 2.5. Every a_{ij} has $\deg_S a_{ij} = 0$: if i,j are both in S or both outside, a_{ij} is homogeneous of degree 0 in u or constant; otherwise numerator and denominator have degree 1. Next, $\deg_S x_A = |A \cap S| \leq \min(k,j)$. Rescaling some variables by the constant s does not change u-degrees, so the last factor has $\deg_S = D_j(F)$. Since $E_k F$ is a polynomial, its $\deg_S$ is its monomial degree. □

4.2. **Support, diagonal coefficient, and the value at $s = 1$.**

Proof of the support statement. Starting from $D_j(1) = 0$ and applying Lemma 4.5 once for each part, $D_j(e^\star_\lambda) \le \sum_i \min(\lambda_i, j) = \lambda'_1 + \cdots + \lambda'_j$. By Lemmas 4.3 and 4.4, $e^\star_\lambda \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \lambda'\} = \operatorname{span}\{e_\nu : \nu \trianglerighteq \lambda\}$. The support part of Corollary 4.2 is the same argument with one step, starting from $D_j(e_\mu) = \sum_i \min(\mu_i, j)$. □

Fix integers $w_1 > w_2 > \cdots > w_m > 0$ and for $\rho = (\rho_1 \ge \cdots \ge \rho_m \ge 0)$ put $W(\rho) = \sum_i w_i \rho_i$.

Lemma 4.6. *If $\operatorname{sort}(\alpha) = \kappa \trianglelefteq \rho$, then $\sum_i w_i \alpha_i \le W(\rho)$, with equality only if $\alpha = \rho$. Hence if $P \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ has m_ρ-coefficient c, the coefficient of $u^{W(\rho)}$ in $P(u^{w_1}x_1, \dots, u^{w_m}x_m)$ is $c\,x^\rho$, and no higher power of u occurs.*

Proof. $\sum_i w_i\alpha_i \le \sum_i w_i\kappa_i$ by the rearrangement inequality, with equality only if $\alpha = \kappa$ (the w_i are distinct). By Abel summation, $\sum_i w_i\kappa_i = \sum_j (w_j - w_{j+1})(\kappa_1 + \cdots + \kappa_j)$ with $w_{m+1} := 0$; all weights are positive, so this is at most the same expression for ρ, with equality only if $\kappa = \rho$. □

Lemma 4.7 (Lead step)**.** *Let $F \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ with m_ρ-coefficient c_F. Then $E_k F \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho + 1^k\}$, with $m_{\rho+1^k}$-coefficient $s^{\rho_1+\cdots+\rho_k} c_F$.*

Proof. Membership is Lemmas 4.4 and 4.5. Substitute $x_i \mapsto u^{w_i}x_i$ and expand each term R_A in u^{-1}. For $i < j$, $a_{ij} = 1 + O(u^{-1})$, and for $i > j$, $a_{ij} = t + O(u^{-1})$, so $c_A = t^{\operatorname{inv}(A)}(1 + O(u^{-1}))$ with $\operatorname{inv}(A) = \#\{i \in A, j \notin A : i > j\}$. Next, x_A becomes $u^{\sum_{a\in A} w_a} x_A$. Finally, $F|_{x_a \mapsto s x_a\,(a\in A)}$ has the monomials of F with x^α rescaled by $s^{\sum_{a\in A}\alpha_a}$, so by Lemma 4.6 it is $u^{W(\rho)}\big(c_F s^{\sum_{a\in A}\rho_a} x^\rho + O(u^{-1})\big)$. The exponent $W(\rho) + \sum_{a\in A} w_a$ is uniquely maximal for $A = [k]$, where $\operatorname{inv} = 0$. So the coefficient of $u^{W(\rho+1^k)}$ in E_kF is $s^{\rho_1+\cdots+\rho_k} c_F\, x^{\rho+1^k}$; conclude by Lemma 4.6 for $\rho + 1^k$. □

Proof of the diagonal coefficient. Apply the operators in any order. If the parts applied so far form μ, the current function lies in $\operatorname{span}\{m_\kappa : \kappa \trianglelefteq \mu'\}$, and applying E_k multiplies the $m_{\mu'}$-coefficient by $s^{\mu'_1+\cdots+\mu'_k} = s^{\sum_i \min(\mu_i, k)}$ (Lemma 4.7; note $\mu' + 1^k = (\mu \cup k)'$). Each unordered pair of parts contributes its minimum once, so the total exponent is $\sum_{i<p} \min(\lambda_i, \lambda_p) = \sum_p (p-1)\lambda_p = n(\lambda)$. By the triangularity in the proof of Lemma 4.3, the only e_ν with $\nu \trianglerighteq \lambda$ containing $m_{\lambda'}$ is e_λ, with coefficient 1. Hence $c_{\lambda\lambda} = s^{n(\lambda)}$. The same computation gives the leading term in Corollary 4.2. □

Proof of integrality and of $c_{\lambda\mu}|_{s=1} = 0$. Let $V = \prod_{i<j}(x_i - x_j)$. Multiplying the subset formula by V gives

$$V\,E_kF = \sum_{|A|=k} \varepsilon_A \prod_{\substack{i<j \\ \text{both in or both out of } A}} (x_i - x_j) \prod_{i\in A,\, j\notin A} (x_i - t x_j)\; x_A\, F|_{x_a \mapsto s x_a\,(a\in A)}, \qquad \varepsilon_A = \pm 1.$$

If $F \in \mathbb{Q}[s,t][x]$, the right side lies in $\mathbb{Q}[s,t][x]$. As V has content 1 and E_kF is a polynomial, Gauss's lemma gives $E_kF \in \mathbb{Q}[s,t][x]$. By induction $e^\star_\lambda \in \mathbb{Q}[s,t][x]^{S_m}$, and its e-coefficients lie in $\mathbb{Q}[s,t]$. Now set $s = 1$ (a ring map on $\mathbb{Q}[s,t][x]$): every $F|_{x_a\mapsto s x_a}$ becomes $F|_{s=1}$, so $V\,(E_kF)|_{s=1} = F|_{s=1} \cdot V\,E_k(1) = V\,e_k\,F|_{s=1}$ by Lemma 2.7. Hence $(E_kF)|_{s=1} = e_kF|_{s=1}$, $e^\star_\lambda|_{s=1} = e_\lambda$, and $c_{\lambda\mu}|_{s=1} = \delta_{\lambda\mu}$. □

4.3. **The exact s-adic valuation.** It remains to show that $s^{n(\mu)}$ divides $c_{\lambda\mu}$ and that $(c_{\lambda\mu}/s^{n(\mu)})|_{s=t=0} = 1$ for all $\mu \trianglerighteq \lambda$; in particular $c_{\lambda\mu} \ne 0$. Put $\omega(\alpha) := \sum_j \binom{\alpha_j}{2}$ for $\alpha \in \mathbb{Z}^m_{\ge 0}$ and $B_\mu := \{\beta : \operatorname{sort}(\beta) \trianglelefteq \mu'\}$, the monomial support of $e^\star_\mu$. Write $e^\star_\lambda = \sum_\beta a^\lambda_\beta x^\beta$; a^λ_β is symmetric in β. The source of s^n is the identity

$$\omega(\beta + 1_A) = \omega(\beta) + 1_A \cdot \beta. \tag{4.1}$$

Let $\mathbb{K} = \mathbb{Q}(t)(s^{1/2})$ (or $\mathbb{Q}(s^{1/2})$ when $t = 0$) with the s-adic valuation v, extended to $\mathbb{K}(x)$ by the Gauss valuation $v(\sum a_\beta x^\beta) = \min_\beta v(a_\beta)$. For $c \in (\frac{1}{2}\mathbb{Z})^m$ let σ_c be the automorphism $x_j \mapsto s^{-c_j} x_j$, so that $v(\sigma_c P) = \min_\beta(v(a_\beta) - c \cdot \beta)$.

Lemma 4.8. *For generic t, and also at $t = 0$, $v(a^\lambda_\alpha) \geq \omega(\alpha)$ for all α.*

Proof. We show $v(\sigma_c e^\star_\lambda) \geq \min_{\beta \in B_\lambda}(\omega(\beta) - c \cdot \beta)$ for all c, by induction along $e^\star_\mu \mapsto E_k e^\star_\mu = e^\star_{\mu \cup k}$, the case $e^\star_\varnothing = 1$ being clear. In a term R_A: $v(\sigma_c a_{ij}) = 0$ for generic t, and at $t = 0$, $v(\sigma_c a_{ij}) = \max(c_i, c_j) - c_i \geq 0$; $\sigma_c x_A = s^{-c \cdot 1_A} x_A$; and $\sigma_c(F|_{x_a \mapsto s x_a\,(a \in A)}) = \sigma_{c - 1_A} F$. Hence, by induction and (4.1),

$$v(\sigma_c R_A) \geq -c \cdot 1_A + \min_{\beta \in B_\mu}\big(\omega(\beta) - (c - 1_A) \cdot \beta\big) = \min_{\beta \in B_\mu}\big(\omega(\beta + 1_A) - c \cdot (\beta + 1_A)\big),$$

and $\beta + 1_A \in B_{\mu \cup k}$ because the top-j sums grow by at most $\min(j, k)$ and $\mu' + 1^k = (\mu \cup k)'$. Now take $c = \alpha - \frac{1}{2}\mathbf{1}$. Then $\omega(\beta) - c \cdot \beta = \sum_j((\beta_j - \alpha_j)^2 - \alpha_j^2)/2$ is uniquely minimised over $\mathbb{Z}^m$ at $\beta = \alpha$, so $v(a^\lambda_\alpha) - c \cdot \alpha \geq v(\sigma_c e^\star_\lambda) \geq \omega(\alpha) - c \cdot \alpha$. □

Proof that $s^{n(\mu)} \mid c_{\lambda\mu}$. Write $c_{\lambda\nu} = s^{n(\nu)} d_\nu$ and suppose $\delta := \min_\nu v(d_\nu) < 0$. Choose μ dominance-minimal with $v(d_\mu) = \delta$. The coefficient of $x^{\mu'}$ is $a^\lambda_{\mu'} = \sum_{\nu \trianglelefteq \mu} c_{\lambda\nu} M_{\nu\mu'}$ with $M_{\mu\mu'} = 1$. Since $\nu \triangleleft \mu$ implies $n(\nu) > n(\mu)$ (as $n(\nu) = \sum_{i \geq 1}(n - \nu_1 - \cdots - \nu_i)$), every term with $\nu \neq \mu$ has valuation $> n(\mu) + \delta$, so $v(a^\lambda_{\mu'}) = n(\mu) + \delta < n(\mu) = \omega(\mu')$, contradicting Lemma 4.8. So d_ν is regular at $s = 0$; as $c_{\lambda\nu} \in \mathbb{Q}[s, t]$, $s^{n(\nu)}$ divides $c_{\lambda\nu}$ in $\mathbb{Q}[s, t]$. The same argument at $t = 0$ shows $d_\mu|_{s=t=0} = [s^{\omega(\mu')}]\, a^\lambda_{\mu'}|_{t=0}$. □

For the value at $s = t = 0$ we need the initial forms. For $R \in \mathbb{K}(x)$ with $v(R) \geq v_0$, let $\mathrm{in}_{v_0}(R)$ be the image of $s^{-v_0} R$ in the residue field of $\mathbb{K}(x)$, which is $\mathbb{Q}(t)(x)$ (or $\mathbb{Q}(x)$ when $t = 0$); it is additive at a common level and multiplicative. For $\kappa = \mathrm{sort}(\alpha)$ let peel_k subtract 1 from the k largest parts.

Lemma 4.9 (Peel Lemma)**.** *If $\kappa \trianglelefteq \rho$, $|\kappa| = |\rho|$ and $k \leq \ell(\rho)$, then $\mathrm{peel}_k \kappa \trianglelefteq \mathrm{peel}_k \rho$.*

Proof. Note $\ell(\kappa) \geq \ell(\rho) \geq k$. Counting parts $\geq c$ gives $(\mathrm{peel}_k \kappa)'_c = \kappa'_c - \min(k, \kappa'_c) + \min(k, \kappa'_{c+1})$, so with S_C the sum of the first C parts and $\kappa'_1 \geq k$, $S_C((\mathrm{peel}_k \kappa)') = S_C(\kappa') - k + \min(k, \kappa'_{C+1})$, and likewise for ρ. Let $D_C = S_C(\kappa') - S_C(\rho') \geq 0$ (by [Mac95, I (1.11)]) and $\bar{m}(y) = \min(k, y)$. If $\bar{m}(\kappa'_{C+1}) \geq \bar{m}(\rho'_{C+1})$ we are done. Otherwise $\kappa'_{C+1} < k$ and $\kappa'_{C+1} < \rho'_{C+1}$, and $D_C = D_{C+1} + \rho'_{C+1} - \kappa'_{C+1} \geq \bar{m}(\rho'_{C+1}) - \bar{m}(\kappa'_{C+1})$. So $(\mathrm{peel}_k \kappa)' \trianglerighteq (\mathrm{peel}_k \rho)'$. □

Lemma 4.10. *At $t = 0$, $\Lambda^0_\lambda(\alpha) := [s^{\omega(\alpha)}]\, a^\lambda_\alpha$ equals 1 if $\mathrm{sort}(\alpha) \trianglelefteq \lambda'$ and 0 otherwise.*

Proof. Induction along E_k; the base case is clear, and $\Lambda^0_{\mu \cup k}(\alpha) = 0$ outside $B_{\mu \cup k}$ by the support statement. Let $\mathrm{sort}(\alpha) \trianglelefteq (\mu \cup k)'$, $c = \alpha - \frac{1}{2}\mathbf{1}$ and $v_0 = \omega(\alpha) - c \cdot \alpha$. By the uniqueness in the proof of Lemma 4.8, $\mathrm{in}_{v_0}(\sigma_c e^\star_{\mu \cup k}) = \Lambda^0_{\mu \cup k}(\alpha)\, x^\alpha$. If $\alpha - 1_A$ has a negative entry or lies outside B_μ, then $v(\sigma_c R_A) > v_0$ by that uniqueness and the term R_A does not contribute. Otherwise, with $\beta_0 = \alpha - 1_A$, the levels split as in (4.1) and

$$\mathrm{in}_{v_0}(\sigma_c R_A) = \mathrm{in}_0(\sigma_c c_A) \cdot \Lambda^0_\mu(\beta_0)\, x^\alpha.$$

At $t = 0$, $\mathrm{in}_0(\sigma_c a_{ij})$ is 1 if $\alpha_i > \alpha_j$, 0 if $\alpha_i < \alpha_j$, and $x_i/(x_i - x_j)$ if $\alpha_i = \alpha_j$. So only sets A that are upper sets for α contribute: with v the k-th largest entry of α and $L = \{j : \alpha_j = v\}$, $A = \{j : \alpha_j > v\} \sqcup B$ with $B \subseteq L$ of size $r = k - \#\{j : \alpha_j > v\}$. As $\ell(\mathrm{sort}\,\alpha) \geq \ell((\mu \cup k)') \geq k$, $v \geq 1$. For all such A, $\mathrm{sort}(\alpha - 1_A) = \mathrm{peel}_k(\mathrm{sort}\,\alpha)$, and Λ^0_μ is symmetric, so

$$\Lambda^0_{\mu \cup k}(\alpha) = \Lambda^0_\mu(\mathrm{peel}_k\,\mathrm{sort}\,\alpha) \cdot \sum_{B \subseteq L,\ |B| = r} \prod_{i \in B,\ j \in L \setminus B} \frac{x_i}{x_i - x_j} = \Lambda^0_\mu(\mathrm{peel}_k\,\mathrm{sort}\,\alpha).$$

The last sum is the case $t=0$ of $\sum_{|B|=r}\prod_{i\in B,j\in L\setminus B}\frac{x_i-tx_j}{x_i-x_j}=\begin{bmatrix}|L|\\ r\end{bmatrix}_t$, which follows from the Poincaré identity [Mac95, III (1.4)] by grouping $S_{|L|}$ according to the image of $\{1,\dots,r\}$. Finally, by Lemma 4.9 with $\rho=(\mu\cup k)'=\mu'+1^k$, $\mathrm{peel}_k(\mathrm{sort}\,\alpha)\trianglelefteq\mathrm{peel}_k(\mu'+1^k)=\mu'$, so $\Lambda^0_\mu(\mathrm{peel}_k\,\mathrm{sort}\,\alpha)=1$. □

End of proof of Theorem 4.1. By the last two proofs, $(c_{\lambda\mu}/s^{n(\mu)})|_{s=t=0}=\Lambda^0_\lambda(\mu')=1$ for $\mu\trianglerighteq\lambda$, since $\mu'\trianglelefteq\lambda'$. In particular $c_{\lambda\mu}\neq0$ and its s-adic valuation is $n(\mu)$. □

Remark 4.11. The leading coefficients $d_{\lambda\mu}(t)=[s^{n(\mu)}]c_{\lambda\mu}$ depend on t; for instance $d_{(1^4),(2,1,1)}=1+3t$. They are identified in Corollary 5.2. The degree count above is the same kind of triangularity argument as for Macdonald's operators [Mac95, VI (3.6)–(3.10)].

5. The $s\to0$ edge

Since $s=q^{-1}$, the edge $s\to0$ is Hikita's $q\to\infty$, where the unrescaled product degenerates [Hik25, Lemma 6.3]. After the rescaling of Theorem 4.1 it becomes Hall–Littlewood multiplication. Throughout this section t is an indeterminate, $b_\mu:=s^{n(\mu)}e_\mu$, and we work in Λ_m with m at least the degree (Lemma 2.8). We keep the notation $\mathbb{K}$, v, σ_c, ω, in_{v_0} of Section 4, with $\mathbb{K}=\mathbb{Q}(t)(s^{1/2})$; so initial forms take values in $\mathbb{Q}(t)(x)$.

Theorem 5.1 (The $s\to0$ edge)**.** *Let $\mathcal{O}=\mathbb{Q}(t)[s]_{(s)}$ and $\mathcal{L}=\bigoplus_\mu\mathcal{O}\,b_\mu$.*

1. *$E_kb_\mu\in\bigoplus_\nu\mathbb{Q}[s,t]\,b_\nu$; in particular $E_k\mathcal{L}\subseteq\mathcal{L}$.*
2. *Let L_k be the map induced by E_k on the quotient $\mathcal{L}/s\mathcal{L}=\bigoplus_\mu\mathbb{Q}(t)\,\bar b_\mu$, and let $\phi(\bar b_\mu):=t^{-n(\mu')}P_{\mu'}(x;t^{-1})$. Then $\phi\circ L_k=t^{-\binom{k}{2}}e_k\cdot\phi$.*
3. *Explicitly, with $\rho=\mu'$ and $\kappa=\nu'$,*

$$L_k\bar b_\mu=\sum_{\kappa/\rho\ \text{vertical k-strip}}t^{\mathrm{inv}(\kappa/\rho)}\prod_{v\ge1}\begin{bmatrix}m_v(\kappa)\\ r_v\end{bmatrix}_t\bar b_\nu,$$

where r_v is the number of rows of κ of length v that end in a box of the strip, and $\mathrm{inv}(\kappa/\rho)=\sum_{v<w}r_v\,(m_w(\kappa)-r_w)$.

Corollary 5.2 (The s-leading coefficients)**.** *For $\lambda,\mu\vdash n$, $d_{\lambda\mu}(t):=[s^{n(\mu)}]\,c_{\lambda\mu}$ satisfies*

$$d_{\lambda\mu}(t)=t^{\,n(\mu')-n(\lambda')}\,[P_{\mu'}(x;t^{-1})]\,e_\lambda=t^{-n(\lambda')}\sum_\nu K_{\nu'\lambda}\,\widetilde{K}_{\nu\mu'}(t),$$

where $\widetilde{K}_{\nu\kappa}(t)=t^{n(\kappa)}K_{\nu\kappa}(t^{-1})$ is the cocharge Kostka–Foulkes polynomial. Hence $d_{\lambda\mu}\in\mathbb{N}[t]$, and $d_{\lambda\mu}(1)$ is the number of 0-1 matrices with row sums λ and column sums μ'.

5.1. **Admissible functions.** Call a homogeneous $F=\sum_\beta a_\beta x^\beta\in\mathbb{K}[x]^{S_m}$ *admissible* if $v(a_\beta)\ge\omega(\beta)$ for all β. As in the proof of Lemma 4.8, F is admissible if and only if $v(\sigma_cF)\ge\min_{\beta\in\mathbb{Z}^m_{\ge0}}(\omega(\beta)-c\cdot\beta)$ for all $c\in(\frac12\mathbb{Z})^m$; the backward direction uses $c=\alpha-\frac12\mathbf{1}$, for which $\beta=\alpha$ is the unique minimiser. For admissible F let $\mathrm{in}(F)(\alpha)\in\mathbb{Q}(t)$ be the residue of $s^{-\omega(\alpha)}a_\alpha$; it depends only on $\mathrm{sort}(\alpha)$. With $c=\alpha-\frac12\mathbf{1}$ and $v_0=\omega(\alpha)-c\cdot\alpha$,

$$\mathrm{in}_{v_0}(\sigma_cF)=\mathrm{in}(F)(\alpha)\,x^\alpha.\tag{5.1}$$

Lemma 5.3. *A homogeneous $F=\sum_\mu c_\mu e_\mu$ of degree $\le m$ is admissible if and only if $v(c_\mu)\ge n(\mu)$ for all μ. In that case $\mathrm{in}(F)(\mu')$ is the residue of $s^{-n(\mu)}c_\mu$.*

Proof. The coefficient of x^β in b_μ is $s^{n(\mu)}M_{\mu\kappa}$ with $\kappa=\mathrm{sort}(\beta)$. If $M_{\mu\kappa}\neq0$ then $\kappa\trianglelefteq\mu'$, so $\kappa'\trianglerighteq\mu$ and $\omega(\beta)=n(\kappa')\le n(\mu)$, with equality only for $\kappa=\mu'$, where $M_{\mu\mu'}=1$. So b_μ is admissible with $\mathrm{in}(b_\mu)(\kappa)=[\kappa=\mu']$, which gives the backward direction and the formula. The

forward direction is the argument in the proof that $s^{n(\mu)} \mid c_{\lambda\mu}$ in Section 4, with a^λ replaced by the coefficients of F. □

Lemma 5.4. $\sum_{B\subseteq[N],|B|=r} \prod_{i\in B, j\notin B} a_{ij} = \begin{bmatrix} N \\ r \end{bmatrix}_t$.

Proof. Group the Poincaré identity $\sum_{w\in S_N} w(\prod_{i<j} a_{ij}) = [N]_t!$ [Mac95, III (1.4)] by $B = w(\{1,\dots,r\})$; the stabiliser $S_r \times S_{N-r}$ contributes $[r]_t![N-r]_t!$. □

5.2. **The recursion.**

Proposition 5.5. *Let F be admissible of degree n, with $m \geq n+k$. Then E_kF is admissible, and for every $\kappa \vdash n+k$*

$$\mathrm{in}(E_kF)(\kappa) = \sum_{\rho:\ \kappa/\rho \text{ vertical } k\text{-strip}} t^{\mathrm{inv}(\kappa/\rho)} \prod_{v\geq 1} \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix}_t \mathrm{in}(F)(\rho). \tag{5.2}$$

Proof. Fix c and a term R_A of Theorem 2.5. Since t is a unit, $v(\sigma_c a_{ij}) = 0$, and the initial form of $\sigma_c a_{ij}$ is 1 if $c_i > c_j$, t if $c_i < c_j$, and a_{ij} if $c_i = c_j$. As in Lemma 4.8, using (4.1),

$$v(\sigma_c R_A) \geq \min_{\gamma \geq 1_A} (\omega(\gamma) - c\cdot\gamma) \geq \min_{\gamma\geq 0} (\omega(\gamma) - c\cdot\gamma),$$

so E_kF is admissible. Now fix α with $\mathrm{sort}(\alpha) = \kappa$, $c = \alpha - \frac{1}{2}\mathbf{1}$, $v_0 = \omega(\alpha) - c\cdot\alpha$; all terms have $v \geq v_0$, so initial forms add, and by (5.1) $\mathrm{in}(E_kF)(\alpha)x^\alpha = \sum_A \mathrm{in}_{v_0}(\sigma_c R_A)$. If $A \not\subseteq \mathrm{supp}(\alpha)$, then $\gamma = \alpha$ is excluded from $\{\gamma \geq 1_A\}$ and $v(\sigma_c R_A) > v_0$. If $A \subseteq \mathrm{supp}(\alpha)$, put $\beta_0 = \alpha - 1_A$; then $c - 1_A = \beta_0 - \frac{1}{2}\mathbf{1}$, the levels add by (4.1), and

$$\mathrm{in}_{v_0}(\sigma_c R_A) = \mathrm{in}_0(\sigma_c c_A)\cdot \mathrm{in}(F)(\beta_0)\, x^\alpha.$$

Let $L_v = \{j : \alpha_j = v\}$ $(v \geq 1)$, so $|L_v| = m_v(\kappa) =: m_v$, and let $B_v = A \cap L_v$, $r_v = |B_v|$. Then

$$\mathrm{in}_0(\sigma_c c_A) = t^{\#\{i\in A,\, j\notin A:\ \alpha_i<\alpha_j\}} \prod_{v\geq 1} \prod_{i\in B_v,\, j\in L_v\setminus B_v} a_{ij}, \qquad \#\{\cdots\} = \sum_{v<w} r_v(m_w - r_w).$$

The partition $\mathrm{sort}(\beta_0)$ is obtained from κ by lowering r_v of its parts equal to v, for each v; this is the ρ with κ/ρ a vertical k-strip and data (r_v), and every such strip arises from exactly one (r_v). So both $\mathrm{in}(F)(\beta_0)$ and the power of t depend only on (r_v), and summing over the B_v with Lemma 5.4 gives $\prod_v \begin{bmatrix} m_v \\ r_v \end{bmatrix}_t$. □

At $t = 0$ every t-binomial is 1 and t^{inv} vanishes unless every part larger than a lowered part is lowered; so (5.2) reduces to the recursion in the proof of Lemma 4.10.

5.3. **Proof of Theorem 5.1 and Corollary 5.2.** *(1).* b_μ is admissible, so E_kb_μ is admissible (Proposition 5.5) and its b_ν-coordinates have $v \geq 0$ (Lemma 5.3). By the integrality part of Theorem 4.1's proof, $E_ke_\mu \in \bigoplus_\nu \mathbb{Q}[s,t]e_\nu$, so the b_ν-coordinate is $p/s^{n(\nu)-n(\mu)}$ with $p \in \mathbb{Q}[s,t]$; having $v \geq 0$ over $\mathbb{Q}(t)$, it lies in $\mathbb{Q}[s,t]$.

(3). By Lemma 5.3, the $\bar{b}_\nu$-coordinate of $L_k\bar{b}_\mu$ is $\mathrm{in}(E_kb_\mu)(\nu')$; apply (5.2) with $\mathrm{in}(b_\mu)(\rho) = [\rho = \mu']$.

(2). Let κ/ρ be a vertical k-strip with data (r_v), obtained by lowering the parts indexed by A. Writing $n(\kappa) = \sum_{\text{pairs of parts}} \min$, a pair of parts both in A loses 1; a pair with one part of value v in A and one of value w not in A loses 1 if and only if $v \leq w$; other pairs lose nothing. Hence

$$n(\kappa) - n(\rho) = \binom{k}{2} + \mathrm{inv}(\kappa/\rho) + \sum_v r_v(m_v - r_v). \tag{5.3}$$

Since $\begin{bmatrix} m \\ r \end{bmatrix}_t\big|_{t\mapsto t^{-1}} = t^{-r(m-r)}\begin{bmatrix} m \\ r \end{bmatrix}_t$, (5.3) gives $t^{\mathrm{inv}}\prod_v \begin{bmatrix} m_v \\ r_v \end{bmatrix}_t = t^{n(\kappa)-n(\rho)-\binom{k}{2}}\psi'_{\kappa/\rho}(t^{-1})$ with $\psi'_{\kappa/\rho}(u) := \prod_v \begin{bmatrix} m_v(\kappa) \\ r_v \end{bmatrix}_u$ (the $q=0$ case of the vertical-strip coefficient $\psi'_{\kappa/\rho}(q,t)$ of [Mac95, VI (6.24)(iv)]; it is not the horizontal-strip $\psi_{\lambda/\mu}$ of [Mac95, III (5.8′)]). The Hall–Littlewood Pieri rule [Mac95, III (3.2)] reads $e_k P_\rho(x;u) = \sum_\kappa \psi'_{\kappa/\rho}(u) P_\kappa(x;u)$, summed over vertical k-strips (there $m_v(\kappa) = \kappa'_v - \kappa'_{v+1}$ and $r_v = \kappa'_v - \rho'_v$). For $X \in \mathcal{L}$ put $h(\kappa) = t^{-n(\kappa)}\mathrm{in}(X)(\kappa)$, so that $\phi(\bar X) = \sum_\kappa h(\kappa) P_\kappa(x;t^{-1})$ by Lemma 5.3. By (5.2), $t^{-n(\kappa)}\mathrm{in}(E_k X)(\kappa) = t^{-\binom{k}{2}}\sum_\rho \psi'_{\kappa/\rho}(t^{-1})h(\rho)$, hence $\phi(L_k\bar X) = t^{-\binom{k}{2}}\sum_\rho h(\rho)\, e_k P_\rho(x;t^{-1}) = t^{-\binom{k}{2}} e_k\, \phi(\bar X)$.

Corollary 5.2. Since $1 = b_\varnothing$ and $\phi(\bar b_\varnothing) = 1$, iterating (2) gives $\phi(e^\star_\lambda \bmod s) = t^{-\sum_i \binom{\lambda_i}{2}} e_\lambda = t^{-n(\lambda')} e_\lambda$. By Theorem 4.1, $e^\star_\lambda \equiv \sum_\mu d_{\lambda\mu}\bar b_\mu$, so comparing coefficients of $P_{\mu'}(x;t^{-1})$ gives the first formula. Expanding $e_\lambda = \omega h_\lambda = \sum_\nu K_{\nu'\lambda} s_\nu$ [Mac95, I §6] and $s_\nu = \sum_\kappa K_{\nu\kappa}(u) P_\kappa(x;u)$ [Mac95, III (2.6)] at $u = t^{-1}$ gives the second. The $K_{\nu\kappa}(t)$ have nonnegative coefficients [Mac95, III (6.5)], and $d_{\lambda\mu} \in \mathbb{Q}[t]$ by Theorem 4.1, so $d_{\lambda\mu} \in \mathbb{N}[t]$. At $t=1$, $P_\kappa(x;1) = m_\kappa$ [Mac95, III §2], so $d_{\lambda\mu}(1) = [m_{\mu'}]e_\lambda = M_{\lambda\mu'}$. □

Remark 5.6 (Prior work; no novelty claim for the matrix). The matrix $d_{\lambda\mu}(t)$ is the classical $e \to$Hall–Littlewood transition matrix: writing $e_\lambda = \sum_\kappa M(e,P)_{\lambda\kappa} P_\kappa(x;t)$, Corollary 5.2 says $d_{\lambda\mu}(t) = t^{n(\mu')-n(\lambda')} M(e,P)_{\lambda\mu'}(t^{-1})$. Its factorisation $M(e,P)_{\lambda\kappa} = \sum_\nu K_{\nu\lambda}K_{\nu'\kappa}(t)$ (our second formula after $t \mapsto t^{-1}$), its positivity and its value at $t=1$ are in [Kir98, §3.2, Thm. 3.4], where the count of 0-1 matrices is credited to Knuth and Thm. 3.4 is proved, as Theorem 5.1(3) is, by iterating the vertical-strip Pieri rule [Mac95, III (3.2)] [Kir98, §4.1]. What is new is only the identification of this matrix with the s-leading coefficients of $\star$, i.e. Theorem 5.1, which we prove from the subset formula. It could also be reached from Theorem 6.2, which is implicit in [DFK19], by the limit argument indicated in Remark 6.5 (not developed here); but the only specialisations of the operators studied in [DFK19] are $t \to \infty$ and $N \to \infty$ (§§4.2–4.3, 7 there), Hall–Littlewood polynomials do not occur there, and we have not found Theorem 5.1 stated elsewhere. One corner is Hikita's: [Hik25, Lemma 6.3] (= [Hik25, Thm. C(ii)]) computes the unrescaled limit $q \to \infty$, and since $n(\mu) > 0$ for $\mu \neq (n)$ that limit is $d_{\lambda,(n)}(t)\, e_n$; his value $[n]_t!/\prod_i [\lambda_i]_t!$ is the case $\mu = (n)$ of Corollary 5.2.

Remark 5.7 (Other edges). The edges $t \to \infty$, $s \to \infty$ and $t = 0$ of the (s,t)-square, and the reflection $(s,t) \mapsto (1/s, 1/t)$, can be treated through the conjugation of $e_k\star(\cdot)$ by the Macdonald-diagonal operator $\mathcal{N}$ of Theorem 6.2, which is implicit in [DFK19]. We do not develop this here and use none of it: the only other edge that enters below is $t = 0$, and for it we cite [DFK18] (Remark 7.7).

6. Hikita's product as a Macdonald-diagonal transport

In this section we show that $\star$ is the ordinary product transported by an operator that is diagonal on Macdonald polynomials (Theorem 6.2). **This identity is implicit in [DFK19]** (Remark 6.4); we give a self-contained proof via a nonsymmetric Gaussian. Nothing outside this section depends on it. The proof uses three standard facts about Cherednik operators: commutativity, Bernstein's relations, and triangularity with simple spectrum. For completeness, Appendix A proves them in our conventions, citing only Bernstein's presentation of the extended affine braid group in the form [Hai06, Thm. 4.2].

Throughout, $\mathsf{P}_\nu := P_\nu(x; s, t^{-1})$ is the Macdonald polynomial [Mac95, VI §4] with parameters $(q,t) = (s, t^{-1})$, in m variables, for $\ell(\nu) \le m$. The P_ν with $\ell(\nu) \le m$ form a basis of Λ_m, and $\mathsf{P}_{(1^k)} = e_k$, because $m_{(1^k)} = e_k$ is minimal in dominance among the monomial functions of degree k.

Definition 6.1. Let $\mathcal{N}$ be the linear operator on Λ_m with $\mathcal{N}\,\mathsf{P}_\nu = t^{n(\nu)}s^{n(\nu')}\,\mathsf{P}_\nu$.

Theorem 6.2 (∇-transport)**.** *For all $m \geq 1$ and $1 \leq k \leq m$, as operators on Λ_m,*

$$E_k = t^{-\binom{k}{2}}\,\mathcal{N} \circ e_k \circ \mathcal{N}^{-1}.$$

Equivalently, $F \star G = \mathcal{N}(\mathcal{N}^{-1}F \cdot \mathcal{N}^{-1}G)$ for all $F, G \in \Lambda_m$, and $e^\star_\lambda = t^{-n(\lambda')}\mathcal{N}e_\lambda$.

Proof of the equivalence. Put $F \star' G := \mathcal{N}(\mathcal{N}^{-1}F \cdot \mathcal{N}^{-1}G)$, a commutative associative product on Λ_m. Since $\mathcal{N}e_k = t^{\binom{k}{2}}e_k$, the displayed identity says $e_k \star F = E_kF = e_k \star' F$ for all F (Theorem 2.5). Both products are associative, so $G \star F = G \star' F$ whenever G is a polynomial in the e_k for either product. Iterating, $e^\star_\lambda = e_{\lambda_1} \star' \cdots \star' e_{\lambda_\ell} = \mathcal{N}(\mathcal{N}^{-1}e_{\lambda_1} \cdots \mathcal{N}^{-1}e_{\lambda_\ell}) = t^{-n(\lambda')}\mathcal{N}e_\lambda$, using $\sum_i \binom{\lambda_i}{2} = n(\lambda')$. These elements span Λ_m as λ runs over the partitions with $\lambda_1 \leq m$, so $\star = \star'$. Conversely, $\star = \star'$ gives $E_kF = e_k \star F = t^{-\binom{k}{2}}\mathcal{N}(e_k\,\mathcal{N}^{-1}F)$. □

Remark 6.3 (The ∇ form). Let ∇ be the operator with $\nabla\widetilde{H}_\nu = t^{n(\nu)}q^{n(\nu')}\widetilde{H}_\nu$ on the modified Macdonald polynomials, which we normalise by $\widetilde{H}_\nu(x;q,t) := t^{n(\nu)}J_\nu[X/(1-t^{-1});q,t^{-1}]$, with J_ν the integral form, a scalar multiple of P_ν [Mac95, VI §8]. Put $\varphi(F) := F[-\varepsilon X/(1-t)]$, so $\varphi^{-1}(G) = G[-\varepsilon X(1-t)]$; here $p_r[-\varepsilon X] = (-1)^{r-1}p_r$. Then $X/(1-t^{-1})$ evaluated at $-\varepsilon X(1-t)$ is εtX, so

$$\varphi^{-1}\widetilde{H}_\nu(x;s,t) = t^{n(\nu)}J_\nu[\varepsilon tX;s,t^{-1}] = (-1)^{|\nu|}\,t^{|\nu|+n(\nu)}\,J_\nu(x;s,t^{-1}).$$

This is a scalar multiple of P_ν, so $\mathcal{N} = \varphi^{-1} \circ \nabla_{q=s} \circ \varphi$. Here the right side acts on Λ; it is diagonal on the P_ν, so it preserves the kernel of the restriction $\Lambda \to \Lambda_m$, which is spanned by the P_ν with $\ell(\nu) > m$, and the identity holds for the induced operator on Λ_m. Theorem 6.2 therefore reads $t^{n(\lambda')}e^\star_\lambda = \varphi^{-1}\nabla_{q=s}\varphi(e_\lambda)$. Since $F[\varepsilon X] = (-1)^d F$ for F of degree d, φ agrees up to this sign with the plethysm $\Sigma\colon X \mapsto X/(t-1)$ of [DFK19, Thm. 4.7], so $\varphi^{-1}\nabla\varphi = \Sigma^{-1}\nabla\Sigma$. With the dictionary of Remark 6.4, the identity $\mathcal{N} = \varphi^{-1}\nabla_{q=s}\varphi$ is [DFK19, Rem. 4.8].

Remark 6.4 (Attribution). We use the numbering of arXiv:1704.00154, and take (q,t) of [DFK19] to be (s,t^{-1}). Then $E_k = t^{k(m-k)}M_{k;1}$, where $M_{k;1}$ (our notation) is the generalised Macdonald operator $\mathcal{M}_{k;1}$ of [DFK19, (1.5)] with $N = m$. By [DFK19, Lemma 2.12, Lemma 2.16, Thm. 2.17], $M_{k;1}$ is a scalar multiple of the image of $e_k(Y)$ under the automorphism τ_+, restricted to symmetric functions. (In [DFK19] lemmas and equations are numbered by separate counters: Lemma 2.12 is the τ_+ lemma, which is not equation (2.12).) By [DFK19, Lemma 2.13, Rem. 2.14, (4.14)] (we use only the statement of Lemma 2.13, not its one-line derivation), the automorphism τ_- is conjugation by an N-variable version $\nabla^{(N)}$ of ∇, which agrees with $\mathcal{N}$ up to a constant and a factor depending only on the degree. After the conventions are matched, the $SL_2(\mathbb{Z})$ relations [DFK19, (6.3), (6.5), (6.8)] give $\tau_-(p_j(X)) = \tau_+(p_j(Y))$ for $j \geq 1$, hence $\tau_-(e_k(X)) = \tau_+(e_k(Y))$. Together these give Theorem 6.2, and the scalars compose to exactly $t^{-\binom{k}{2}}$: the constant of $\nabla^{(N)}$ cancels in the conjugation, and the remaining factors (the $t^{k(m-k)}$ above, the powers of $t^{1/2}$ and $s^{1/2}$ from [DFK19, (2.11) in Thm. 2.17, Lemma 2.16], and the degree factor of [DFK19, (4.14)]) combine to $t^{k(m-k)} \cdot t^{-k(m-k)/2}s^{-k/2} \cdot t^{-k(m-1)/2}s^{k/2} = t^{-\binom{k}{2}}$. The product $\star$ does not appear in [DFK19], and the identity is not stated there for $k \geq 2$. Neither Theorem 6.2 nor its equivalent forms are claimed as new. What we add is an elementary route, which makes the normalisations explicit: one telescoping commutator (Lemma 6.6) and a Gaussian on the nonsymmetric eigenbasis (Proposition 6.7).

Remark 6.5 (What uses Theorem 6.2). No other result of this paper uses Theorem 6.2. In particular, Theorems 4.1, 5.1 and 7.6 and the results of Sections 8 to 10 are proved without it. It does explain the shape of Theorem 5.1. If $e_\mu = \sum_{\nu \trianglelefteq \mu'} a_{\mu\nu}\mathsf{P}_\nu$, then the coefficient of P_ν in

$\mathcal{N}^{-1}b_\mu$ carries the factor $s^{n(\mu)-n(\nu')}$. For $\nu \trianglelefteq \mu'$ we have $n(\nu') \le n(\mu)$, with equality only for $\nu = \mu'$. So modulo s only $\mathsf{P}_{\mu'}$ survives, as in Theorem 5.1(2). The same conjugation governs the edges in Remark 5.7. We do not develop either limit argument here.

6.1. **Untwisted operators.** Let $\mathrm{Pol} := \mathbb{Q}(s,t)[x_1,\dots,x_m]$ and let Pol_d be its homogeneous part of degree d. Besides the operator π of Section 2, we use Cherednik's untwisted shift and the corresponding operators

$$\pi_\circ f(x_1,\dots,x_m) := f(x_2,\dots,x_m,sx_1), \qquad Y_i^\circ := t^{m-i}\,T_{i-1}\cdots T_1\,\pi_\circ\,T_{m-1}^{-1}\cdots T_i^{-1}.$$

So $\pi = x_1\pi_\circ$, and hence $Y_1 = x_1Y_1^\circ$. Besides the relations of Section 2, we use the following elementary relations, proved in Appendix A.1:

(R1) $T_ix_iT_i = t\,x_{i+1}$, and T_i commutes with multiplication by x_j for $j \notin \{i, i+1\}$;
(R2) $\pi_\circ x_j = x_{j+1}\pi_\circ$ for $j < m$, and $\pi_\circ x_m = s\,x_1\pi_\circ$;
(R3) $\pi_\circ T_j = T_{j+1}\pi_\circ$ for $1 \le j \le m-2$;
(R4) $T_i^{-1} = t^{-1}T_i - (1-t^{-1})$.

From (R1) and (R4) we get

$$T_j^{-1}x_j = x_{j+1}T_j^{-1} - (1-t^{-1})\,x_j, \qquad T_1x_2 = x_1T_1 + (t-1)\,x_2, \qquad x_{j+1}T_j^{-1} = t^{-1}T_jx_j. \tag{6.1}$$

For the first, rewrite (R1) as $T_jx_j = t\,x_{j+1}T_j^{-1}$ and substitute $T_j = tT_j^{-1} + (t-1)$. For the second, rewrite (R1) as $x_1T_1 = tT_1^{-1}x_2 = (T_1 - (t-1))x_2$. The third is (R1) multiplied by $t^{-1}T_j^{-1}$ on the right.

6.2. **The commutator.**

Lemma 6.6. *On* Pol, $\left[\,Y_1^\circ + \cdots + Y_m^\circ,\ x_1\right] = (s-1)\,x_1Y_1^\circ$.

Proof. The terms $j \ge 2$. The tail $T_{m-1}^{-1}\cdots T_j^{-1}$ commutes with x_1, and $\pi_\circ x_1 = x_2\pi_\circ$. By (6.1), $T_1x_2 = x_1T_1 + (t-1)x_2$, and $T_{j-1}\cdots T_2$ commutes with x_1. Hence

$$[Y_j^\circ, x_1] = (t-1)\,L_j, \qquad L_j := t^{m-j}\,T_{j-1}\cdots T_2\,x_2\,\pi_\circ\,T_{m-1}^{-1}\cdots T_j^{-1}.$$

The term $j = 1$. Moving x_1 to the left with the first identity of (6.1), one factor at a time, and using that x_i commutes with $T_{i+1}^{-1},\dots,T_{m-1}^{-1}$, we get

$$T_{m-1}^{-1}\cdots T_1^{-1}x_1 = x_mT_{m-1}^{-1}\cdots T_1^{-1} - (1-t^{-1})\sum_{i=1}^{m-1} x_i\,T_{m-1}^{-1}\cdots T_{i+1}^{-1}\,T_{i-1}^{-1}\cdots T_1^{-1}.$$

Apply $t^{m-1}\pi_\circ$ and use (R2). The first term becomes $s\,x_1Y_1^\circ$, so

$$[Y_1^\circ, x_1] = (s-1)\,x_1Y_1^\circ - (t-1)\sum_{i=1}^{m-1} R_i, \qquad R_i := t^{m-2}\,x_{i+1}\,\pi_\circ\,T_{m-1}^{-1}\cdots T_{i+1}^{-1}\,T_{i-1}^{-1}\cdots T_1^{-1}.$$

Telescoping. The factors $T_{i-1}^{-1}\cdots T_1^{-1}$ commute with $T_{m-1}^{-1}\cdots T_{i+1}^{-1}$, and by (R3) they pass through $\pi_\circ$ as $T_i^{-1}\cdots T_2^{-1}$. By the third identity of (6.1), applied $i-1$ times, $x_{i+1}T_i^{-1}\cdots T_2^{-1} = t^{-(i-1)}T_i\cdots T_2\,x_2$. Hence $R_i = L_{i+1}$. Summing over j, the terms $(t-1)L_j$ for $j \ge 2$ cancel the terms $-(t-1)R_i$. □

6.3. **The nonsymmetric Gaussian.** By Corollary A.5, Pol_d has a basis $\{\mathcal{E}_\lambda\}$, indexed by the compositions $\lambda \in \mathbb{Z}^m_{\geq 0}$ with $|\lambda| = d$, such that

$$Y_i^\circ\, \mathcal{E}_\lambda = y_i(\lambda)\, \mathcal{E}_\lambda, \qquad y_i(\lambda) = s^{\lambda_i} t^{\mathrm{rk}_i(\lambda)}, \qquad \mathrm{rk}_i(\lambda) = \#\{j : \lambda_j > \lambda_i\} + \#\{j < i : \lambda_j = \lambda_i\},$$

and $i \mapsto \mathrm{rk}_i(\lambda)$ is a bijection $[m] \to \{0, \dots, m-1\}$. Two consequences are used repeatedly. First, the vector $y(\lambda)$ determines λ. Second, the sum $\sum_i y_i(\lambda) = \sum_{r=0}^{m-1} s^{a_r} t^r$ determines the multiset $\{y_i(\lambda)\}$, because its exponents of t are distinct. Define $\theta(s^a t^r) := s^{\binom{a}{2}} t^{ar}$, so that

$$\theta(s\, y) = y\, \theta(y) \qquad (y = s^a t^r), \tag{6.2}$$

and let $\widehat{\gamma}$ be the operator on Pol with $\widehat{\gamma}\, \mathcal{E}_\lambda = \gamma(\lambda)\, \mathcal{E}_\lambda$, where $\gamma(\lambda) := \prod_i \theta(y_i(\lambda))$.

Proposition 6.7 (Nonsymmetric Gaussian)**.** *For $1 \leq i \leq m$, as operators on* Pol, $\widehat{\gamma}\, x_i\, \widehat{\gamma}^{-1} = Y_i$, *where Y_i are the operators of Section 2. Consequently the Y_i commute, and $e_k(Y) = \widehat{\gamma}\, e_k(x)\, \widehat{\gamma}^{-1}$.*

Proof. $i = 1$. Write $x_1 \mathcal{E}_\lambda = \sum_\mu c_{\mu\lambda} \mathcal{E}_\mu$, with $|\mu| = |\lambda| + 1$. Applying Lemma 6.6 to $\mathcal{E}_\lambda$ and comparing coefficients gives, for each μ with $c_{\mu\lambda} \neq 0$,

$$\textstyle\sum_j y_j(\mu) = \sum_j y_j(\lambda) + (s-1)\, y_1(\lambda).$$

Write $\sum_j y_j(\lambda) = \sum_r s^{a_r} t^r$, and similarly $\sum_r s^{a'_r} t^r$ for μ. Comparing coefficients of t^r gives $a'_r = a_r$ for $r \neq \mathrm{rk}_1(\lambda)$ and $a'_r = a_r + 1$ for $r = \mathrm{rk}_1(\lambda)$. So the multiset $\{y_j(\mu)\}$ is $\{y_j(\lambda)\}$ with $y_1(\lambda)$ replaced by $s\, y_1(\lambda)$, and $\gamma(\mu) = y_1(\lambda)\gamma(\lambda)$ by (6.2). Hence $\widehat{\gamma} x_1 \widehat{\gamma}^{-1} \mathcal{E}_\lambda = y_1(\lambda)\, x_1 \mathcal{E}_\lambda = x_1 Y_1^\circ \mathcal{E}_\lambda = Y_1 \mathcal{E}_\lambda$.

$\widehat{\gamma}$ commutes with each T_i. Fix d. Let $S_d \subset \mathbb{Q}(s,t)^m$ be the finite set of all $w \cdot y(\lambda)$, with $|\lambda| = d$ and $w \in S_m$. Choose a polynomial g that agrees with $(y_1, \dots, y_m) \mapsto \prod_i \theta(y_i)$ on S_d. Since that function and S_d are S_m-invariant, the symmetrisation f_d of g still agrees with it on S_d. Then $\widehat{\gamma}|_{\mathrm{Pol}_d} = f_d(Y^\circ)|_{\mathrm{Pol}_d}$, which commutes with T_i by Lemma A.1 and Theorem A.2. Since T_i preserves Pol_d, $\widehat{\gamma}$ commutes with T_i.

All i. By (R1) and Lemma A.1(a) for π, $x_{i+1} = t^{-1} T_i x_i T_i$ and $Y_{i+1} = t^{-1} T_i Y_i T_i$. Conjugate by $\widehat{\gamma}$ and induct on i. The consequences follow because the x_i commute. □

Lemma 6.8. *$\widehat{\gamma}$ preserves $\Lambda_m \subset$ Pol, and $\widehat{\gamma}|_{\Lambda_m} = \mathcal{N}$.*

Proof. The proofs of Propositions 2.2 and 2.4 use only (H5) and the fact that $\pi^n F$ is symmetric in $x_1, \dots, x_n$ and in $x_{n+1}, \dots, x_m$. Both also hold for $\pi_\circ$: (H5) is (R3), and $\pi_\circ^n F = F(x_{n+1}, \dots, x_m, sx_1, \dots, sx_n)$. For $k = 1$ and symmetric F this gives

$$e_1(Y^\circ) F = \textstyle\sum_a c_{\{a\}}\, F|_{x_a \mapsto s x_a} = t^{m-1} \sum_a \prod_{j \neq a} \frac{t^{-1} x_a - x_j}{x_a - x_j}\, F|_{x_a \mapsto s x_a},$$

which is t^{m-1} times Macdonald's operator D^1_m [Mac95, VI (3.4)] with $(q,t) = (s, t^{-1})$. Hence $e_1(Y^\circ) \mathsf{P}_\nu = \sum_i s^{\nu_i} t^{i-1}\, \mathsf{P}_\nu$ [Mac95, VI §§3–4]. Expand $\mathsf{P}_\nu = \sum_\lambda u_\lambda \mathcal{E}_\lambda$. Each λ with $u_\lambda \neq 0$ has $\sum_i y_i(\lambda) = \sum_i s^{\nu_i} t^{i-1}$, so its spectral multiset is $\{s^{\nu_i} t^{i-1}\}_i$. Therefore

$$\gamma(\lambda) = \textstyle\prod_i \theta(s^{\nu_i} t^{i-1}) = s^{\sum_i \binom{\nu_i}{2}}\, t^{\sum_i (i-1)\nu_i} = s^{n(\nu')} t^{n(\nu)},$$

so $\widehat{\gamma} \mathsf{P}_\nu = t^{n(\nu)} s^{n(\nu')} \mathsf{P}_\nu$. □

Proof of Theorem 6.2. Let $F \in \Lambda_m$. By Theorem 2.5, Proposition 6.7 and Lemma 6.8,

$$E_k F = t^{-\binom{k}{2}} e_k(Y) F = t^{-\binom{k}{2}}\, \widehat{\gamma}\, e_k\, \widehat{\gamma}^{-1} F = t^{-\binom{k}{2}}\, \widehat{\gamma}(e_k \mathcal{N}^{-1} F) = t^{-\binom{k}{2}}\, \mathcal{N}\, e_k\, \mathcal{N}^{-1} F,$$

since $e_k \mathcal{N}^{-1} F$ is symmetric. □

Computer checks. Lemma 6.6 was checked for $m \le 4$ on all monomials of degree ≤ 3. Proposition 6.7 was checked exactly, with scalar 1 and for all i, for $m = 2$ (degree ≤ 3) and $m = 3$ (degree ≤ 2). Theorem 6.2 was checked for all ν and k with $|\nu| + k \le 5$ at a rational point (s, t). Each of the three wrong Gaussians $s^{\pm\binom{a}{2}} t^{\pm ar}$ (all sign choices other than θ) fails already in degree 1. The two displayed identities of Remark 6.3 were checked symbolically for $|\nu| \le 4$, with $\widetilde{H}_\nu$ computed independently from the Garsia–Haiman triangularities. The identity $\mathcal{N} = \varphi^{-1} \nabla_{q=s} \varphi$ on all e_λ, and the last sentence of the remark, with $e^\star_\lambda$ computed from the operators of Section 2 in $m = |\lambda|$ variables, were checked symbolically for $|\lambda| \le 4$. Replacing $-\varepsilon$ by $+1$ in φ, exchanging $n(\nu)$ and $n(\nu')$, or exchanging s and t makes each identity fail.

7. The block valuation law at $s = 1$

7.1. **Taylor pieces.** We write v for the $(s-1)$-adic valuation on $\mathbb{Q}(t)[s]$, $I = (e_1, e_2, \dots)$ for the augmentation ideal (so I^b is spanned by the e_μ with $\ell(\mu) \ge b$), and $\mathrm{lin}_e G := [e_n]G$ for G homogeneous of degree n; lin_e vanishes on I^2. For ρ a partition, a homogeneous symmetric function is *of type* ρ if it lies in $\mathrm{span}\{e_\nu : \nu \trianglerighteq \rho\}$; by Lemmas 4.3 and 4.4 this means $D_j(P) \le \sum_i \min(\rho_i, j)$ for all j. For a set C of positions, λ_C is the partition formed by the parts λ_i, $i \in C$.

In Theorem 2.5 the variable s enters only through $F|_{x_i \mapsto s x_i \,(i \in A)} = s^{\Delta_A} F$, $\Delta_A = \sum_{i \in A} x_i \partial_i$. Since $s^{\Delta_A} x^\alpha = s^{\alpha \cdot 1_A} x^\alpha = \sum_p (s-1)^p \binom{\Delta_A}{p} x^\alpha$,

$$E_k = \sum_{p \ge 0} (s-1)^p E_k^{(p)}, \qquad E_k^{(p)} = \sum_{|A| = k} c_A x_A \binom{\Delta_A}{p}. \tag{7.1}$$

As $E_k F$ is a symmetric polynomial in x and s (Theorem 4.1), each $E_k^{(p)}$ preserves Λ_m. Since $\binom{\Delta_A}{p}$ is a polynomial of degree p in a derivation, and multiplication by $c_A x_A$ preserves this, $E_k^{(p)}$ is a differential operator of order $\le p$: all $(p+1)$-fold commutators $[\cdots [E_k^{(p)}, a_0], \dots, a_p]$ with multiplication operators vanish. Moreover $E_k^{(0)}$ is multiplication by e_k (by the proof of Theorem 4.1, $(E_k F)|_{s=1} = e_k F|_{s=1}$), and $E_k^{(p)}(1) = 0$ for $p \ge 1$.

7.2. **First order.**

Theorem 7.1 (First order is a biderivation)**.** *Let $D_k := E_k^{(1)}$ and $B(f, g) := \partial_s (f \star g)|_{s=1}$ for $f, g \in \Lambda_m$ with coefficients in $\mathbb{Q}(t)$ (independent of s). Then D_k is a derivation of Λ_m, and*

$$B(f, g) = \sum_{k,l} M_{kl} \frac{\partial f}{\partial e_k} \frac{\partial g}{\partial e_l}, \qquad M_{kl} := D_k(e_l) = B(e_k, e_l) = M_{lk}.$$

So B is a symmetric biderivation, $B(f, g) = L(fg) - fL(g) - gL(f)$ for $L = \frac{1}{2} \sum M_{kl} \partial^2 / \partial e_k \partial e_l$.

Proof. $D_k = \sum_A c_A x_A \Delta_A$ is a sum of derivations. By Theorem 4.1 the $e^\star_\rho$ form a basis whose transition matrix to the e_ρ is unitriangular at $s = 1$; write $f = \sum_\rho a_\rho(s) e^\star_\rho$ with a_ρ regular at $s = 1$. By associativity, $f \star g = \sum_\rho a_\rho(s) E_{\rho_1} \cdots E_{\rho_\ell} g$. By (7.1),

$$B(f, g) = \sum_\rho a'_\rho(1) e_\rho g + \sum_\rho a_\rho(1) \sum_i \Big(\prod_{j \ne i} e_{\rho_j} \Big) D_{\rho_i} g.$$

Applying the same computation to $f = \sum_\rho a_\rho(s) E_{\rho_1} \cdots E_{\rho_\ell}(1)$ itself, which does not depend on s, and using $D_k(1) = 0$, gives $\sum_\rho a'_\rho(1) e_\rho = 0$. Since $f|_{s=1} = \sum_\rho a_\rho(1) e_\rho$, the remaining sum is $\sum_k \frac{\partial f}{\partial e_k} D_k(g)$, and $D_k(g) = \sum_l \frac{\partial g}{\partial e_l} D_k(e_l)$. Symmetry of M is commutativity of $\star$ [Hik25]. □

Proposition 7.2 (The matrix M)**.** *For $k \geq r \geq 0$, with $L(a,b) := \dfrac{(1-t^{a+b})(t^{ab}-1)}{(1-t^a)(1-t^b)}$,*

$$M_{kr} = B(e_k, e_r) = r\, e_k e_r + \sum_{j=1}^{r} L(k-r+j,\, j)\, e_{k+j} e_{r-j}.$$

Consequently $[(s-1)^1]e^\star_\lambda = \sum_{i<j} M_{\lambda_i\lambda_j}\, e_{\lambda\setminus\{\lambda_i,\lambda_j\}}$.

Proof. $M_{kr} = [(s-1)^1]\, e_k \star e_r$. Use the form of the Pieri rule in Corollary 9.6 (Section 9; its proof does not use Proposition 7.2): $e_k \star e_r = \sum_{y=0}^{r} s^y \mathcal{C}_{k-y,r-y} e_{k+r-y} e_y$. Since $\pi_a(b)|_{s=1} = \delta_{b0}$, every $\mathcal{C}_{a,b}$ with $b \geq 1$ vanishes at $s=1$; so the $y=r$ term $s^r e_k e_r$ contributes $r\, e_k e_r$, and the term $y = r-j$, $j \geq 1$, contributes $\partial_s \mathcal{C}_{a,j}|_{s=1}$ with $a = k-r+j$. Now $\partial_s \pi_a(j)|_{s=1} = [u^j] \sum_{m<a} \frac{ut^m}{1+ut^m} = (-1)^{j-1}[a]_{t^j}$, so $\partial_s \mathcal{C}_{a,j}|_{s=1} = -\frac{[a+j]}{[a]}[a]_{t^j} = L(a,j)$. For the second statement, $[(s-1)^1]E_{\lambda_1}\cdots E_{\lambda_\ell}(1) = \sum_i \prod_{j<i} e_{\lambda_j}\, D_{\lambda_i}\big(\prod_{j>i} e_{\lambda_j}\big)$; expand with the Leibniz rule. □

7.3. **Blocks.**

Lemma 7.3 (Polarization)**.** *Let D be a linear operator on a commutative ring and $g_1, \ldots, g_r$ elements of it. For $S \subseteq [r]$ let*

$$\delta_S := [\cdots[D, g_{s_1}], \ldots, g_{s_q}](1) = \sum_{T\subseteq S} (-1)^{|S\setminus T|} \prod_{i\in S\setminus T} g_i\, D\Big(\prod_{i\in T} g_i\Big).$$

Then

$$D(g_1\cdots g_r) = \sum_{S\subseteq[r]} \Big(\prod_{i\notin S} g_i\Big)\delta_S,$$

and if D has order $\leq p$, then $\delta_S = 0$ for $|S| > p$, and for $|S| = p$ the map $(g_i)_{i\in S} \mapsto \delta_S$ is a derivation in each argument.

Proof. Substituting the second expression for δ_S, the right side is

$$\sum_T \prod_{i\notin T} g_i\, D\Big(\prod_{i\in T} g_i\Big) \sum_{S\supseteq T} (-1)^{|S\setminus T|},$$

and the inner sum is $[T = [r]]$. Vanishing is the definition of order $\leq p$. For $|S| = p$, $D' := [\cdots[D, g_{s_2}], \ldots, g_{s_p}]$ has order ≤ 1, so $D'(ab) = aD'(b) + bD'(a) - abD'(1)$, i.e. $[D', ab](1) = a[D', b](1) + b[D', a](1)$; and the iterated commutator is symmetric in its arguments. □

Proposition 7.4 (Block expansion)**.** *Let λ have ℓ parts and $p_1, \ldots, p_\ell \geq 0$ with $\sum_i p_i = m$. Then*

$$E^{(p_1)}_{\lambda_1} \cdots E^{(p_\ell)}_{\lambda_\ell}(1) = \sum_{\pi} \prod_{C\in\pi} g^\pi_C,$$

a finite sum over set partitions π of $[\ell]$ with at least $\ell - m$ blocks, where each g^π_C is homogeneous of type λ_C.

Proof. Induction on the number of operators applied, from the right; the empty product 1 has no blocks. Apply $D = E^{(p)}_k$, $k = \lambda_j$, $p = p_j$, to a term $\prod_{C\in\pi} g_C$. By Lemma 7.3 the result is $\sum_{S\subseteq\pi, |S|\leq p} \prod_{C\notin S} g_C \cdot \delta_S$, and we let $\pi' = (\pi \setminus S) \cup \{C'\}$ with $C' = \{j\} \cup \bigcup S$. Then $|\pi'| \geq |\pi| + 1 - p$, which gives the block count. Each δ_S is a combination of products of the g_C and of D applied to such products. Rescaling monomials does not enlarge monomial supports, so the degree step Lemma 4.5 holds for $E^{(p)}_k$ as well, and D_j is additive on products; hence $D_j(\delta_S) \leq \sum_{C\in S} \sum_{i\in C} \min(\lambda_i, j) + \min(\lambda_j, j)$, i.e. δ_S has type $\lambda_{C'}$. □

Definition 7.5. For $\lambda, \mu \vdash n$, $\kappa(\lambda,\mu)$ is the largest r such that $\lambda = \lambda^1 \sqcup \cdots \sqcup \lambda^r$ and $\mu = \mu^1 \sqcup \cdots \sqcup \mu^r$ as multisets of parts, with every block nonempty and $\mu^i \trianglerighteq \lambda^i$ (so $|\mu^i| = |\lambda^i|$); $\kappa = -\infty$ if $\mu \not\trianglerighteq \lambda$. μ is a *coarsening* of λ if $\kappa(\lambda,\mu) = \ell(\mu)$.

7.4. **The block valuation law.**

Theorem 7.6 (Block valuation law)**.** *For all $\lambda, \mu \vdash n$ and all t, $v(c_{\lambda\mu}) \geq \ell(\lambda) - \kappa(\lambda,\mu)$. For $\mu \trianglerighteq \lambda$, $\mu \neq \lambda$, equality holds over $\mathbb{Q}(t)$ and at $t = 0$:*

$$v(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda,\mu).$$

At $t = 0$ the leading coefficient is $(-1)^{\ell-\kappa} N(\lambda,\mu)$ where $N(\lambda,\mu) \geq 1$ is the number of $(k_{ij})_{1\leq i<j\leq \ell}$ in $\mathbb{Z}_{\geq 0}$ with exactly $\ell - \kappa$ nonzero entries such that $\lambda + \sum k_{ij}(e_i - e_j)$ is a rearrangement of μ (e_i the unit vectors; these are the minimal raising-operator configurations of the proof).

Proof of the lower bound. By (7.1), $[(s-1)^m] e^\star_\lambda = \sum_{\sum p_i = m} E^{(p_1)}_{\lambda_1} \cdots E^{(p_\ell)}_{\lambda_\ell}(1)$. By Proposition 7.4 each term is a sum of products $\prod_{C\in\pi} g_C$ with $|\pi| \geq \ell - m$ and g_C of type λ_C, and such a product lies in $\operatorname{span}\{e_{\sqcup_C \nu^C} : \nu^C \trianglerighteq \lambda_C\}$. So e_μ can occur only if $\kappa(\lambda,\mu) \geq |\pi| \geq \ell - m$. The argument works over $\mathbb{Q}(t)$ and for every specialisation $t \in \mathbb{Q}$. □

Remark 7.7 (Input at $t = 0$). The $t = 0$ edge $e^\star_\mu|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \widetilde{H}_\mu(x;s)$, where $\widetilde{H}_\mu(x;s) = s^{n(\mu)} Q'_\mu(x; s^{-1})$, is **not ours**: it is assembled from [DFK18] eq. (5.15) (with $M_{k,1} = E_k|_{t=0}$), Cor. 5.8 in its level-one form (5.25), Cor. 5.18 (eq. (5.27)) (numbering of arXiv:1505.01657v2), and [Mac95, VI (5.1), (4.14)(iv)]. In particular Theorem 7.6 does not depend on Cherednik's nonsymmetric theory; only Section 6 and Appendix A use it, and nothing outside them depends on them.

Proof of equality. By Remark 7.7, $c_{\lambda\mu}(s,0) = [h_\mu]\, s^{n(\lambda)} Q'_\lambda(x; s^{-1})$. By [Mac95, III (2.15′) with §5 Ex. 7(a)], $Q'_\lambda(x;u) = \prod_{i<j\leq\ell} \frac{1-R_{ij}}{1-uR_{ij}} h_\lambda$, where $R_{ij} h_\alpha = h_{\alpha + e_i - e_j}$, $h_\alpha = h_{\alpha_1} h_{\alpha_2} \cdots$ and $h_r = 0$ for $r < 0$. Expand $\frac{1-R}{1-uR} = 1 + \sum_{a\geq 1} (u-1) u^{a-1} R^a$; at $u = s^{-1}$ the coefficient of R^a is $-(s-1)s^{-a}$. A *configuration* is a function $\mathbf{m}$ on pairs $i < j \leq \ell$ with values in $\mathbb{Z}_{\geq 0}$; its edge set is $E(\mathbf{m}) = \{ij : \mathbf{m}_{ij} \geq 1\}$ and its result is $\alpha(\mathbf{m}) = \lambda + \sum \mathbf{m}_{ij}(e_i - e_j)$. Then

$$c_{\lambda\mu}(s,0) = \sum_{\mathbf{m}:\ \alpha(\mathbf{m})\geq 0,\ \operatorname{sort}\alpha(\mathbf{m}) = \mu} (-1)^{|E(\mathbf{m})|} (s-1)^{|E(\mathbf{m})|} s^{a(\mathbf{m})}, \qquad a(\mathbf{m}) \in \mathbb{Z},$$

a finite sum (the conditions $\alpha_\ell \geq 0$, $\alpha_{\ell-1} \geq 0, \ldots$ bound the $\mathbf{m}_{ij}$ successively). Every configuration contributes $(-1)^{|E|}$ at order $(s-1)^{|E|}$, so there is no cancellation at the lowest order: $v(c_{\lambda\mu}(s,0)) = E_{\min}$, the least $|E(\mathbf{m})|$ over admissible $\mathbf{m}$, with leading coefficient $(-1)^{E_{\min}} N(\lambda,\mu)$, N the number of minimisers.

$E_{\min} \geq \ell - \kappa$: let C be a connected component of the graph $([\ell], E(\mathbf{m}))$. Transfers stay inside C and move mass from a larger index to a smaller one, and λ is weakly decreasing; so the partial sums of $\alpha|_C$ in index order dominate those of λ_C, and sorting only increases them. Hence the components form a compatible block decomposition, there are at most κ of them, and $|E| \geq \ell - \kappa$.

$E_{\min} \leq \ell - \kappa$: take a decomposition with κ blocks. By maximality each block $(\rho,\nu) = (\lambda_C, \mu^C)$ has no compatible refinement. Let $c_1 < \cdots < c_L$ be the positions of C, pad ν with zeros to length L, and put $B_r = \nu_1 + \cdots + \nu_r$, $P_r = \rho_1 + \cdots + \rho_r$. Then $B_r > P_r$ for $1 \leq r < L$: $\geq$ holds by dominance, and $B_r = P_r$ would split (ρ,ν) into the compatible pieces $(\rho_{\leq r}, \nu_{\leq r})$ and $(\rho_{>r}, \nu_{>r})$, both of positive size. Put $\mathbf{m}_{c_r c_{r+1}} = B_r - P_r \geq 1$. Position c_r then ends with $\rho_r + (B_r - P_r) - (B_{r-1} - P_{r-1}) = \nu_r \geq 0$. This uses $L - 1$ edges per block, $\ell - \kappa$ in all.

Finally, $c_{\lambda\mu} \in \mathbb{Q}[s,t]$ (Theorem 4.1); write $c_{\lambda\mu} = \sum_j (s-1)^j a_j(t)$. The lower bound gives $a_j = 0$ for $j < \ell - \kappa$, and $a_{\ell-\kappa}(0) = (-1)^{\ell-\kappa} N(\lambda,\mu) \neq 0$, so $a_{\ell-\kappa} \neq 0$. □

Remark 7.8 (Novelty). We have not found the block law stated anywhere. Its $t = 0$ case is classical in substance: the raising-operator formula follows from [DLT94, eq. (11)] and Jacobi–Trudi, after which the valuation count above is a short exercise. What is new is the lower bound for all t (from the subset formula alone), the exactness over $\mathbb{Q}(t)$, and the merge weights below.

7.5. **Coarsenings and merge weights.**

Theorem 7.9 (Coarsenings: merge-history expansion)**.** *Let $\mu \neq \lambda$ be a coarsening of λ and $m = \ell(\lambda) - \ell(\mu)$. Then $v(c_{\lambda\mu}) = m$, and $\mathrm{Lead}_{\lambda\mu}(t) := [(s-1)^m]c_{\lambda\mu}$ equals*

$$\mathrm{Lead}_{\lambda\mu}(t) = \sum_{\text{tight histories } H \text{ ending at } \mu} \prod_{\text{merges in } H} W_k(J), \qquad \mathrm{Lead}_{\lambda\mu}(0) = (-1)^m \#\{H\} \neq 0,$$

where a (tight) history processes $\lambda_\ell, \dots, \lambda_1$, each part k either opening a block of size k or merging a set of $p \geq 1$ existing (distinguishable) blocks of sizes $J = (j_1, \dots, j_p)$ into one block of size $k + |J|$ via the order-p piece $E_k^{(p)}$, it is tight if the orders add up to m, it ends at μ if the final block sizes are μ, and $W_k(J) := \mathrm{lin}_e\,[\cdots[E_k^{(p)}, e_{j_1}], \dots, e_{j_p}](1)$.

Proof. Since $\ell(\mu) = \ell - m$, the coefficient of e_μ in $[(s-1)^m]e_\lambda^\star$ can be read modulo $I^{\ell-m+1}$. In Proposition 7.4, a product over $|\pi| > \ell - m$ blocks of positive-degree factors lies in $I^{\ell-m+1}$; so only the terms where every step merges exactly p_j blocks survive, i.e. tight histories. At a tight step, δ_S is a derivation in each argument (Lemma 7.3), so $\delta_S(bc, \dots) \in I^2$ for $b, c \in I$, and modulo I^2, $\delta_S(g_1, \dots, g_p)$ depends only on the linear parts $g_i \equiv \gamma_i e_{n_i}$: it is $\prod_i \gamma_i\, W_k(n_1, \dots, n_p)\, e_n$. An opening step contributes e_k, with $\gamma = 1$. The final product of blocks is $\prod_C \gamma_C e_{|C|}$ modulo $I^{\ell-m+1}$, which gives the formula. By Theorem 7.10 below, $W_k(J)|_{t=0} = (-1)^{|J|}$, so every tight history contributes $(-1)^m$ at $t = 0$. Histories exist: in each block of the coarsening, let the last-processed part merge all the others, opened earlier as singletons. So $\mathrm{Lead}_{\lambda\mu}(0) \neq 0$, and $v(c_{\lambda\mu}) = m$ by the lower bound in Theorem 7.6. □

Let $T_k f := \sum_{|A|=k} c_A x_A f(x_A)$ for $f \in \Lambda_k$, where $f(x_A)$ is f evaluated at the k variables x_a, $a \in A$. (This is unrelated to the Hecke operators of Section 2.)

Theorem 7.10 (Merge weights)**.** *With $n = k + |J|$ and $p = |J|$,*

$$W_k(J) = (-1)^p \frac{[n]_t}{[k]_t} \prod_{j \in J} [k]_{t^j}.$$

Consequently $(-1)^m \mathrm{Lead}_{\lambda\mu}(t) > 0$ for coarsenings and every real $t > 0$.

Lemma 7.11. *For $f \in \Lambda_k$ homogeneous of degree d and $n = k + d$:*

$$\mathrm{lin}_e\, T_k(f) = (-1)^d \frac{[n]_t}{[k]_t} f(1, t, \dots, t^{k-1}).$$

Remark 7.12 (Novelty). Lemma 7.11 is a short corollary of [Mac95, III.7 Ex. 2, III (2.2)] and is probably known.

Lemma 7.13. *For $\ell(\rho) \leq k \leq m$, $T_k(P_\rho(x_1, \dots, x_k; t)) = P_{\rho+1^k}(x_1, \dots, x_m; t)$.*

Proof. By [Mac95, III (2.2)], $P_\lambda = \sum_{w \in S_m/S_m^\lambda} w\big(x^\lambda \prod_{i<j,\, \lambda_i > \lambda_j} \frac{x_i - t x_j}{x_i - x_j}\big)$, S_m^λ the stabiliser of λ. For $\lambda = \rho + 1^k$ the positive entries are the first k; a coset is an image set $A = w([k])$ together with a coset of S_k/S_k^ρ; the pairs with $\lambda_i > \lambda_j$ are those with $i \leq k < j$, giving c_A, and those with $i, j \leq k$, $\rho_i > \rho_j$; and $x^\lambda = x_{[k]} x^\rho_{[k]}$. Grouping by A, the inner sum is [Mac95, III (2.2)] for P_ρ in the variables x_A. □

Lemma 7.14. *For $\lambda \vdash n$ with $\ell(\lambda) = k$, $\operatorname{lin}_e P_\lambda(x;t) = (-1)^{n-k}\frac{[n]_t}{[k]_t} P_{\lambda-1^k}(1,t,\dots,t^{k-1};t)$.*

Proof. Since $[p_n]e_n = (-1)^{n-1}/n$ and e_ν with $\ell(\nu) \geq 2$ has no p_n term, $\operatorname{lin}_e G = (-1)^{n-1}\langle G, p_n\rangle = (-1)^{n-1}(1-t^n)\langle G, p_n\rangle_t$. By $\langle P_\lambda, Q_\mu\rangle_t = \delta_{\lambda\mu}$ and $Q_\lambda = b_\lambda P_\lambda$ [Mac95, III (4.9)], $\langle P_\lambda, p_n\rangle_t = [P_\lambda]p_n/b_\lambda(t)$, and $[P_\lambda]p_n = t^{n(\lambda)}\varphi_{k-1}(t^{-1}) = (-1)^{k-1}t^{n(\lambda)-\binom{k}{2}}\varphi_{k-1}(t)$ [Mac95, III.7 Ex. 2]. On the other side, with $\rho = \lambda - 1^k$, [Mac95, III.2 Ex. 1] gives $P_\rho(1,\dots,t^{k-1}) = t^{n(\rho)}\varphi_k/\prod_{i\geq 0}\varphi_{m_i(\rho)}$ with $m_0(\rho) = k - \ell(\rho)$; here $m_i(\rho) = m_{i+1}(\lambda)$, so the denominator is b_λ, and $n(\rho) = n(\lambda) - \binom{k}{2}$. Since $\frac{[n]}{[k]}\varphi_k = (1-t^n)\varphi_{k-1}$, the two sides agree. ☐

Proof of Lemma 7.11. The $P_\rho(x_1,\dots,x_k;t)$ with $\ell(\rho) \leq k$ form a basis of Λ_k [Mac95, III (2.7)]; for $f = P_\rho$, $\rho \vdash d$, combine Lemmas 7.13 and 7.14. ☐

Proof of Theorem 7.10. Let $J = (j_1,\dots,j_p)$, $d = |J|$. First, $W_k(J) = \operatorname{lin}_e E_k^{(p)}(e_{j_1}\cdots e_{j_p})$: in Lemma 7.3, $\delta_\varnothing = E_k^{(p)}(1) = 0$, and for $\varnothing \neq S \neq [p]$ the term $\prod_{i\notin S} e_{j_i}\,\delta_S$ lies in I^2. Next, with $E(u) = \sum_r e_r u^r$ and $R(u) := \prod_{a\in A}\frac{1+sux_a}{1+ux_a}$ read as a function of x_A,

$$E_k\Big(\prod_{i=1}^{p} E(u_i)\Big) = \prod_i E(u_i)\cdot T_k\Big(\prod_i R(u_i)\Big), \qquad \log R(u) = \sum_{r\geq 1}(-1)^{r-1}\frac{s^r-1}{r}u^r p_r(x_A).$$

Take $[u_1^{j_1}\cdots u_p^{j_p}]$ and $[(s-1)^p]$. Terms in which some e_{r_i}, $r_i \geq 1$, is taken from $\prod E(u_i)$ lie in I^2. In the remaining term, each $[u_i^{j_i}]R(u_i)$ has $(s-1)$-valuation ≥ 1, with $(s-1)^1$-coefficient $(-1)^{j_i-1}p_{j_i}(x_A)$. Hence $W_k(J) = (-1)^{d-p}\operatorname{lin}_e T_k(p_J)$, and Lemma 7.11 with $p_j(1,\dots,t^{k-1}) = [k]_{t^j}$ gives the formula. For $t > 0$ every $W_k(J)$ has sign $(-1)^{|J|}$, and the orders along a tight history add up to m. ☐

8. Leading coefficients: graph cumulants and blocks

Throughout this section μ is a coarsening of λ, $\ell = \ell(\lambda)$, $m = \ell - \ell(\mu)$, and the parts of λ are indexed by positions $[\ell]$. Everything rests on Theorems 7.9 and 7.10.

Lemma 8.1 (Histories are increasing forests)**.** *Tight histories of λ are in bijection with increasing forests on $[\ell]$ (every child larger than its parent): when position i merges existing blocks, i becomes the parent of their roots. The trees are the final blocks, and the orders automatically add up to $\ell - \#\text{trees}$.*

Proof. Every existing block at step i was built from positions $> i$, so all edges increase. Conversely, an increasing forest determines the history: at step i, merge the subtrees rooted at the children of i, which are complete by then. Each edge accounts for one unit of order. ☐

Lemma 8.2 (Telescoping)**.** *For an increasing tree T on a set C of positions, let N_v be the λ-mass of the subtree at v. The product of the merge weights along the history of T is*

$$\prod_v W_{\lambda_v}\big((N_c)_{c\ \text{child of}\ v}\big) = \frac{1-t^{|\lambda_C|}}{\prod_{i\in C}(1-t^{\lambda_i})}\prod_{\text{edges } v\to c}(t^{\lambda_v N_c}-1).$$

Proof. By Theorem 7.10, the merge at v contributes $(-1)^{\#\text{children}}\frac{1-t^{N_v}}{1-t^{\lambda_v}}\prod_c\frac{1-t^{\lambda_v N_c}}{1-t^{N_c}}$ (a leaf contributes 1). Every non-root c is a child exactly once, so its factor $1-t^{N_c}$ cancels against its own numerator, leaving the root mass $|\lambda_C|$; and $(-1)^{\#\text{edges}}\prod(1-x) = \prod(x-1)$. ☐

Lemma 8.3 (Tree–graph identity)**.** *For indeterminates $x_{ij} = x_{ji}$ $(i \neq j \in [\ell])$,*

$$\sum_{\text{increasing trees } T \text{ on } [\ell]}\ \prod_{\text{edges } v\to c}\Big(\prod_{j\in T_c} x_{vj} - 1\Big) = \sum_{\text{connected graphs } H \text{ on } [\ell]}\ \prod_{e\in H}(x_e - 1),$$

where T_c is the vertex set of the subtree at c.

Proof. Map a connected graph H to an increasing tree recursively: the root is $\min V$, the children are the minima of the components of $H - \min V$, and recurse on each component. Fix T and sum over the H mapping to it. At each vertex v and child c, the edges from v into T_c form an arbitrary nonempty set $S \subseteq T_c$ (nonempty since T_c must be attached to v; otherwise free, since they do not change the components inside T_c); there are no edges between different child sets; and the edges inside T_c are accounted for recursively. So the fibre sum factors, and $\sum_{\varnothing \neq S \subseteq T_c} \prod_{j \in S}(x_{vj} - 1) = \prod_{j \in T_c} x_{vj} - 1$. □

Theorem 8.4 (Full-merge lead)**.** *For $\lambda \vdash n$ with $\ell = \ell(\lambda) \geq 2$,*

$$\mathrm{Lead}_{\lambda,(n)}(t) = \frac{1 - t^n}{\prod_i (1 - t^{\lambda_i})} K_\lambda(t), \qquad K_\lambda(t) := \sum_{H \text{ connected on } [\ell]} \prod_{ij \in H} (t^{\lambda_i \lambda_j} - 1).$$

K_λ is the connected part (joint cumulant) of $t^{e_2(\lambda)} = \prod_{i<j}(1 + (t^{\lambda_i \lambda_j} - 1))$.

Proof. By Theorem 7.9 with $\mu = (n)$ and Lemma 8.1, $\mathrm{Lead}_{\lambda,(n)}$ is a sum over increasing trees on $[\ell]$; by Lemma 8.2 each tree contributes $\frac{1-t^n}{\prod_i (1-t^{\lambda_i})} \prod_{v \to c}(t^{\lambda_v N_c} - 1)$; and Lemma 8.3 with $x_{ij} = t^{\lambda_i \lambda_j}$, for which $\prod_{j \in T_c} x_{vj} = t^{\lambda_v N_c}$, converts the tree sum into K_λ. The cumulant statement is the exponential formula. □

Corollary 8.5. *(a) $\lambda = 1^n$: $\mathrm{Lead} = (-1)^{n-1}[n]_t\, I_n(t)$ with I_n the tree inversion enumerator [MR68]. (b) $t = 0$: $\mathrm{Lead} = (-1)^{\ell-1}(\ell - 1)!$. (c) $t \to 1$: $\mathrm{Lead} \to (-1)^{\ell-1} n^{\ell-1}$ (weighted Cayley). (d) $\operatorname{sgn} \mathrm{Lead} = (-1)^{\ell-1}$ for all $t > 0$.*

Proof. (a) For $\lambda = 1^n$ the prefactor is $[n]_t/(1-t)^{n-1}$ and $K_\lambda = \sum_{H \text{ conn.}} (t-1)^{|H|} = (t-1)^{n-1} I_n(t)$ [MR68]. (b) At $t = 0$ the prefactor is 1, and in the tree form every edge factor is -1; there are $(\ell - 1)!$ increasing trees. (c) Put $\varepsilon = t - 1$. In K_λ only spanning trees contribute to the lowest order $\varepsilon^{\ell-1}$, with $t^{\lambda_i \lambda_j} - 1 = \lambda_i \lambda_j \varepsilon + O(\varepsilon^2)$, and $\sum_T \prod_{ij \in T} \lambda_i \lambda_j = \sum_T \prod_i \lambda_i^{\deg_T(i)} = (\prod_i \lambda_i)\, n^{\ell-2}$ by the vertex-weighted Cayley formula (each i occurs $\deg_T(i) - 1$ times in the Prüfer code). The prefactor is $(-n\varepsilon)/((-\varepsilon)^\ell \prod_i \lambda_i) \cdot (1 + O(\varepsilon))$. (d) is the case $\mu = (n)$, $m = \ell - 1$, of the sign statement in Theorem 7.10: each merge contributes $(-1)^p$ times a positive quantity, $\sum p = m$ along every history, and histories exist. □

Remark 8.6 (Prior art — graph half). The tree, connected-graph and cumulant forms of K_λ are a special case of Dołęga [Doł19, Prop. 2.1 and (11), Lemma 2.3] ($K_\lambda = (t-1)^{\ell-1} T(1, t)$ for the Tutte polynomial of the multigraph with $\lambda_i \lambda_j$ edges between i and j), cf. also [GS96; MR68]; step (3) is a Penrose-type tree–graph identity [Pen67]. Ours: the identification of the ⋆-lead with prefactor·K_λ (Theorems 7.9, 7.10 and 8.4) and Theorems 8.7 and 8.8. Haglund–Tewari's single-row cumulant [HT26, Def. 7.1, Thm. 7.3] (their q is our t) is the same graph half: since $\langle \widetilde{H}_{(m)}, e_m \rangle = q^{\binom{m}{2}}$, their cumulant $\kappa(\lambda)$ (we write C_λ) satisfies $\langle C_\lambda, e_n \rangle = t^{n(\lambda')}(t-1)^{1-\ell} K_\lambda(t)$. *Separator.* Full-merge leads $L(\lambda)$ of any such theory are only defined up to the gauge $L(\lambda) \mapsto L(\lambda)\, a^{\ell(\lambda)} g(|\lambda|) \prod_i f(\lambda_i)$ (rescaling generators, degree-wise normalisation, rescaling the expansion parameter). The first gauge invariant appears at $n = 4$: $J := L(1^4)L(2,2)/L(2,1,1)^2$ (the exponents of a, $g(4)$, $f(1)$ and $f(2)$ all cancel). For ⋆, Theorem 8.4 gives $J = \frac{(t^2+1)(t^3+3t^2+6t+6)}{(t+1)(t^2+t+2)^2}$, and Haglund–Tewari's $\langle C_\lambda, e_n \rangle$ gives the same value. Dołęga's ⊕-cumulant of columns (lowest $(q-1)$-coefficient of $[g_n]\widetilde{H}_{\lambda'}$, $g_k = \widetilde{H}_{1^k}$) gives, by computation, $J \equiv 3/2 = J_\star(0)$, whereas $J_\star(1) = 1$, the Cayley value. So up to gauge Dołęga's ⊕ is t-independent and agrees with ⋆ only at $t = 0$, where the leads are partition-lattice Möbius values (Corollary 8.5(b)), while Haglund–Tewari's is the same graph object without ⋆.

Theorem 8.7 (Coarsening leads factor)**.** *For μ a coarsening of λ,*

$$\mathrm{Lead}_{\lambda\mu} = \sum_{\pi} \prod_{B\in\pi} \mathrm{Lead}_{\lambda_B,(|\lambda_B|)},$$

summed over set partitions π of the positions $[\ell]$ whose sorted block sums equal μ (singletons contribute 1). There is no interleaving factor.

Proof. By Lemma 8.1, tight histories ending at μ are increasing forests whose trees have sorted masses μ, i.e. a set partition π with block sums μ together with an increasing tree on each block. By Lemma 8.2 the weight is a product over the trees, each depending only on its own parts; summing over the trees on each block gives $\mathrm{Lead}_{\lambda_B,(|\lambda_B|)}$ by Theorem 8.4 (and 1 for a singleton). □

Theorem 8.8 (Block multiplicativity)**.** *Let $\kappa = \kappa(\lambda,\mu) \geq 1$, $\ell = \ell(\lambda)$, $m = \ell - \kappa$. Then*

$$[(s-1)^m]\, c_{\lambda\mu} = \sum_{\pi} \sum_{(\nu^C)_{C\in\pi}} \prod_{C\in\pi} [(s-1)^{|C|-1}]\, c_{\lambda_C,\nu^C},$$

where π runs over set partitions of $[\ell]$ into exactly κ blocks and (ν^C) over tuples with $\bigsqcup_C \nu^C = \mu$ and $\nu^C \trianglerighteq \lambda_C$; every factor then has $\kappa(\lambda_C,\nu^C) = 1$. With Theorem 7.6 this is $\mathrm{Lead}_{\lambda\mu}$.

Proof. Expand $[(s-1)^m]e^\star_\lambda = \sum_{\sum p_i = m} E^{(p_1)}_{\lambda_1} \cdots E^{(p_\ell)}_{\lambda_\ell}(1)$ step by step with Lemma 7.3, as in Proposition 7.4: this is a sum over histories H (step i chooses a set S_i of current blocks with $|S_i| \leq p_i$), each giving $\prod_{C\in\pi(H)} g^H_C$ with g^H_C of type λ_C, and $|\pi(H)| = \ell - \sum |S_i| \geq \ell - m = \kappa$, with equality iff $|S_i| = p_i$ for all i (tight). A non-tight term has more than κ blocks, so its e_μ-coefficient vanishes by maximality of κ. In a tight history, step i only acts, through δ_{S_i}, on the blocks it merges; so g^H_C is the function produced by the restricted history on C, which is a single-block tight history of the composition λ_C (positions of C in their order), since $\sum_{i\in C} |S_i| = |C|-1$. Conversely, single-block tight histories on the blocks of π interleave uniquely, by position, into a global tight history. Hence the tight part is $\sum_{|\pi|=\kappa} \prod_{C\in\pi} \mathrm{Conn}(\lambda_C)$, where $\mathrm{Conn}(\lambda_C)$ is the sum over single-block tight histories of λ_C. Applying the same expansion to λ_C with $m = |C| - 1$ shows $[e_\nu]\mathrm{Conn}(\lambda_C) = [e_\nu][(s-1)^{|C|-1}]e^\star_{\lambda_C}$ whenever $\kappa(\lambda_C,\nu) = 1$. Finally, $[e_\mu]\prod_C \mathrm{Conn}(\lambda_C) = \sum_{(\nu^C)} \prod_C [e_{\nu^C}]\mathrm{Conn}(\lambda_C)$ over tuples with $\bigsqcup_C \nu^C = \mu$; factors with $\nu^C \not\trianglerighteq \lambda_C$ vanish by type, and $\kappa(\lambda_C,\nu^C) \geq 2$ would refine π to more than κ compatible blocks. Commutativity of $\star$ [Hik25] identifies the coefficients for the composition λ_C with those for the sorted partition. □

Remark 8.9. Theorem 8.7 is the case where each ν^C is a single part. The factorisation has the shape of a Mayer (cluster) expansion, i.e. moment–cumulant inversion of the graph identity in Theorem 8.4; the $\star$-specific content is that the connected leads are those of Theorem 8.4, with no interleaving correction.

9. Box Complement and the linear coefficient

In this section zero parts are allowed: λ is padded with zeros to a prescribed length, $E_0 = \mathrm{id}$, and $c_{\lambda\mu}$ refers to the nonzero parts. By Lemma 2.8, in N variables (for every N, including N below the degree) $e^\star_\lambda(x_1,\dots,x_N) = \sum_\mu c_{\lambda\mu} e_\mu(x_1,\dots,x_N)$, where $e_\mu = 0$ if $\mu_1 > N$.

Theorem 9.1 (Box Complement)**.** *Let λ,μ have at most ℓ parts (padded with zeros) and all parts $\leq N$; put $N^\ell - \lambda := (N-\lambda_\ell,\dots,N-\lambda_1)$. Then*

$$c_{N^\ell-\lambda,\,N^\ell-\mu}(s,t) = s^{N\binom{\ell}{2}-(\ell-1)|\lambda|}\, c_{\lambda\mu}(s,t).$$

Lemma 9.2 (Inversion)**.** *In N variables, let G be homogeneous of degree g and $F(x) := e_N(x)^m\, G(1/x)$. Then*

$$E_{N-k}F = s^{m(N-k)-g}\, e_N^{m+1}\, (E_kG)(1/x).$$

Proof. Write $y = 1/x$ and index the terms of E_{N-k} by the complement A ($|A| = k$) of the $(N-k)$-subset B. Then $c_B(x) = \prod_{i\in B, j\in A} \frac{y_j - t y_i}{y_j - y_i} = c_A(y)$; $x_B = e_N(x)\, y_A$; $F|_{x_i \mapsto s x_i\,(i\in B)} = s^{m(N-k)} e_N(x)^m\, G|_{y_i\mapsto s^{-1}y_i\,(i\in B)}$; and by homogeneity $G|_{y_i\mapsto s^{-1}y_i\,(i\in B)} = s^{-g}\, G|_{y_a\mapsto s y_a\,(a\in A)}$. Summing over A gives $s^{m(N-k)-g} e_N^{m+1} \sum_A c_A(y) y_A G|_{y_a\mapsto s y_a}$. ☐

Proof of Theorem 9.1. Let $G_j := E_{\lambda_{\ell-j+1}} \cdots E_{\lambda_\ell}(1)$, of degree $D_j = \lambda_\ell + \cdots + \lambda_{\ell-j+1}$, and $F_j := E_{N-\lambda_{\ell-j+1}} \cdots E_{N-\lambda_\ell}(1)$, in N variables. By Lemma 9.2 and induction, $F_j = s^{\sigma_j} e_N^j G_j(1/x)$ with $\sigma_0 = 0$ and $\sigma_{j+1} = \sigma_j + j(N - \lambda_{\ell-j}) - D_j$. Writing $p_i = \lambda_{\ell+1-i}$,

$$\sigma_\ell = N\binom{\ell}{2} - \sum_i (i-1)p_i - \sum_i (\ell - i)p_i = N\binom{\ell}{2} - (\ell-1)|\lambda|.$$

By commutativity of $\star$, $F_\ell = e^\star_{N^\ell - \lambda}$. Since $e_{N-r}(x) = e_N(x) e_r(1/x)$ in N variables, $e_N^\ell e_\mu(1/x) = e_{N^\ell - \mu}(x)$ for $\ell(\mu) \le \ell$, and every μ in the support of $e^\star_\lambda$ has $\ell(\mu) \le \ell(\lambda) \le \ell$ (Theorem 4.1). Hence $e^\star_{N^\ell-\lambda} = s^{\sigma_\ell} \sum_{\mu_1 \le N} c_{\lambda\mu}\, e_{N^\ell - \mu}(x)$. The map $\mu \mapsto N^\ell - \mu$ is a bijection between partitions with at most ℓ parts, all $\le N$; every ν in the support of $e^\star_{N^\ell-\lambda}$ has at most ℓ parts; and the e_ν with $\nu_1 \le N$ form a basis of Λ_N. Compare coefficients. ☐

Corollary 9.3 (Column Lemma; lead invariance)**.** $c_{\lambda + 1^\ell, \mu + 1^\ell} = s^{\binom{\ell}{2}} c_{\lambda\mu}$. *Hence v and* Lead *are invariant under $(\lambda, \mu) \mapsto (\lambda + 1^\ell, \mu + 1^\ell)$ and $(\lambda, \mu) \mapsto (N^\ell - \lambda, N^\ell - \mu)$; e.g. $(2,2,2) \to (3,3)$ and $(2,2,2) \to (4,1,1)$ have the lead $[3]_t(t+2)$ of $(1,1,1) \to (3)$.*

Proof. Complement with N and then with $N+1$: $(N+1)^\ell - (N^\ell - \lambda) = \lambda + 1^\ell$, and the exponents add up to $\big(N\binom{\ell}{2} - (\ell-1)|\lambda|\big) + \big((N+1)\binom{\ell}{2} - (\ell-1)(N\ell - |\lambda|)\big) = \binom{\ell}{2}$. Since s^σ is a unit of $\mathbb{Q}[[s-1]]$ with constant term 1, valuations and leading coefficients are preserved. For the example, $(2,2,2) \to (4,1,1)$ is $(1,1,1) \to (3,0,0)$ plus a column, and $(2,2,2) \to (3,3)$ is the complement with $N = 3$ of $(1,1,1) \to (3,0,0)$; the lead of $(1,1,1) \to (3)$ is $[3]_t(t+2)$ by Theorem 8.4. ☐

Remark 9.4 (Prior art). The inversion lemma follows in a few lines from [DFK19, Rem. 3.3] ($S\Gamma_I S = \Gamma_I^{-1}$ under $x \mapsto x^{-1}$) plus subset complement; stability is [Hik25, Prop. 3.8, Cor. 3.10]. We have not found Theorem 9.1 or Corollary 9.3 for $\star$ in the literature.

Theorem 9.5 (Plethystic linear coefficient)**.** *For G homogeneous of degree d: $\mathrm{lin}_e(e_k \star G) = (-1)^d \frac{[k+d]_t}{[k]_t}\, G[(s-1)[k]_t]$, where $G \mapsto G[(s-1)[k]_t]$ is the ring map $p_r \mapsto (s^r - 1)[k]_{t^r}$.*

Proof. By linearity take $G = e_{j_1} \cdots e_{j_p}$, $d = \sum j_i$. As in the proof of Theorem 7.10, $E_k(e_J) = \sum_{w \le J} e_{J-w}\, T_k\big([u^w] \prod_i R(u_i)\big)$. For $w \ne J$ the term is a product of an element of I and a symmetric function of positive degree, hence lies in I^2. For $w = J$, Lemma 7.11 applied to the symmetric function $f = \prod_i [u_i^{j_i}] R(u_i)$ of x_A gives $(-1)^d \frac{[k+d]}{[k]} f(1, t, \ldots, t^{k-1})$, and $[u^j]R(u)$ at $x_A = (1, t, \ldots, t^{k-1})$ is $\pi_k(j) := [u^j] \prod_{m=0}^{k-1} \frac{1+sut^m}{1+ut^m}$. Taking logarithms, $e_j \mapsto \pi_k(j)$ is the ring map $p_r \mapsto (s^r - 1)[k]_{t^r}$. ☐

Corollary 9.6 (Pieri rule, all s)**.** $e_k \star e_r = \sum_{y=0}^{\min(k,r)} s^y \mathcal{C}_{k-y, r-y}\, e_{k+r-y} e_y$ *with* $\mathcal{C}_{a,0} = \mathcal{C}_{0,b} = 1$, $\mathcal{C}_{a,b} = (-1)^b \frac{[a+b]}{[a]} \pi_a(b)$, $\pi_a(j) = [u^j] \prod_{m=0}^{a-1} \frac{1+sut^m}{1+ut^m}$.

Proof. By Theorem 4.1 and commutativity, the support of $e_k \star e_r$ consists of $e_{k+r-y}e_y$ with $y \le \min(k,r)$. Applying Corollary 9.3 with $\ell = 2$ y times, $c_{(k,r),(k+r-y,y)} = s^y\, c_{(k-y,r-y),(k+r-2y,0)} = s^y \lim_e(e_{k-y} \star e_{r-y})$. If $k - y = 0$ this is s^y; otherwise Theorem 9.5 with $G = e_{r-y}$ gives $s^y(-1)^{r-y}\frac{[k+r-2y]}{[k-y]}\pi_{k-y}(r-y) = s^y\mathcal{C}_{k-y,r-y}$. □

Remark 9.7. This proof of the Pieri rule is independent of the one in Section 3: it uses neither the parabolic kernel computation of Proposition 2.2 beyond the subset formula, nor the recursion of Theorem 3.2. For $r \ge k$ the monomials $e_b e_{r+k-b}$, $0 \le b \le k$, are distinct, so comparing with Corollary 3.3 gives $F_n(t^m) = (-1)^m\frac{[n+m]}{[n]}\pi_n(m)$ for $1 \le n \le m$. Hikita's [Hik25, Thm. 3.12] is the case $k = 1$.

10. Second order: every valuation-2 lead

By Theorem 8.8, valuation-2 leads reduce to connected leads with $\ell(\lambda) = 3$, $\kappa = 1$; by Corollary 9.3 to two-part μ. With T_a as in Lemma 7.11, put

$$\Gamma_a(e_b,e_c) := E_a^{(2)}(e_be_c) - e_bE_a^{(2)}(e_c) - e_cE_a^{(2)}(e_b) = \sum_{1\le r\le b,\ 1\le q\le c} (-1)^{r+q} e_{b-r}e_{c-q}\,T_a(p_rp_q).$$

Indeed, with $R(u)$ as in the proof of Theorem 7.10, $R(u) = \prod_{x\in A}(1+(s-1)y_x(u))$ with $y_x(u) = \frac{ux_x}{1+ux_x}$, so $[(s-1)^2]R(u)R(v) = e_2(Y_u)+e_2(Y_v)+p_1(Y_u)p_1(Y_v)$ in the alphabets $Y_u = \{y_x(u)\}_{x\in A}$, Y_v; hence $E_a^{(2)}(E(u)E(v)) - E(u)E_a^{(2)}E(v) - E(v)E_a^{(2)}E(u) = E(u)E(v)\,T_a(p_1(Y_u)p_1(Y_v))$, and $p_1(Y_u) = \sum_{r\ge1}(-1)^{r-1}u^r p_r(x_A)$. Take $[u^bv^c]$.

Proposition 10.1. *If (a,b,c) is any ordering of λ and $\kappa(\lambda,\mu) = 1$, then $[(s-1)^2]c_{\lambda\mu} = [e_\mu](\Gamma_a(e_b,e_c) + D_aD_b(e_c))$.*

Proof. As $\star$ is commutative and associative (Section 2), $e^\star_\lambda = E_aE_bE_c(1)$ for every ordering (a,b,c) of λ; fix one. Since $E^{(p)}(1) = 0$ for $p \ge 1$, $[(s-1)^2]E_aE_bE_c(1) = E_a^{(2)}(e_be_c) + D_aD_b(e_c) + e_aE_b^{(2)}(e_c)$, and $E_a^{(2)}(e_be_c) = \Gamma_a(e_b,e_c)+e_bE_a^{(2)}(e_c)+e_cE_a^{(2)}(e_b)$. So the surviving expansion terms are $E_a^{(2)}(e_be_c)$, $e_aE_b^{(2)}(e_c)$ and $D_aD_b(e_c)$, while the three terms to discard, $e_aE_b^{(2)}(e_c)$, $e_bE_a^{(2)}(e_c)$, $e_cE_a^{(2)}(e_b)$, are not these three: they are one term discarded from the expansion plus the two introduced by the Γ_a packaging (the two triples share one element), and this list is complete. Each of them has support in $\kappa(\lambda,\mu) \ge 2$. Indeed, $E_c(1) = e_c$, so $E_a(e_c) = E_aE_c(1) = e^\star_{(a,c)}$ (by commutativity) and $E_a^{(2)}(e_c) = [(s-1)^2]\,e^\star_{(a,c)}$. By Theorem 4.1, $e^\star_{(a,c)}$ is supported on $\{e_\nu : \nu \trianglerighteq (a,c)\}$, hence so is $E_a^{(2)}(e_c)$, and $e_bE_a^{(2)}(e_c)$ is supported on $\{e_{\nu\sqcup(b)}\}$. For such $\mu = \nu \sqcup (b)$, the blocks $\lambda^1 = \mu^1 = \{b\}$, $\lambda^2 = \{a,c\}$, $\mu^2 = \nu$ form a two-block decomposition, so $\kappa(\lambda,\mu) \ge 2$. The other two terms are identical with the letters permuted. □

Order-independence in Proposition 10.1 is a conclusion, not a hypothesis: the left side does not depend on the ordering, and $\Gamma_a(e_b,e_c)$ is symmetric in b,c, but $D_aD_b(e_c)$ is not manifestly so. Comparing the orderings (a,b,c) and (a,c,b) gives $[e_\mu]D_aD_b(e_c) = [e_\mu]D_aD_c(e_b)$ whenever $\kappa(\lambda,\mu) = 1$.

Theorem 10.2 (Constant-term adjoint formula)**.** *For $g \in \Lambda_a$ and $F \in \Lambda$,*

$$\varphi_a(t)\,\langle T_a g, F\rangle_t = \mathrm{CT}\big[Z\,g(z)\,F(z_1^{-1},\dots,z_a^{-1})\,\Omega(z)\big],$$

where $Z = z_1\cdots z_a$ and $\Omega(z) = \prod_{1\le i<j\le a}\frac{z_j-z_i}{z_j-tz_i}$, expanded in z_i/z_j $(i<j)$.

Proof. Let $G = T_a g$. In exactly a variables only $A = [a]$ survives and $c_{[a]} = 1$, so $G(z) = Zg(z)$; and $G \in \operatorname{span}\{P_\mu : \ell(\mu) = a\}$ by Lemma 7.13. We read CT of a Laurent polynomial times Ω either formally or as the integral over the torus $|z_i| = 1$ for $|t| < 1$; the two agree because the expansion of Ω converges there, and both sides of the theorem are rational in t.

Step 1. For $\alpha \in \mathbb{Z}^a_{\geq 0}$, $\langle G, q_\alpha\rangle_t = [y^\alpha]G(y_1, \dots, y_a)$, with $q_\alpha = \prod q_{\alpha_i}$: pair G with the Cauchy identity $\sum_\mu Q_\mu(x)P_\mu(y) = \prod_{i,j} \frac{1-tx_iy_j}{1-x_iy_j} = \sum_\alpha q_\alpha(x)y^\alpha$ [Mac95, III (4.2), (4.4)], using $\langle P_\lambda, Q_\mu\rangle_t = \delta_{\lambda\mu}$ [Mac95, III (4.9)].

Step 2. For $\ell(\lambda) \leq a$ (padded), the raising-operator formula [Mac95, III (2.15′)] gives $Q_\lambda = \sum_\beta c_\beta q_{\lambda+\beta}$ with $\sum_\beta c_\beta y^{-\beta} = \prod_{i<j} \frac{1-y_j/y_i}{1-ty_j/y_i}$; operators involving positions beyond $\ell(\lambda)$ only produce negative entries, where q and $[y^{\cdot}]$ both vanish. With step 1 and the relabelling $z_k = y_{a+1-k}$,

$$\langle G, Q_\lambda\rangle_t = \mathrm{CT}\big[G(z)\, z^{-\lambda^{\mathrm{rev}}}\, \Omega(z)\big], \qquad \lambda^{\mathrm{rev}} = (\lambda_a, \dots, \lambda_1).$$

Step 3. Let $\Delta = \prod_{i\neq j} \frac{1-z_i/z_j}{1-tz_i/z_j}$ and $\theta = \prod_{i<j} \frac{z_i - tz_j}{z_i - z_j}$, so that $\Delta\theta = \Omega$. For symmetric f, $f\Omega$ is analytic on the torus and $f\Delta$ is symmetric, so averaging over S_a and using $\sum_w w(\theta) = v_a(t) = [a]_t!$ [Mac95, III (1.4)] gives $\mathrm{CT}[f\Omega] = \frac{v_a}{a!}\mathrm{CT}[f\Delta]$.

Step 4. By [Mac95, III (2.1)–(2.2)], $Q_\lambda(z^{-1}) = \frac{b_\lambda}{v_\lambda}\sum_w w(z^{-\lambda}\tilde\theta)$ with $\tilde\theta = \prod_{i<j} \frac{z_j - tz_i}{z_j - z_i}$ and $v_\lambda = \prod_{i\geq 0} v_{m_i(\lambda)}$, $m_0 = a - \ell(\lambda)$. Since $\tilde\theta\Delta = \prod_{i<j} \frac{z_i - z_j}{z_i - tz_j}$, step 3, symmetry, and the relabelling $z_k \leftrightarrow z_{a+1-k}$ give

$$\mathrm{CT}\big[G\, Q_\lambda(z^{-1})\, \Omega\big] = v_a \frac{b_\lambda}{v_\lambda}\, \mathrm{CT}\big[G\, z^{-\lambda^{\mathrm{rev}}}\Omega\big] = v_a \frac{b_\lambda}{v_\lambda}\langle G, Q_\lambda\rangle_t.$$

Step 5. Write $F = \sum_\lambda F_\lambda Q_\lambda$. Terms with $\ell(\lambda) > a$ vanish on both sides, and $\langle G, Q_\lambda\rangle_t = 0$ unless $\ell(\lambda) = a$, in which case $m_0 = 0$ and $v_a\, b_\lambda/v_\lambda = v_a(1-t)^a = \varphi_a$. Hence $\mathrm{CT}[GF(z^{-1})\Omega] = \varphi_a\langle G, F\rangle_t$. □

Theorem 10.3 (Two-point formula)**.** *Let $g \in \Lambda_a$ be homogeneous of degree d, $n = a+d$, $x, y \geq 1$, $x + y = n$, and $\Phi_a(g; x, y) := \langle T_a g, p_x p_y\rangle$ (Hall pairing, not the $\langle\cdot,\cdot\rangle_t$ of Theorem 10.2). For $1 \leq A \leq a-1$ put $B = a - A$ and $G_A(w) = \sum_k G_{A,k} w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$. Then*

$$\begin{aligned}\Phi_a(g; x, y) = (-1)^a(1-t^x)(1-t^y)\Bigg\{\sum_{A=1}^{a-1} t^{-AB}\Big[&\frac{G_{A,y-B}}{(1-t^A)(1-t^B)} \\ &+ \frac{1}{1-t^a}\sum_{m\geq 1}(t^{-Am} - t^{Bm})\, G_{A,y-B-m}\Big] \\ &- \frac{g(1, t, \dots, t^{a-1})}{1-t^a}\sum_{j=0}^{a-1} t^{-jy}\Bigg\}.\end{aligned}$$

Proof. Let $S(u) := \sum_{i=1}^a \frac{1}{uz_i - 1} = \sum_{x\geq 1} u^{-x} p_x(z^{-1})$. By Theorem 10.2 and homogeneity,

$$\mathcal{R}(u, v) := \frac{1}{\varphi_a}\mathrm{CT}\big[g\, Z\, S(u)S(v)\, \Omega\big] = \sum_{x+y=n} \frac{\Phi_a(g; x, y)}{(1-t^x)(1-t^y)} u^{-x}v^{-y} = u^{-n}\rho(w), \qquad w = u/v,$$

with ρ a polynomial, so $\Phi_a(g; x, y) = (1-t^x)(1-t^y)[w^y]\rho$.

String expansion. Take $|t|$ small, u, v generic, and radii $\max(|u|^{-1}, |v|^{-1}) < r_1 < \cdots < r_a$; the constant term is the integral of $g\, S(u)S(v)\,\Omega \prod_m dz_m/(2\pi i)$ over $|z_m| = r_m$. Expand $S(u)S(v)$ into its a^2 summands and integrate $z_1, \dots, z_a$ in turn. When z_m is integrated, its poles inside $|z_m| = r_m$ are at $1/u$ or $1/v$ if z_m carries the u- or v-factor, and at tz_l for $l < m$ (from the factor

$\frac{z_m - z_l}{z_m - t z_l}$); the poles z_j/t, $j > m$, lie outside, and g is a polynomial. The numerator $\prod_{l<m}(z_m - z_l)$ kills the residue at tz_l if some earlier $z_{l'}$ already equals tz_l. Hence the evaluation points form *t-strings* $q, qt, \dots, qt^{A-1}$ (at increasing positions), each started by a *seed* at $1/u$ or $1/v$, and position 1 is always a seed. The surviving configurations are: (i) a word $\omega \in \{U, V\}^a$ containing both letters, the u-factor at the first U and the v-factor at the first V, the $(\alpha+1)$-st U at t^α/u and the $(\beta+1)$-st V at t^β/v; (ii) the word U^a, all points t^{m-1}/u, with the v-factor at any position $j+1$, where it is simply evaluated; (iii) the mirror image V^a.

String weight. A string of length A with base q contributes $(-1)^{A-1}\varphi_{A-1}(t)\, q^A$: the seed residue is q; a successor pair contributes the residue $qt^\alpha(t-1)$; a non-successor pair at string distance $\delta \geq 2$ contributes $\frac{t^\delta - 1}{t^\delta - t}$; and $\prod_{\delta=2}^{A-1}\big(\frac{1-t^\delta}{t(1-t^{\delta-1})}\big)^{A-\delta} = t^{-\binom{A-1}{2}}\varphi_{A-1}/(1-t)^{A-1}$.

Interaction. For a U-string of length A and a V-string of length $B = a - A$, the cross factors are $\frac{wt^{\beta-\alpha}-1}{wt^{\beta-\alpha}-t}$ (U_α before V_β) and $\frac{1-wt^{\beta-\alpha}}{1-wt^{\beta-\alpha+1}}$ (V_β before U_α), and their sum over shuffles is the function $\mathrm{Sh}_{A,B}(w)$ of Lemma 10.4.

Assembling. Since $g(z^*) = u^{-d}G_A(w)$ for configuration (i) and $u^{-d}g(\pi_a)$ for (ii), where $\pi_a = (1, t, \dots, t^{a-1})$, and $v^{-B} = u^{-B}w^B$,

$$\varphi_a \rho(w) = \sum_{A=1}^{a-1} (-1)^a \varphi_{A-1}\varphi_{B-1}\, w^B\, \mathrm{Sh}_{A,B}(w)\, G_A(w) + (-1)^{a-1}\varphi_{a-1}\, g(\pi_a) \sum_{j=0}^{a-1} \frac{w}{t^j - w} + O(w^n),$$

the last term coming from (iii). This is an identity of rational functions of w, each term analytic at $w = 0$, so $[w^y]$ may be taken termwise. By Lemma 10.4, $\mathrm{Sh}_{A,B}(w) = \begin{bmatrix} a \\ A \end{bmatrix}_t t^{-AB}(1 + \kappa_{AB}\sum_{m\geq 1}(t^{-Am} - t^{Bm})w^m)$ with $\kappa_{AB} = \frac{(1-t^A)(1-t^B)}{1-t^a}$, and $\begin{bmatrix} a \\ A \end{bmatrix}_t \varphi_{A-1}\varphi_{B-1} = \frac{\varphi_a}{(1-t^A)(1-t^B)}$, $\frac{\varphi_{a-1}}{\varphi_a} = \frac{1}{1-t^a}$. For $1 \leq y \leq n-1$ this gives the formula. □

Lemma 10.4 (Shuffle identity)**.**

$$\mathrm{Sh}_{A,B}(w) = \begin{bmatrix} A+B \\ A \end{bmatrix}_t t^{-AB}\, \frac{(1-w)(1-t^{B-A}w)}{(1-t^{-A}w)(1-t^B w)},$$

$$\frac{(1-w)(1-t^{B-A}w)}{(1-t^{-A}w)(1-t^B w)} = 1 + \kappa_{AB}\sum_{m\geq 1}(t^{-Am} - t^{Bm})w^m.$$

Proof. Induction on $A + B$; for $A = 0$ or $B = 0$ both sides are 1. Condition on the last letter. If it is U_{A-1}, its cross factors telescope to $\frac{1-wt^{1-A}}{1-wt^{B+1-A}}$; if it is V_{B-1}, to $t^{-A}\frac{1-wt^{B-1}}{1-wt^{B-1-A}}$. So $\mathrm{Sh}_{A,B} = \frac{1-wt^{1-A}}{1-wt^{B+1-A}}\mathrm{Sh}_{A-1,B} + t^{-A}\frac{1-wt^{B-1}}{1-wt^{B-1-A}}\mathrm{Sh}_{A,B-1}$. With $X = t^A$, $Y = t^B$, $\begin{bmatrix} A+B-1 \\ A-1 \end{bmatrix}_t / \begin{bmatrix} A+B \\ A \end{bmatrix}_t = \frac{1-X}{1-XY}$ and $\begin{bmatrix} A+B-1 \\ A \end{bmatrix}_t / \begin{bmatrix} A+B \\ A \end{bmatrix}_t = \frac{1-Y}{1-XY}$, the claimed closed form satisfies this recursion if and only if $Y(1-X)(1-w/X) + (1-Y)(1-Yw) = (1-XY)(1-wY/X)$, and both sides equal $1 - XY - wY/X + wY^2$. For the expansion: the left side is 1 at $w = 0$, tends to 1 at ∞, and has simple poles with residue $-t^A\kappa_{AB}$ at $w = t^A$ and $t^{-B}\kappa_{AB}$ at $w = t^{-B}$; so does the right side. □

Remark 10.5 (Relation to Green polynomials). Write $p_\mu = \sum_\lambda \mathcal{X}^\lambda_\mu(t) P_\lambda(x;t)$. For $g = P_\rho$ and $\lambda = \rho + 1^a$ one has $\Phi_a(P_\rho; x, y) = (1-t^x)(1-t^y)\mathcal{X}^\lambda_{(x,y)}(t)/b_\lambda(t)$, so Theorem 10.3 evaluates Green polynomials at two-part classes; the two t-strings are the Frobenius orbits of a maximal torus of type (x, y) [Gre55]. Theorem 10.2 is, up to the normalising factors $b_\lambda(t)$ and $\prod_i(1-t^{\mu_i})$, Jing's constant-term expression for the matrix element $\langle Q_\lambda, p_\mu\rangle$ [Jin91], cf. [JL21, (2.17), (2.20)]. Jing and Liu [JL21, Thm. 2.7] expand the same matrix element as a nested sum valid for all μ, extracting one variable at a time; at two-part μ that sum does not close up, since its inner sums run over classes of arbitrary length. Their closed forms [JL21, Thm. 2.10, (2.37)–(2.40)]

restrict the Hall–Littlewood index λ, not the class μ; their Murnaghan–Nakayama rule [JL21, Thm. 3.2] generalises a formula of Morris [Mor77][1], its case $\ell(\lambda) = 2$. We evaluate by residues instead, and at $\ell(\mu) = 2$ the poles organise into two t-strings. The formula is explicit, with no sum over classes, and linear in the two-string specialisation G_A of g; for general ρ, G_A is itself a finite (branching) sum, so Theorem 10.3 is not a product formula.

Example 10.6 (Two rows)**.** For $a = 2$, only $A = B = 1$ occurs, and $G_1(w) = P_\rho(1, w; t) = w^{\rho_2}\frac{(1-tw)-w^r(w-t)}{1-w}$ with $r = \rho_1 - \rho_2 \geq 1$ ($G_1 = w^{\rho_2}$ if $r = 0$). Substituting into Theorem 10.3 gives, for $\lambda = (\lambda_1, \lambda_2)$, $y \leq x$, and $m_{xy} = 2$ if $x = y$, 1 otherwise,

$$\mathcal{X}^{(\lambda_1,\lambda_2)}_{(x,y)}(t) = \begin{cases} (t-1)\,t^{\lambda_2-1-y}(1+t^y), & y < \lambda_2, \\ m_{xy} - (1-t)\,t^{\lambda_2-1}, & y = \lambda_2, \\ (t-1)\,t^{\lambda_2-1}, & y > \lambda_2, \end{cases}$$

in agreement with a Kostka–Foulkes computation for $|\lambda| \leq 10$. This is not new: for $\lambda = (n-k, k)$ and every class ρ, Jing and Liu [JL21, Thm. 2.6, unnumbered display after (2.32), same in v1 and v2] give $\mathcal{X}^\lambda_\rho = \pi_k + (t-1)\sum_{j<k} t^{k-1-j}\pi_j$, with π_j the number of j-subsets fixed by ρ (their $D^{(j)}(\rho)$). Their display is written for $\lambda = (m, n)$: their m is our $\lambda_1 = n - k$ and their n is our $\lambda_2 = k$; swapping the two gives a wrong formula. A short proof: $\mathcal{X}^\lambda_\rho = \sum_\nu \chi^\nu_\rho K_{\nu\lambda}(t)$ [Mac95, III (7.6′)], where $K_{(n-j,j),\lambda} = t^{k-j}$ (one tableau, and K is monic of degree $n(\lambda) - n(\nu)$ [Mac95, III (6.5)]) and $\chi^{(n-j,j)}_\rho = \pi_j - \pi_{j-1}$ by Young's rule. On the diagonal class $(x, y) = \lambda$ it gives $t^{\lambda_2} - t^{\lambda_2-1} + m_{xy}$, which for $\lambda_1 > \lambda_2$ with $\lambda_2 \geq 3$ odd equals -1 at $t = -1$ and 1 at $t = 0$. A root in $(-1, 0)$ rules out any product or quotient of cyclotomic factors and powers of t, so Green polynomials at two-part classes admit no product formula even for two rows; for expressions $t^c\prod_i(1-t^{d_i})^{e_i}$ with $e_i \in \mathbb{Z}$ (products and quotients) this was observed independently in [Cli26, Thm. D].

Theorem 10.7 (Closed $\ell = 3$, $\kappa = 1$ leads)**.** *Let $\lambda = (a, b, c)$ be any ordering of a length-3 partition, $\mu = (x, y)$ with $\kappa(\lambda, \mu) = 1$, $n = x + y$, $m_{xy} = 2$ if $x = y$ and 1 otherwise. Put*

$$\Xi_a(r, q) := \operatorname{lin}_e T_a(p_r p_q) = (-1)^{r+q}\frac{[a+r+q]}{[a]}[a]_{t^r}[a]_{t^q}$$

and $U_a(b, c; x, y) := \big((-1)^n\Phi_a(p_b p_c; x, y) - \Xi_a(b, c)\big)/m_{xy}$. Then

$$[(s-1)^2]\,c_{\lambda\mu} = (-1)^{b+c}U_a(b, c; x, y) + \sum_{\substack{1\leq r<b \\ \{b-r,\,a+r+c\}=\{x,y\}}} (-1)^{r+c}\Xi_a(r, c) + \sum_{\substack{1\leq q<c \\ \{c-q,\,a+b+q\}=\{x,y\}}} (-1)^{b+q}\Xi_a(b, q) + [e_x e_y]\,D_a(M_{bc}),$$

with D_a the derivation $D_a(e_j) = M_{aj}$ of Proposition 7.2.

Lemma 10.8. *For G homogeneous of degree $n = x+y$, $\langle G, p_x p_y\rangle = (-1)^n\big(m_{xy}[e_x e_y]G + \operatorname{lin}_e G\big)$.*

Proof. $\langle e_\nu, p_x p_y\rangle = \langle \Delta e_\nu, p_x \otimes p_y\rangle$ since p_x, p_y are primitive, and $\langle e_m, p_m\rangle = (-1)^{m-1}$. Each tensor factor must be a single e, so $\ell(\nu) \geq 3$ gives 0, $\langle e_n, p_x p_y\rangle = (-1)^n$, $\langle e_x e_y, p_x p_y\rangle = (-1)^n m_{xy}$, and other two-part ν give 0. □

[1] We have not seen [Mor77] first-hand; its scope is as described in [JL21, p. 12, before Thm. 3.2].

Proof of Theorem 10.7. By Proposition 10.1, $[(s-1)^2]c_{\lambda\mu} = [e_x e_y](\Gamma_a(e_b, e_c) + D_a(M_{bc}))$. In $\Gamma_a(e_b, e_c) = \sum_{r,q}(-1)^{r+q} e_{b-r} e_{c-q} T_a(p_r p_q)$, terms with $r < b$ and $q < c$ have at least three factors in I and do not contribute. For $r < b$, $q = c$, only the single-e part $\Xi_a(r, c)e_{a+r+c}$ of $T_a(p_r p_c)$ (Lemma 7.11) contributes; symmetrically for $r = b$, $q < c$. For $r = b$, $q = c$ the contribution is $(-1)^{b+c}[e_x e_y]T_a(p_b p_c) = (-1)^{b+c}U_a(b, c; x, y)$ by Lemma 10.8 and $\mathrm{lin}_e\, T_a(p_b p_c) = \Xi_a(b, c)$. □

Corollary 10.9. *Every lead at valuation 2 ($\kappa = \ell - 2$, any ℓ) has a closed formula: by Theorem 8.8 it is a sum of products of pair leads (Proposition 7.2) and $\ell = 3$, $\kappa = 1$ leads (Theorem 8.4, Corollary 9.3, Theorem 10.7).*

Example 10.10. $(3, 3, 3) \to (7, 2)$: Lead $= 2t^{13} + 3t^{12} + 3t^{11} + 6t^{10} + 6t^9 + 6t^8 + 9t^7 + 6t^6 + 3t^5 + 9t^4 + 6t^3 + 3t + 4$ (agrees with direct computation). Leads are *not* positive in general: $(4, 4, 2) \to (7, 3)$ has Lead $= (t + 1)(t^2 + 1)(2t^8 + t^7 + t^6 + 3t^4 + t^3 + t^2 - t + 2)$.

Computer checks. All statements in this section were checked against an independent implementation of Hikita's subset formula: Theorem 10.7 on 16 pairs with $n \le 12$ (both examples among them) and on 27 pairs in all 3 orderings with $n \le 10$; Theorem 10.3 in 229 cases ($a \le 5$); and the Green-polynomial dictionary symbolically for $|\lambda| \le 8$ and numerically for $|\lambda| \le 9$.

11. Open problems

(1) **Valuation ≥ 3.** Connected leads with $\ell \ge 4$, $\kappa = 1$. Hierarchy: $v = 1 \leftrightarrow p_n \leftrightarrow$ one t-string (Lemma 7.11); $v = 2 \leftrightarrow p_x p_y \leftrightarrow$ two strings (Theorem 10.3). Does the three-string shuffle sum factor pairwise?

(2) **Zero set at generic t.** At $t = -2$ seven pairs with $n = 6$ have $v > \ell - \kappa$; for coarsenings this is cancellation between histories, e.g. $\mathrm{Lead}_{(1,1,1),(3)} = [3]_t(2 + t)$. Describe the zero set of $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$ in t.

(3) **Counting.** A formula for $N(\lambda, \mu)$ in Theorem 7.6; for $\ell(\lambda) = 3$ and $\mu = (n-1, 1)$ with $\kappa = 1$ we observe $\mathrm{Lead}_{\lambda\mu}(1) = 2n^2 - 6n + 3$, independent of λ ($n \le 10$; computed only).

(4) **Positivity.** Leads are not t-positive (Example 10.10), so no "t-count of minimal flows" can hold. Is there a signed or cancellation-free model?

(5) **Cumulant link.** Is there a Macdonald-cumulant construction whose leads are the $\star$-leads for all t, i.e. cumulants of $e_k \mapsto t^{\binom{k}{2}}$?

Appendix A. Cherednik operators in our conventions

This appendix proves the facts about the operators T_i, $\pi_\circ$, Y_i° used in Section 6. These are textbook facts. For completeness we include short proofs in our conventions, and we claim nothing new here. The only external input is Bernstein's presentation, cited from [Hai06, Thm. 4.2]. Notation is as in Sections 2 and 6, and $K := \mathbb{Q}(s, t)$.

A.1. **Elementary relations.** Fix i and let $R_i \subset \mathrm{Pol}$ be the subring of s_i-invariant polynomials. Then T_i is R_i-linear, and Pol is a free R_i-module with basis $\{1, x_i\}$. From the definition, $T_i 1 = t$ and $T_i x_i = x_{i+1}$, so $T_i x_{i+1} = T_i(x_i + x_{i+1}) - T_i x_i = t x_i + (t-1)x_{i+1}$.

- *Quadratic relation.* $T_i^2 = (t-1)T_i + t$ holds on 1 (since $t^2 = (t-1)t + t$) and on x_i (both sides give $t x_i + (t-1)x_{i+1}$). This gives (R4).
- *(R1).* Both sides of $T_i x_i T_i = t x_{i+1}$ are R_i-linear. On 1 both give $t x_{i+1}$; on x_i both give $t x_i x_{i+1}$. If $j \notin \{i, i+1\}$, then $x_j \in R_i$.
- *(R2)* is immediate from the definition.
- *(R3)* follows as for (H5): $\pi_\circ$ intertwines s_j with s_{j+1} and multiplication by $x_{j+1}/(x_j - x_{j+1})$ with multiplication by $x_{j+2}/(x_{j+1} - x_{j+2})$, for $j \le m - 2$.

- *Braid relations.* We use these as stated in Section 2. Far commutation holds because T_i and T_k, $|i-k| \geq 2$, act on disjoint pairs of variables, each linearly over polynomials in the other pair. For $T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}$, both sides are linear over the polynomials that are symmetric in x_i, x_{i+1}, x_{i+2}. Over that ring, $\mathbb{Q}[x_i, x_{i+1}, x_{i+2}]$ is free on the six monomials $x_i^a x_{i+1}^b$ with $a \leq 2$, $b \leq 1$, so the relation reduces to a finite computation on these six monomials. We carried it out symbolically in s and t.

A.2. **Bernstein relations.**

Lemma A.1. *Let ϖ be an operator on* Pol *with $\varpi T_k = T_{k+1}\varpi$ for $1 \leq k \leq m-2$; this covers $\varpi = \pi$ by (H5) and $\varpi = \pi_\circ$ by (R3). Define $Y_i' := t^{m-i}T_{i-1}\cdots T_1 \varpi T_{m-1}^{-1}\cdots T_i^{-1}$. Then for $1 \leq i \leq m-1$:*

(a) $T_iY_i'T_i = tY_{i+1}'$;
(b) $T_iY_i' = Y_{i+1}'T_i - (t-1)Y_{i+1}'$ *and* $T_iY_{i+1}' = Y_i'T_i + (t-1)Y_{i+1}'$; *hence* $[T_i, Y_i' + Y_{i+1}'] = 0$ *and* $T_iY_i'Y_{i+1}' = Y_{i+1}'Y_i'T_i$;
(c) $T_iY_j' = Y_j'T_i$ *for* $j \notin \{i, i+1\}$.

If moreover the Y_i' commute, then $T_i f(Y') = f(Y')T_i$ for every $f \in K[y_1, \ldots, y_m]^{S_m}$.

Proof. (a) The trailing T_i^{-1} of Y_i' cancels against T_i. The T_i on the left extends $T_{i-1}\cdots T_1$ to $T_i\cdots T_1$, so the result is tY_{i+1}'.

(b) Multiply (a) on the right, respectively on the left, by $T_i^{-1} = t^{-1}T_i - (1-t^{-1})$. Adding the two relations cancels the $(t-1)Y_{i+1}'$ terms. Applying the first relation and then the second gives $T_iY_i'Y_{i+1}' = Y_{i+1}'T_iY_{i+1}' - (t-1)Y_{i+1}'^2 = Y_{i+1}'Y_i'T_i$.

(c) Case $j \geq i+2$. By the braid and far relations, $T_i\,T_{j-1}\cdots T_1 = T_{j-1}\cdots T_1\,T_{i+1}$. Then $T_{i+1}\varpi = \varpi T_i$, and T_i commutes with $T_{m-1}^{-1}, \ldots, T_j^{-1}$.

Case $j \leq i-1$. T_i commutes with $T_{j-1}, \ldots, T_1$, and $T_i\varpi = \varpi T_{i-1}$. The braid relation in the form $T_{i-1}T_i^{-1}T_{i-1}^{-1} = T_i^{-1}T_{i-1}^{-1}T_i$ gives $T_{i-1}\,T_{m-1}^{-1}\cdots T_j^{-1} = T_{m-1}^{-1}\cdots T_j^{-1}\,T_i$.

Finally, a symmetric f is a polynomial in $y_i + y_{i+1}$, y_iy_{i+1} and the y_j with $j \neq i, i+1$. If the Y' commute, then $f(Y')$ is the same polynomial in the corresponding operators, and each of these commutes with T_i by (b) and (c). □

A.3. **Commutativity.**

Theorem A.2. *The operators $Y_1^\circ, \ldots, Y_m^\circ$ commute pairwise on* Pol.

What we cite. Let $W_e = Y \rtimes W_0 = \Omega \ltimes W_a$ be the extended affine Weyl group, where Ω is the stabiliser of the fundamental alcove. Let $B(W_e) = \Omega \ltimes B(W_a)$ be the extended affine braid group, in which Ω acts on $B(W_a)$ through its action on the Coxeter generators [Hai06, §§3.9–3.13, 4.1]. For $\varpi \in \Omega$ write $\varpi = y^{\lambda_\varpi}v_\varpi$ with $v_\varpi \in W_0$ [Hai06, (13)]. Let $B(Y, W_0)$ be the right affine braid group of [Hai06, §2.9]. It is generated by the finite braid group $B(W_0)$ and the abelian group Y, subject to the relations [Hai06, (8)–(9)]

$$T_iy^\lambda T_i^{-1} = y^\lambda \ \text{ if } \langle \lambda, \alpha_i^\vee \rangle = 0, \qquad T_i^{-1}y^\lambda T_i^{-1} = y^{s_i\lambda} \ \text{ if } \langle \lambda, \alpha_i^\vee \rangle = 1.$$

Bernstein's theorem [Hai06, Thm. 4.2] gives an isomorphism $\Phi\colon B(Y, W_0) \to B(W_e)$ that is the identity on $B(W_0)$ and sends $y^{\lambda_\varpi} \mapsto \varpi T_{v_\varpi^{-1}}$. We use only one consequence of it: *the elements $\Phi(y^\lambda)$, $\lambda \in Y$, commute in $B(W_e)$.*

The dictionary for GL_m. Take $Y = \mathbb{Z}^m$ and $\alpha_i = \alpha_i^\vee = e_i - e_{i+1}$ [Hai06, Ex. 2.5]. The fundamental alcove is $\{x_1 \geq \cdots \geq x_m \geq x_1 - 1\}$. Let v be the cyclic permutation $e_j \mapsto e_{j+1}$, indices mod m. Then $y^{e_1}v$ sends x to $(x_m+1, x_1, \ldots, x_{m-1})$, so it preserves the alcove. Hence it is the generator ϖ of $\Omega \cong \mathbb{Z}$ with $\lambda_\varpi = e_1$ and $v_\varpi = v = s_1s_2\cdots s_{m-1}$. Its inverse $v^{-1} = s_{m-1}\cdots s_1$ is reduced: in one-line notation it is $(m, 1, \ldots, m-1)$, of length $m-1$. So $\Phi(y^{e_1}) = \varpi T_{m-1}\cdots T_1$.

Conjugation by ϖ sends $s_i \mapsto s_{i+1}$ for $i \le m-2$, sends $s_{m-1} \mapsto y^{e_1-e_m} s_{e_1-e_m} = s_0$, and sends $s_0 \mapsto s_1$. So in $B(W_e)$ we have $\varpi T_j \varpi^{-1} = T_{j+1}$, indices mod m. By the second displayed relation with $\lambda = e_j$, induction on i gives

$$\Phi(y^{e_i}) = T_{i-1}^{-1} \cdots T_1^{-1}\, \varpi\, T_{m-1} \cdots T_i.$$

The assignment $T_j \mapsto T_j^{-1}$ $(0 \le j \le m-1)$, $\varpi \mapsto \varpi$ preserves all defining relations of $B(W_e)$. Indeed, $ABA = BAB$ implies $A^{-1}B^{-1}A^{-1} = B^{-1}A^{-1}B^{-1}$, and the relation $\varpi T_j \varpi^{-1} = T_{j+1}$ is preserved. So it defines an involutive automorphism ι. It maps $\Phi(y^{e_i})$ to $W_i := T_{i-1} \cdots T_1\, \varpi\, T_{m-1}^{-1} \cdots T_i^{-1}$. Hence *the W_i commute in $B(W_e)$*. As a consistency check, ι applied to $T_j^{-1} y^{e_j} T_j^{-1} = y^{e_{j+1}}$ gives $W_{j+1} = T_j W_j T_j$, which is Lemma A.1(a) for $Y_i^\circ = t^{m-i} W_i$.

The representation. For $m \ge 3$, $B(W_e)$ is generated by $T_0, \dots, T_{m-1}, \varpi^{\pm 1}$, subject to the braid relations of type $\tilde{A}_{m-1}$ (indices mod m) and $\varpi T_j \varpi^{-1} = T_{j+1}$. For $m = 2$ there is no relation between T_0 and T_1. Set $\rho(T_j) = T_j$ for $1 \le j \le m-1$, $\rho(\varpi) = \pi_\circ$ and $\rho(T_0) := \pi_\circ T_{m-1} \pi_\circ^{-1}$. These operators are invertible: the T_j by (R4), and $\pi_\circ$ because it is a K-algebra automorphism of Pol. It remains to check the relations.

- Among $T_1, \dots, T_{m-1}$ they are the braid relations of Appendix A.1.
- $\pi_\circ T_j \pi_\circ^{-1} = T_{j+1}$ for $j \le m-2$ is (R3).
- $\pi_\circ^2 T_{m-1} \pi_\circ^{-2} = T_1$: the automorphism $\pi_\circ^2$ maps $x_{m-1} \mapsto s x_1$ and $x_m \mapsto s x_2$, so it intertwines s_{m-1} with s_1. It maps $x_m/(x_{m-1} - x_m)$ to $s x_2/(s x_1 - s x_2) = x_2/(x_1 - x_2)$.
- The relations involving T_0 follow by conjugation. $\pi_\circ T_0 \pi_\circ^{-1} = T_1$ is the previous item. Conjugating by $\pi_\circ^{-1}$, respectively $\pi_\circ^{-2}$, turns the (T_{m-1}, T_0) and (T_0, T_1) braid relations into the (T_{m-2}, T_{m-1}) braid relation. It turns $T_0 T_j = T_j T_0$ $(2 \le j \le m-2)$ into a far relation among $T_1, \dots, T_{m-1}$.

So ρ is a representation of $B(W_e)$ on Pol, and $\rho(W_i) = t^{-(m-i)} Y_i^\circ$. This proves Theorem A.2. □

Remark A.3 (The twisted operators). For $\pi = x_1 \pi_\circ$ the same relations hold on the Laurent polynomial ring, on which π is invertible. On Pol, however, π is not surjective ($1 \notin \pi(\mathrm{Pol})$). So Pol is a module only for the subalgebra generated by the $T_j^{\pm 1}$ and π, and not for π^{-1}. The words W_i use π but never π^{-1}. Hence the Y_i of Section 2 commute on Laurent polynomials, and therefore on Pol. This also follows from Proposition 6.7.

A.4. **Triangularity and simple spectrum.** For a composition $\mu \in \mathbb{Z}_{\ge 0}^m$ write $P_q(\mu) = \mu_1 + \cdots + \mu_q$. Let $\mu \le_c \lambda$ mean $P_q(\mu) \le P_q(\lambda)$ for all q. For compositions of the same size, define $\mu \prec \lambda$ if either $\mathrm{sort}(\mu) \lhd \mathrm{sort}(\lambda)$, or $\mathrm{sort}(\mu) = \mathrm{sort}(\lambda)$, $\mu \ne \lambda$ and $\mu \le_c \lambda$. This is a strict partial order. Within one S_m-orbit it puts the decreasing rearrangement on top.

Theorem A.4. *For all i and all compositions λ, $Y_i^\circ x^\lambda = s^{\lambda_i} t^{\mathrm{rk}_i(\lambda)} x^\lambda + \sum_{\mu \prec \lambda} c_\mu x^\mu$, with rk_i as in Section 6.*

T on monomials. Write (a, b) for the exponents of (x_j, x_{j+1}); the other variables are spectators. Let $B(a,b) := \sum_{\min(a,b) < c < \max(a,b)} x_j^c x_{j+1}^{a+b-c}$ be the sum of the "between" terms. Summing the geometric series in the definition of T_j gives:

$$T_j x^{(a,b)} = \begin{cases} x^{(b,a)} + (1-t)B, & a > b, \\ t x^{(b,a)} + (t-1) x^{(a,b)} + (t-1)B, & a < b, \\ t x^{(a,a)}, & a = b; \end{cases}$$

$$T_j^{-1} x^{(a,b)} = \begin{cases} t^{-1} x^{(b,a)} - (1-t^{-1}) x^{(a,b)} + t^{-1}(1-t)B, & a > b, \\ x^{(b,a)} + (1-t^{-1})B, & a < b, \\ t^{-1} x^{(a,a)}, & a = b. \end{cases}$$

The second table follows from the first by (R4).

The dominance filtration. If $b < c < a$, then replacing two entries a, b of a composition by $c, a+b-c$ makes its sorted version strictly smaller in dominance. Indeed, $P_k(\operatorname{sort} v)$ is the maximum of $\sum_{p\in S} v_p$ over k-sets S. A k-set containing exactly one of the two changed positions can trade that position for the position of a without decreasing the old sum. So no partial maximum increases, and since the multisets differ, some partial maximum strictly decreases. Let F_ν, respectively $F_{\lhd\nu}$, be the span of the x^μ with $\operatorname{sort}(\mu) \trianglelefteq \nu$, respectively $\lhd \nu$. By the tables, $T_j^{\pm 1}$ preserve both spaces, and so does $\pi_\circ$, which acts on monomials by

$$\pi_\circ x^\mu = s^{\mu_m} x^{(\mu_m,\mu_1,\dots,\mu_{m-1})}.$$

So Y_i° acts on $\mathrm{gr}_\nu = F_\nu/F_{\lhd\nu}$, which has basis the x^μ with $\operatorname{sort}(\mu) = \nu$. On gr_ν the between terms vanish. Let (p, q) be the entries in positions $j, j+1$. Then T_j^{-1} swaps them with coefficient 1 if $p < q$, and multiplies by t^{-1} if $p = q$. If $p > q$, it swaps them with coefficient t^{-1} and also "stays" with coefficient $-(1-t^{-1})$. Likewise, T_j swaps with coefficient 1 if $p > q$, multiplies by t if $p = q$, and if $p < q$ swaps with coefficient t and stays with coefficient $t-1$. (When $p = q$ we count the move as a swap.)

Paths. Expand $Y_i^\circ x^\lambda$ in $\mathrm{gr}_{\operatorname{sort}(\lambda)}$ as a sum over paths, each binary step choosing "swap" or, where allowed, "stay". The operator T_i^{-1} acts first, then $T_{i+1}^{-1}, \dots, T_{m-1}^{-1}$, then $\pi_\circ$, then $T_1, \dots, T_{i-1}$.

- *Phase 1* (T_j^{-1}, $j = i, \dots, m-1$). Before step j, position j holds a carried value w_j, with $w_i = \lambda_i$, and position $j+1$ still holds λ_{j+1}. A swap deposits $D_j = \lambda_{j+1}$ at position j and sets $w_{j+1} = w_j$. A stay is allowed only if $w_j > \lambda_{j+1}$; it deposits $D_j = w_j$ and sets $w_{j+1} = \lambda_{j+1} < w_j$. In both cases $D_j + w_{j+1} = w_j + \lambda_{j+1}$, and w is non-increasing. Put $e := w_m$.
- $\pi_\circ$ moves e to position 1 with the factor s^e, and shifts the other entries one place to the right.
- *Phase 2* (T_j, $j = 1, \dots, i-1$). Before step j, position j holds u_j, with $u_1 = e$, and position $j+1$ holds λ_j. A swap deposits $g_j = \lambda_j$ and sets $u_{j+1} = u_j$. A stay is allowed only if $u_j < \lambda_j$; it deposits $g_j = u_j$ and sets $u_{j+1} = \lambda_j > u_j$. In both cases $g_j + u_{j+1} = u_j + \lambda_j$, and u is non-decreasing.

The path outputs $\mu = (g_1, \dots, g_{i-1}, u_i, D_i, \dots, D_{m-1})$. For $q \le i-1$, telescoping phase 2 gives $P_q(\mu) = P_q(\lambda) - (u_{q+1} - e) \le P_q(\lambda)$. For $q \ge i$, telescoping phase 1 gives $\sum_{j=q}^{m-1} D_j = \sum_{p>q} \lambda_p + (w_q - e)$, hence $P_q(\mu) = P_q(\lambda) - (w_q - e) \le P_q(\lambda)$. Equality for all q forces u and w to be constant. Every stay changes the carried value, so equality forces the all-swap path, which outputs $\mu = \lambda$. The all-swap path has coefficient $t^{m-i} s^{\lambda_i} \prod_{j>i,\ \lambda_j \le \lambda_i} t^{-1} \prod_{j<i,\ \lambda_j \ge \lambda_i} t$, and

$$(m-i) - \#\{j > i : \lambda_j \le \lambda_i\} + \#\{j < i : \lambda_j \ge \lambda_i\} = \mathrm{rk}_i(\lambda).$$

Lifting from gr to $F_{\operatorname{sort}(\lambda)}$ adds only terms with $\operatorname{sort}(\mu) \lhd \operatorname{sort}(\lambda)$. This proves Theorem A.4. □

For example, take $m = 3$, $\lambda = (2, 0, 1)$ and $Y_1^\circ = t^2 \pi_\circ T_2^{-1} T_1^{-1}$. In gr the result is $s^2 x^{(2,0,1)} - s(t-1)x^{(1,0,2)} - t(t-1)x^{(0,2,1)}$. The partial sums $(1, 1, 3)$ and $(0, 2, 3)$ are both $\le (2, 2, 3)$. In Pol there is one more term, $-s(t-1)x^{(1,1,1)}$, which is strictly lower in dominance.

Corollary A.5. *On each* Pol_d *the* Y_i° *are simultaneously diagonalisable, with simple joint spectrum. There is an eigenbasis* $\mathcal{E}_\lambda = x^\lambda + \sum_{\mu \prec \lambda} (\cdots)$ *with* $Y_i^\circ \mathcal{E}_\lambda = s^{\lambda_i} t^{\mathrm{rk}_i(\lambda)} \mathcal{E}_\lambda$. *For each* λ, $i \mapsto \mathrm{rk}_i(\lambda)$ *is a bijection* $[m] \to \{0, \dots, m-1\}$.

Proof. Rank the indices j by "larger λ_j first, ties broken by smaller j". Then $\mathrm{rk}_i(\lambda)$ is the number of indices ranked before i, which gives the bijection. Since the s-degree of $y_i(\lambda)$ is λ_i, the vectors $y(\lambda)$ are pairwise distinct. Choose $c \in \mathbb{Q}$ such that the values $z(\lambda) = \sum_i c^{i-1} y_i(\lambda)$, $|\lambda| = d$, are

pairwise distinct; each pair excludes at most $m-1$ values of c. By Theorem A.4, $Z := \sum_i c^{i-1} Y_i^\circ$ is triangular with respect to a linear extension of $\prec$, with distinct diagonal entries $z(\lambda)$. So Z is diagonalisable, with one-dimensional eigenspaces $K\mathcal{E}_\lambda$, where $\mathcal{E}_\lambda = x^\lambda + \sum_{\mu \prec \lambda} (\cdots)$. By Theorem A.2 each Y_i° commutes with Z, so it preserves each line $K\mathcal{E}_\lambda$. Its eigenvalue there is the coefficient of x^λ in $Y_i^\circ \mathcal{E}_\lambda$, which is $y_i(\lambda)$ by Theorem A.4. □

Remark A.6. Theorem A.4 and the path expansion were compared with a direct implementation: the path sum agrees with the actual same-orbit part of $Y_i^\circ x^\lambda$ coefficient by coefficient, for $m = 2, 3, 4, 5$ in degrees $\leq 5, 4, 3, 2$. The order $\leq_c$ is what the proof controls. The reversed order within an orbit is violated already for $m = 3$, $d = 3$.

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